



TRAINING MANUAL ON HANDS ON TRAINING ON DISEASE MANAGEMENT AND DIAGNOSTIC IN BRACKISHWATER AQUACULTURE

13-17 February, 2012



काकद्वीप शोध केन्द्र

केन्द्रिय खारा जलजीव पालन आनुसंधान संस्थान
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Kakdwip Research Centre

Central Institute of Brackishwater Aquaculture

(Indian Council of Agricultural Research)

Kakdwip, 24 Pgs (S), W.B. – 743347, India

TRAINING MANUAL
ON
HANDS ON TRAINING ON DISEASE MANAGEMENT AND DIAGNOSTIC IN
BRACKISHWATER AQUACULTURE

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Sponsored by
National Fisheries Development Board (NFDB), Hyderabad



Course Coordinator
Dr. R. Ananda Raja

Course Co-coordinator
Dr. Sujeet Kumar

Edited by
Dr. S. M. Pillai
Dr. K. P. Jithendran
Dr. P. K. Patil
Dr. S. K. Otta
Dr. R. Ananda Raja
Dr. Sujeet Kumar



Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture

Indian Council of Agricultural Research

Kakdwip, South 24 Parganas, West Bengal - 743 347. India



Published by
Dr. A. G. Ponniah
Director
CIBA, Chennai

Dr. R. Ananda Raja
Course Coordinator
Scientist, KRC of CIBA

Dr. Sujeet Kumar
Course Co-coordinator
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Central Institute of Brackishwater Aquaculture



Dr. A. G. PONNIAH
Director

FOREWORD

Aquaculture, the fastest growing food producing sector in India with enormous possibilities for further increase in production through horizontal expansion by harnessing the all existing water bodies and productivity per unit area. Brackishwater aquaculture remains at the centre stage in the economy of coastal states of the country. Shrimp culture took a shape of booming industry in early eighties. But due to outbreak of diseases, particularly by White spot virus (WSSV) and Monodon baculovirus (MBV), the industry reached to virtual collapse stage within few years after outbreak. The bacterial diseases like luminescent Vibriosis remain a major problem in hatcheries. With the growing demand and start of culture of *Litopenaeus vannamei* many other deadly viruses of shrimp such as Yellow head virus and Taura syndrome are expected to emerge in future. Similarly, for quite long time it was considered that finfishes are somewhat resistant to diseases compared to shrimp. But often parasitic diseases have been reported to be problematic for fishes. Viral diseases like Viral nervous necrosis and Iridovirus infection have also been reported from many parts of the world, including India. These diseases appear to have significant impact considering the growing interest among farming community for the culture of Asian seabass with availability of inputs and commercialized technologies in Central Institute of Brackishwater Aquaculture, Chennai.

After ban on uses of antibiotics in aquaculture, there is a strategic shift towards disease prevention, rather than control. Adoption of many preventive strategies at farm level such as zero water exchange, application of disinfectants and probiotics, and adoption of biosecurity measures help in disease prevention. This signifies the importance of timely diagnosis of diseases by adopting conventional and molecular approaches. Fortunately, considerable developments have taken place in last few decades in diagnostic which help us to diagnose disease with precision and on time. CIBA has the credit at its helm to develop a PCR based diagnostic kit for the diagnosis of WSD and many other diseases.

I am grateful to the National Fisheries Development Board (NFDB), Hyderabad for sponsoring the training program on “Hands on Training on Disease Management and Diagnostic in Brackishwater Aquaculture” during 13th to 17th February, 2012. I hope that it will help the trainees from different parts of the country in enhancing their skills for disease diagnosis, prevention and control. The topics covered in the training manual will guide the participants step by step in the area of disease diagnosis and management. I express my best wishes and compliment the Course coordinator, Dr. R. Ananda Raja, Co-coordinator Dr. Sujeet Kumar and their team for conducting the training program in right time at Kakdwip. I wish all success of the training program.

Chennai
13th February, 2012

A. G. Ponniah
Director



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Kakdwip Research Centre
Central Institute of Brackishwater Aquaculture



डॉ. आर. आनन्द राजा

Dr. R. ANANDARAJA

Scientist and AAHED
Course Coordinator

PREFACE

The aquaculture sector has been recognized as a powerful income and employment generator with a number of subsidiary industries, and is a source of cheap and nutritious food besides being a good foreign exchange earner. As the world's population continues to increase exponentially, the demands for food production remain challenging and so also the new innovative sustainable technologies. With the adoption of new technologies in aquaculture, occurrence of disease associated problems are also quite inevitable. Disease outbreaks have been one of the major challenges faced by aquaculture industry. Pioneering research has been carried out at CIBA in the frontier areas of disease diagnostics and therapeutics. While being the sole research centre of CIBA, KRC has made significant contributions in development of Aquaculture Database System in Brackishwater (ADS-BW) containing a complete information including the disease prevalence in brackishwater farms in West Bengal and Odisha. Realizing the importance of disease outbreaks in brackishwater aquaculture, this training programme was proposed to the National Fisheries Development Board (NFDB), Hyderabad with an intension to expose the fellow researchers, academicians, extension specialists and members from coastal NGOs to the latest developments in the field. The training is being organized for duration of five days from 13th to 17th February, 2012 and fully sponsored by NFDB.

This manual is the outcome of the lectures presented by various faculties. I am overwhelmed by the tremendous response received from various institutions to participate in this program. I take this opportunity to thank Dr. A.G. Ponniah, Director, CIBA for his cooperation and guidance in organizing this training successfully. I express my gratitude to Dr. S. M. Pillai, OIC, PME Unit, Dr. K. P. Jithendran, SIC, AAHED and Dr. N. Kalaimani (Rtd), CIBA who have been instrumental at different stages of official formalities and helped in shaping this tome scientifically attractive. My heartfelt thanks are due to Dr. J. K. Sundaray, OIC, Kakdwip Research Centre, Dr. Sujeet Kumar, Co-coordinator and beloved colleagues Dr. T. K. Ghoshal, Dr. Debasis De, Dr. Ashutosh D. Deo and Mrs. Shyne Anand, P.S. who gave me official and moral strength to complete this task.

This program would have been a tough task to complete without the full support of division heads, SICs, all the scientists, administrative, technical and skilled supportive staffs at CIBA and KRC of CIBA. So, I am immensely indebted to all the staffs who worked hard in bringing laurel to KRC of CIBA. I am sure that this manual would enable the participants in enhancing their knowledge and competence in the field of aquaculture disease diagnosis and management.

Kakdwip
13th February, 2012

R. Ananda Raja
Scientist and AAHED
KRC of CIBA
Course Coordinator

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STRATEGIES FOR ENHANCING PRODUCTION OF BRACKISHWATER AQUACULTURE IN INDIA

J. K. Sundaray, G. Biswas, Ashutosh D Deo and P. S. Shyne Anand

Aquaculture has developed rapidly over the last three decades and has become as an important economic activity world wide. The ever increasing population and the rising demand for animal protein is causing pressure on fisheries development globally. Fish and fishery products contribute around 15% of the animal protein supporting the nutritional security. The world fish production is in the order of 130 million metric tons, of which 50% is contributed by aquaculture. The contribution of aquaculture to the total fish production was around 15% in 1990 which has grown about 50% in 2005. More than 250 different species of plants and animals are being cultured world wide. There is a constant increase in the diversification of species every year. The aquaculture development was contributed by the top 10 species in 1990s and in recent years about 25 species account for 90% of the contribution indicating that species diversification is increasing. India is a major maritime state and an important aquaculture country in the world. The sector has high potentials and development is imperative for rural development, domestic nutritional security, employment generation, gender mainstreaming and export earnings. It plays important role in the socio-economic development and human welfare in the country. The 8118 kms long coastline of India has a very rich biodiversity. It has a vast brackishwater area (1.2 million ha) available for aquaculture out of which, till 2002-2003, only 1,52,080 ha has been utilized for shrimp culture, leading to a total production of 1,15,320 MT of shrimps.

The country also has an important role in global fisheries as the third largest producer of fish in the world and higher enhancement levels as compared to world fish production levels (Table 1)

Table 1 Fish Production in World and India

Year	World, million tonnes	% change	India, million tonnes	% change	India's share, %
1950-51	23.50	-	0.75	-	3.19
1960-61	43.60	85.53	1.16	54.67	2.66
1970-71	66.20	51.83	1.76	51.72	2.66
1980-81	72.30	9.21	2.44	38.64	3.37
1990-91	98.26	35.91	3.84	57.38	3.91
2000-01	129.00	32.35	5.66	47.40	4.39
2003-04	140.48	8.90	6.40	13.07	4.56
2006-07	--	--	6.87	--	--

Present status of Brackish water Aquaculture Research in India

With the establishment of Central Institute of Brackishwater Aquaculture in April, 1987 under VII plan, in last 25 years some important milestones have been achieved by the Institute. These are

- Development of technology for improved traditional system of farming in Sunderban with an average production of 1.5 ton of *P.monodon*/ha/120 days.
- Extensive investigation on the white spot virus disease
- Development of nested PCR diagnostic test for shrimp white spot virus and launching.
- Development of technology package for shrimp feed processing and production in commercial scale.
- Domestication of *Lates calcarifer* under captive condition and round the year seed production of seabass.
- Grow out demonstration of seabass in pond in coastal states of India.
- On farm demonstration of polyfarming of fish and shrimps.
- Closing of life cycle of *Marsupenaeus japonicus* under captive condition and production of F 3 generation brood stock
- Grow out demonstrations of *Fenneropenaeus merguensis*.
- Captive brood stock maintenance, induced maturation, breeding and completion of life cycle of crab- *Scylla tranquebarica* & *S.serrata*.
- Field testing of immunostimulant for *Penaeus monodon*
- Development zero water exchanged based shrimp farming system
- Development of Shrimp production technology on Organic principles
- Demonstration of improved traditional system of culture of *P.monodon* in Pokkali fields of Kerala.
- Extensive investigation on the socio economic and environmental impact of shrimp farming in Andhra Pradesh, Orissa, West Bengal and Tamil Nadu.
- Survey of shrimp farming practices under different system to study the problems and constraints of technology adoption faced by the shrimp farmers.
- Transfer of technology through Exhibition, Farmers meet and training etc.

Need for brackishwater aquaculture development in India

The global average per capita consumption of fish is around 15 kg. The present average per capita consumption in India is around 9 kg. Even in countries like Japan and some of the South East Asian countries the average per capita consumption is more than 100 kg. Even to reach the global average of 15 kg, taking into consideration of 50% of Indian population will be fish consumers; by 2020 the domestic requirement itself will be in the order of 9 million tons. The average fish production in India is around 7 million tons equally contributed by fresh water and marine. The maximum sustainable yield is static and the capture fisheries trend is declining. It is necessary that the coastal aquaculture has to make a greater contribution in the fish production in the Indian context. By 2020, the coastal aquaculture is expected to support to the tune of around 3, 50,000 tons of which, from the current production of around 1, 50,000 tons. This implies that a quantum jump has to be made in the ensuing years. Out of this, shrimp is expected to contribute around 250,000 tons and rest has to come through fishes and other non conventional groups.

Status of Brackishwater Aquaculture

Coastal aquaculture is a traditional practice in India. In the low lying fields of Kerala (Pokkali), West Bengal (bheries and gheries), Orissa, Goa (Khazan) and Karnataka (Kar) which experiences influx of salt water, traditional farming of fish/shrimp were practiced. The practice is just allowing juveniles of fish/shrimp in the fields and allowing them to grow, supplementary feeding sometimes, facilitating water exchange through tidal waters and harvesting periodically at 3-4 months. With the improvement of technologies and realizing the importance of aquaculture, these practices were improved with the supplementary stocking and water quality management with moderate to higher production. The technology improvement made in the aquaculture sector opened new areas for the scientific farming which is called as semi-intensive and intensive farming following all the protocols for farming with production as much as 10 tons per ha per culture period of 4-5 months mainly shrimp and brackishwater fish like seabass production of 3 to 4 tonnes/ha/crop in the coastal area. The technology advancement helped in the establishment of more than 390 shrimp and 1 crab hatcheries. The coastal aquaculture witnessed a phenomenal growth during 1980s and in the beginning of 1990s. But the growth is not progressed as visualized from the later part of 1990s due to socio economic, environmental issues coupled with the outbreak of uncontrollable diseases like white spot syndrome virus (WSSV), Monodon slow growth diseases, loose shell syndrome diseases etc on shrimp. One of the reasons attributed for this is the unregulated development and unforeseen disease outbreaks. The coastal aquaculture in

India was also solely dependent on single species Tiger shrimp. The effect of which has brought the pronounced impact on the coastal farming sector questioning the very sustainability of the coastal aquaculture.

Diversification in the Brackishwater aquaculture

The estimated potential area under coastal aquaculture 1.24 million ha out of which only 10% area has been brought under the aquaculture. Considering the production potential of the sector, its production is projected to grow by four-fold by 2020 from the present at 0.14 million tonnes. Development of Indian coastal aquaculture in the country is driven by the technologies for seed production of tiger shrimp, *Penaeus monodon* and the white shrimp *P. indicus*. With the tiger shrimp being almost the single commodity commercially being produced in the area, there are established hatcheries for the purpose. With most hatcheries using wild spawners for seed production, contamination of wild populations of shrimps with white spot virus has been a major cause of concern in recent years. Disease outbreaks due to white spot virus during the last 10-12 years have acutely affected shrimp farming in the country. With the diagnostic kits developed for detecting the white spot virus, PCR-tested seed is available all over the country. Supplementary feeding forms the most important management measure in commercial shrimp farming. Commercial shrimp farming in India largely involves use of formulated pellet feed, constituting a significant share of the input expenditure. While bulk of the feed used was imported from Southeast Asian countries till a decade back, it is at present mainly produced in the country. Non-availability of quality ingredients, especially the fish meal, has been a major constraint faced by these industries, requiring import. The high price of the commercial feed, however, is forcing the small-scale farmers to resort to farm-made feeds. For the sustainable eco-friendly aquaculture practice, diversification to other species is considered as one of the important component. Fishes like Asian seabass (*Lates calcarifer*), Grouper (*Epinephelus tauvina*), Snappers (*Lutjanus* Sp.) which are high value carnivorous fishes and Grey mullet (*Mugil cephalus*), Milk fish (*Chanos chanos*), Pearl spot (*Etroplus suratensis*), Rabbit fish (*Siganus*Sp.) which are herbivorous/omnivorous farming in the coastal eco-system are available. The species like Cobia (*Rachycentron canadum*) and Silver pomfret are being considered as candidate species for farming. Efforts have been made to develop comprehensive technology packages for seed production under controlled conditions and farming for these candidate species. Technologies have been developed else where in the world. In Indian scenario the successful technology has been developed for the seed production of Asian Seabass, *Lates calcarifer* under controlled conditions and farming by the Central Institute of Brackishwater

Aquaculture. The controlled breeding of Groupers *Epinephelus tauvina* and Grey Mullet *Mugil cephalus*, Pearlscale *Etroplus suratensis* has also been successful. In addition to this a new avenues have come by successful breeding and seed production of ornamental fish, spotted scat, *Scatophagus argus*. Marine sea weeds have been a new area for brackishwater farming in different coastal states of India. Recently, some enterprising farmers in Tamil Nadu, Andhra Pradesh and Maharashtra have taken up seabass culture. High export prices of crabs have made fattening of species like *Scylla serrata* and *S. tranquebarica* a remunerative farming practice.

Aquaculture resources and Production systems

India is located in the monsoon belt and has a vast expanse of water resources, both marine and inland. These waters offer immense potential for fisheries and aquaculture, with over 10% of the global fish biodiversity available in the country. While capture fisheries in both marine and inland waters has brought into focus the need for natural resource management, sustainability is presently the concern in aquaculture systems in marine, coastal and freshwater environments. The aquaculture practices are akin to animal husbandry, where fish seed is stocked, fed, fattened and harvested in specially prepared/community ponds. Aquaculture is also practised in different types of waters, e.g. fresh, coastal and marine waters. It is recognized as the major area for future growth. Among these, brackishwater is most important and has major contribution in national fish production. Aquaculture is practiced in different scales and intensities. The intensive and super-intensive aquaculture practices need to ensure sustainability of ecosystem environment.

Strategies for enhancing production from brackishwater area

The issues in fisheries and aquaculture that need to be addressed pertain to biodiversity loss and depletion of fish stocks, excess coastal fishing, oceanic and deep sea fisheries, enhancing fish productivity in all cultivable waters, impact of climate change on fisheries, trans-boundary fisheries issues, inland and coastal pollution, large scale sedimentation of rivers, estuaries and lakes/wetlands, effective compliance of code of conduct of responsible fisheries, lack of diversification in aquaculture practices, water management in aquaculture, quality seed and relevant certification measures, planning for feed provision in farming, introduction of exotics with due quarantine procedures, emergence of new diseases, mechanization in fisheries and aquaculture, cold chain and hygienic fish handling, quality assurance issues in exports, overseas market fluctuations, disaster management, credit and insurance, inadequate database and poor linkage in domestic marketing. Demand projections for fish by 2012 is 9.74 million tonnes at an estimated annual

consumption growth rate of 3.5% and that by 2020 is about 12.5 million tonnes. We need to have different mode of strategies to develop brackishwater aquaculture. Short and long term strategies have also been developed for enhancing the production from brackishwater area which is mentioned below.

Short-term strategies

Enhancing area under cultivation

Presently only 15 % of the potential area has been developed into shrimp farms. The reason for the slow development may be attributed to the following:

- Most of the potential areas are under public lands and are located in areas where the basic infrastructures are not available.
- Land leasing policy in most of the coastal states is not standardized.
- The revenue classification of lands needs reclassification. For example, most of the unproductive agricultural lands could not be converted to shrimp farms.
- The capital investment required for construction of shrimp farms is high at about Rs. 3 to 4 lakhs/ ha.

Creation of infrastructure facilities by the government in the form of ‘Aquaculture Estates’ and reclassifying the land as per their present condition and cluster development through society and cooperative mode will help in increasing the area under culture.

Improving traditional system of farming

Nearly 40% of the total areas under shrimp farming in the country are following traditional practices of farming. The average production in these systems is only 500 kg/ ha/ year, while the national average is 750 kg/ha/year. Coastal Aquaculture Authority has brought out guidelines for improved traditional system of farming, where the production levels can be easily doubled. CIBA at its Kakdwip Research Centre has developed a technology for improved system of farming in tide-fed systems and consistent production of 1 to 1.5 tonnes of *Penaeus monodon*/ha/crop is being achieved. In recent years, another closed system of farming in small inland fresh water ponds is practiced by the farmers of West Bengal, particularly in Contai and South 24Paraganas (South) region and produce 3-4 tons of shrimp per ha/crop at a higher stocking density. PCR (Polymerase chain reaction) tested selective stocking with hatchery produced, disease free seed, strict feed management, health management and water quality management are the highlights of this improved system of farming.

Control of diseases through Good Aquaculture Practices (GAqPs)

During the last few years, globally shrimp production has received a set back due to the outbreak of viral diseases. In India, it is estimated that there is an annual loss of about to tune of Rs. 500-700 crores due to white spot virus infection in shrimps. The viral diseases cannot be cured but could be controlled following **GAqPs**, which will avoid stress to the cultured shrimps. Lack of awareness, poor biosecurity measures and improper design and poor management of farms are the major causative factors for the development of viral diseases. The white spot virus is known to be transmitted both vertically and horizontally. Stocking PCR tested seed will reduce the chance of vertical transmission. Horizontal transmission can be avoided only when all the farmers in an area adopt **GAqPs** and function as a unit under complete cooperation and consideration for fellow farmers.

Long-term strategies

Utilization of inland saline soils

In India, a total of about 8.5 million hectares of salt affected soils are distributed in the inland and coastal areas of the country. Out of the total 6.5 million ha of salt affected soil available in the Inland areas, 4.0 million ha is available in non-maritime states.

In recent years, in countries like USA and Australia technologies have been developed for the culture of shrimps in low-saline inland waters. Such technologies if developed and adopted in the inland salt affected areas in the states of Rajasthan, Punjab, Haryana, UP and Bihar, the production levels of shrimps from the country will increase many fold. ICAR has taken up this as an important thrust area and institutes like Central Institute of Fisheries Education; Mumbai is in the process of testing the technologies in these states. CIBA could successfully complete the demonstration of fish and shrimp farming in inland saline water during 2000 to 2003 with collaboration of state agricultural university.

Organic farming of shrimps

In today's context, where export markets are governed by WTO (world trade organisation) treaties, it becomes increasingly important to follow the quality standards set by the importing countries. Antibiotics residues in the imported shrimps are one such issue which needs to be addressed immediately. Further, the importance of 'ecolabelling' and 'traceability' issues needs to be considered. In this context, developing technologies of organic shrimp farming is of utmost importance. CIBA at its Kakdwip Reserach Centrea has initiated the process for developing shrimp in organic principles.

Genetic improvement of shrimp stock through selective breeding

The important genetic traits like growth and disease resistance can be improved by genetic improvement of shrimp and fish stock and will lead to an increase in the production levels. Development of Specific Pathogen Free (SPF) and Specific Pathogen Resistant (SPR) shrimps and fish are an important area of long-term study.

Improved culture methods

The traditional culture method is improved with stocking of uniform sized seed at specific density and fed with low cost trash fishes/formulated feed of required quantity. Water quality is maintained with exchange periodically. Fishes are allowed to grow to marketable size, harvested and marketed for high unit price. Seabass and other brackishwater fin fish culture can be done in more organized manner as a small-scale/large scale aquaculture in brackishwater and freshwater ponds in cages.

Poly culture method

This is an improvement over the traditional method, where the feed, the live fishes, shrimps are deliberately allowed in to the seabass culture ponds to serve as facilitating feed for the seabass in the pond. In the traditional method there is no control over the quantity and quality of the feed entering the ponds which may or may not be adequate. At times of scarcity for feed, the seabass may resort to cannibalism resulting in low survival and production though few fishes will be large size. Under polyculture method, the feed in the form of forage fishes are produced in the culture ponds itself and made available to the seabass fish to prey upon as and when it requires. Proper technology package on polyculture of fish and shrimp is available and it has been successful in West Bengal.

Cage farming of fishes

Fish culture in cages has been identified as one of the eco-friendly at the same time intensive culture practice for increasing in aquaculture production. Cages can be installed in open sea or in coastal area. The former is yet to be developed in many countries where seabass is cultured but coastal cage culture is an established household activity in the South East Asian countries. There are abundant potential as in India also for cage culture in the lagoons, protected coastal areas, estuaries and creeks. Since, cage culture of seabass has been proved to be a technically feasible and viable proposition this can be taken up in a large scale in suitable areas. Cage culture system allows high stocking density, assures high survival rate. It is natural and eco-friendly and can be adopted to any scale. Feeding can be controlled and cages can be easily managed. Even in areas, where the topography of the bottom is unsuitable

for pond construction, cage can be installed. Diseases can be easily monitored. Fishes in the cages can be harvested as per the requirement of the consumers, which will fetch high unit price. Above all, cage culture has got low capital input and operating costs are minimal. Cages can be relocated whenever necessary to avoid any unfavorable condition.

Farmers and brackishwater aquaculture development

In spite of these research achievements, formulation of strategies in the National Institute and sincere efforts by State Fishery departments and Marine product Export Development Authority (MPEDA) with different schemes to boost production and promotion of marine and brackishwater activity in the country, the progress is not yet satisfactory. As the aquaculture development is closely related with the development of the ability of the farmer's understanding and adoption of new technology, the farmers need to be supplied with recent, useful and practical information related to brackish water aquaculture with special reference to culture of shrimp. The role of extension methodologies will be vital for enhancing the production and productivity from brackishwater farming in the country.

The poor growth and slow rate of adoption of scientific aquaculture technologies is attributable to following socio-economic constraints.

1. Non availability of quality seed
2. High cost of formulated commercial feed
3. Financial constraints and paucity of credit
4. Absence of organized marketing network
5. Weak extension linkage
6. Biological hazard
7. Poor technical skill of farmers
8. Inadequate training facilities for extension staff
9. Lack of research institutes for applied research on specific problems

Apart from these socio economic constraints some issues which are emerging to overcome the constraints can be considered.

1. **Participatory fisheries management:** The resource users and resource managers are to ensure that the fisheries resources provide long term productivity, stability, sustainability, equity and biodiversity.
2. **Public Private Partnership (PPP):** Growing willingness of farmers and other companies has lead develop PPP mode of aquaculture development. We need to grow interest among stake holder to develop brackishwater aquaculture through PPP mode with greater investment.

3. **Significance of Indigenous technical knowledge (ITK) in aquaculture:** Rural people of developing countries have no choice but to rely on traditional knowledge because sophisticated technologies of developed nations are generally irrelevant and expensive. ITK can be incorporated into existing effort to enhance and expand effectiveness for higher production.
4. **Encouraging entrepreneurship:** Aquaculture in freshwater is mainly constituted by Indian major carps and comparatively low valued fishes. In their context only seabass and other shrimp and fish species are considered as a substitute or supporting species for sustaining the aquaculture. The unutilized brackishwater ponds, the open water bodies and the entrepreneurs and farmers, aspiring the good value added species can venture in to farming of new brackishwater fish/crab/shrimp species.

Conclusion

The importance of coastal aquaculture will become more apparent in the coming year with demanding pressure from an ever increasing world population. Coastal aquaculture has great potential for the production of food, alleviation of poverty and generation of wealth for people living in coastal areas, many of whom are under privileged. The assurance of mature and sound techniques in coastal aquaculture is the most important channel towards a sustainable aquaculture industry. Therefore some strategies should be worked out. The first is to speed up the completion of unfinished current research on different aspects of aquafarming. The second is to formulate and implement systematic regulations and self controls. An integrated regulation system must be imposed to ensure that the environment is not harmed by practice of coastal aquaculture, that natural traits of propagated fish are not lost in the process. Thirdly it is important to strengthen regional, national and international communications and information exchange. Exchange of technical know-how is necessary to prevent research duplication for human resource management. Development of technologies should be prioritized and international cooperation is prompted.

SOIL AND WATER QUALITY REQUIREMENTS FOR BRACKISHWATER AQUACULTURE

G. Biswas, P.S. Shyne Anand, J.K. Sundaray and Ashutosh D. Deo

Fish or shrimp growth and health are influenced by the pond bottom soil (sediment) and water quality parameters. Therefore, management of pond sediments and water quality is the key for successful aquaculture practice. As properties of the pond sediment and processes occurring at the bottom soil and in the soil-water interface determine the water quality, management of the former is an advanced approach in modern aquaculture. Healthy pond ecology provides suitable environment for the cultured animal to grow properly and maintenance of optimum culture environment through soil and water quality management excludes the risk of stress and diseases of species under culture.

Soil quality requirements

The pond soil plays a vital role in the mineralization process and supplies the required nutrients to the overlying water. These nutrients help in production of various primary food organisms. Considering this significance, bottom soil is designated as the chemical laboratory of the pond. However, availability of desirable soil quality is often a limiting factor to undertake aquaculture, and several protocols are followed for the purpose of improving pond soils.

Soil texture

Soil texture indicates the particle-size distribution of sand, silt and clay in soil. Many important physico-chemical properties influencing the fertility of fishponds are affected greatly by soil texture. An ideal pond soil should not be too sandy to allow water seepage or too clayey to keep all the nutrients adsorbed on to it. Therefore, soils with moderately heavy texture such as sandy clay, sandy clay loam, clay loam, silty loam are found to be favourable for brackishwater aquaculture.

Soil pH

Soil pH controls most of the chemical reactions in the pond environment and plays an important role in pond productivity. Pond bottom soil pH can range from < 4 to > 9, but the best pH for pond soils is considered to be about neutral to slightly alkaline (7 and little above). Too low soil pH can reduce the availability of key nutrients in the water and lower pond productivity. Maximum availability of soil phosphorus usually occurs at about pH 7. Most soil microorganisms, and especially soil bacteria, function best at pH 7 to 8.

Organic carbon content

Organic carbon (OC) acts as a source of energy for bacteria and other microbes that release nutrients through various biochemical processes. Based on OC content, soils may be of two types: organic soil containing >15% OC and mineral soil containing <15% OC. Organic soils are not good for pond aquaculture and should be avoided. Mineral soils with <0.5% OC is considered unproductive while those in the range of 0.5-1.5% and 1.5-3.0% to have medium and high productivity, respectively. OC content of >3.0% may not be suitable for fish production, since it may lead to excessive bloom of microbes and oxygen depletion in water.

Carbon to nitrogen ratio (C:N)

C:N ratio of soil influences mineralization process by microbes. Mineralization is very fast, moderately fast and slow at C:N ratios in the range of <10, 10-20 and >20, respectively. In general, soil C:N ratios between 10 and 15 are considered favourable for aquaculture and a ratio of 20:1 or narrower gives good results.

Nutrient elements in soil

Nitrogen, phosphorus and potassium are the major nutrients present in soil for phytoplankton production and primary productivity. Generally, small amount of potassium is needed in fish ponds. Single most critical nutrient for pond productivity is phosphorus content of soil and water. Pond soils with 30, 30-60, 60-120 ppm and more than 120 ppm available phosphate (P_2O_5) are considered to have poor, average, good and high productivity, respectively. Ponds with <250 ppm available soil nitrogen are regarded as low productive, while concentrations in the range 250 to 500 ppm and >500 ppm are considered to be medium and highly productive, respectively.

Water quality requirements

There is a Chinese proverb stating “cultivating water is a must for aquaculture” and it emphasizes the importance of water quality management in aquaculture. Water quality parameters influence the pond environment. The pond environment should be optimum for the species to be cultured. Therefore, it is important to know about water quality parameters and their management, which affect the growth and survival of aquatic organisms.

Temperature

Temperature affects the fish metabolism in molecular dynamics and biochemical reaction rates. For best growth of tropical species in brackishwater system temperature between 25°C and 32°C is essential. Reactions in chemical and biological processes are also influenced by temperature and they are doubled for every 10°C increase in temperature. This

means that aquatic organisms will use twice as much dissolved oxygen at 30°C as at 20°C and biochemical reactions will progress twice as fast at 30°C as at 20°C.

Dissolved oxygen (DO)

The optimum DO content of pond water is in the range of 5 ppm to saturation level for good growth of fish. Although DO is a critical parameter in fish culture, generally this variable has less influence in brackishwater farms, but intensive culture needs artificial aeration using paddle wheel aerators.

Salinity

Salinity is the total amount of solid material (g) contained in 1 kg of seawater, when all the carbonate has been converted to oxide, the bromine and iodine replaced by chlorine and all organic matter completely oxidised. Salinity of brackishwater ranges between 0.5 to 30 ppt. Salinity influences the solubility of different dissolved gases. Most cultured brackishwater shrimp and fish species are euryhaline and adapt easily to salinity fluctuation. However, sudden changes in salinity may harm the cultured animals. Cultured species thriving at a particular salinity should not be placed in water with salinity difference without proper acclimatization.

Water pH

pH is a measure of hydrogen ion concentration in water and indicates how much the water is acidic or basic. Water pH affects metabolism and physiological processes of fish. It also exerts considerable influence on toxicity of ammonia (NH₃) and hydrogen sulphide (H₂S) as well as solubility of nutrients and thereby water fertility. NH₃ toxicity may occur at high pH (>8), whereas low pH increases the risk of nitrite and H₂S toxicity. The well buffered brackishwater usually does not pose any threat to the aquatic animals and it remains within 6.5 to 9.0. However, optimum pH range for culture environment should be 7.5-8.5. Liming and applying gypsum (CaSO₄) are done to increase and reduce the pH, respectively.

Turbidity (≈Transparency)

Turbidity is caused in ponds by several factors including suspended soil particles, plankton population and humic substances produced through decomposition of organic matter. In culture systems, turbidity caused by planktonic organisms is a desirable trait, whereas that caused by suspended clay particle is undesirable. It is measured by Secchi disc transparency. Optimum range of transparency in a culture pond is 25-45 cm. If plankton transparency is <25 cm, water treatment is required and if it is >45 cm, fertilization is done to increase plankton production.

Total alkalinity

Alkalinity of water is caused by all the carbonates and bicarbonates of alkaline and alkaline earth metals present in solution. The most common basic ions which are indispensable for fish culture are calcium and magnesium. Water containing alkalinity >40 mg/l as CaCO_3 is found to be productive.

Carbon dioxide (CO_2)

High concentration of CO_2 can be tolerated by fish and shellfish and most species will survive in waters containing up to 60 mg/l. Its concentration in water increases during night hours and eutrophic condition caused by phytoplankton die-off. Application of CaO and Ca(OH)_2 can remove excess carbon dioxide.

Ammonia (NH_3)

Ammonia is a product of protein catabolism in fish and microbial decomposition of organic matter. When NH_3 concentration increases in water, its excretion by aquatic animals diminishes and thereby NH_3 level increases in blood and other tissues. Fish are very sensitive to this unionized ammonia (NH_3) and its desirable range is 0.02-0.05 ppm in pond water. Usually in the case of high DO and high carbon dioxide concentrations, the ammonia toxicity to fish is reduced.

Nitrite (NO_2)

Inorganic nitrogen is present in water mainly in the forms of NH_3 and nitrate (NO_3). NO_2 is an intermediate product of nitrification and NO_3 reduction. NO_2 is highly toxic to fish and less toxic to shrimp, as it oxidizes haemoglobin to form methaemoglobin which is incapable of transporting oxygen in the former's circulatory system. In well oxygenated ponds, NO_2 concentration is negligible. Safe level of nitrate is less than 0.2 ppm.

Hydrogen sulphide (H_2S)

Under anaerobic condition H_2S is produced by heterotrophic bacteria in pond bottom soils and it is highly toxic to fish. In water, H_2S exists in unionised (H_2S) and ionised forms (HS^- and S_2^{2-}). Only the unionised form is toxic and its concentration is mainly affected by pH. With the increase of pH the concentration of H_2S decreases. At concentration of 0.01 ppm of H_2S fish lose their equilibrium and are subjected to sub-lethal stress. Frequent exchange of water can prevent building up of H_2S . Further, increasing water pH through liming can also reduce the H_2S toxicity.

Proper pond management is the key to sustainability in aquaculture. To maintain the desirable pond culture environment, system-based management protocols need to be followed, as pond management steps vary according to the farming systems such as fish or

shrimp culture. Therefore, appropriate procedure for pond management helps to improve environmental conditions, profitability and sustainability.

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BETTER MANAGEMENT PRACTICES IN SHRIMP FARMING

Shyne anand, Sujeet kumar and A. Panigrahi

Aquaculture has evolved as the fastest growing food producing sector and shrimp farming has significant contribution in national food security, export revenue and employment generation. However, worldwide shrimp farming activity is regressed by diseases and unsustainable practices which sometimes affecting the coastal ecosystem and livelihood. Existing and emerging viral diseases is one of the major bottlenecks in success of shrimp farming. Better management practices (BMPs) are innovative, dynamic, and improved farming practices applied to shrimp farming and production systems to help ensure that sustainable development is achieved in an environmentally responsible manner. BMPs protect wildlife and coastal ecosystem as it primarily works to develop vitally needed quality shrimp production lowering the risk of disease outbreak and assuring sustainability, food security and safety.

Good Pond preparation: it is one of the most important steps which determine the success of a shrimp farm. Following measures should be adapted during pond preparation. No mangrove deforastration for shrimp pond construction and conserving the existing mangroves are important BMPs

- a. **Removal of bottom sludge:** This involves the removal of black soil layer which contain high organic content.
- b. **Ploughing of soil:** It exposes the black soil layer to sunlight and atmospheric oxygen which oxidize the organic waste.
- c. **Lime application:** Liming maintains the optimum soil and water Ph (6.5-9). Generally Agricultural lime ,hydrated lime and dolomite are used to maintain pH

Water intake

Intake water must be filtered with fine mesh screen filter bag to prevent the entry of virus carriers such as crabs, wild shrimps and also to avoid entry of fish or crustacean, which may be predator or competitor for shrimp. For every two grow-out ponds one extra pond should be maintained as a water reservoir. Disinfection of water and fertilization for growth of plankton can performed in this reservoir before pumping in to grow out ponds. Pond water must be disinfected with bleaching powder@60 ppm. It reduces the risk of pathogens and carrier.

Seed Selection and Stocking

Quality of post larvae plays a major role in success of shrimp farming. Before purchasing, shrimp post larvae should be checked for their general condition such as activity, color, size, *etc.* If there is any dead and abnormal colored PL in the tank, the entire batch should be rejected. Before stocking at the pond, PL should be treated with formalin at 100 ppm concentration for 30 minutes in well aerated tanks to remove weak PL.

Pond bottom and water quality management

Disease outbreaks in shrimp grow out culture is directly related to pond bottom and water quality. Though water exchange at regular interval maintains the good quality pond water, zero water exchange (no water exchange) or minimal water exchange is advisable as it reduces the incidence of disease outbreak by ensuring the biosecurity. Ponds using aeration tend to have higher shrimp production and low disease outbreak. regular monitoring of soil and water quality parameter like temperature, salinity pH, total suspended solid *etc.* and chemical parameter such as ammonia, nitrate, nitrite, total organic carbon, dissolved oxygen, BOD, COD *etc.*

Better Feed management

Cost of feed accounts for about 40% to 60% of the total production cost. Feed monitoring should be done with check tray evaluation for optimum feed management. Shifting feeding area at least once in 7 to 10 days depending on the bottom condition along feeding area. Reduce feeding during periods of low DO, plankton crash, rain fall, extremes of temperature never over feed. Slightly under feeding is better than over feeding, which saves money and reduce disease risks and during disease outbreaks. Proper storage

Shrimp Health Monitoring

Shrimps should be sampled once in a week by cast netting and should be checked for their general health conditions, like external appearance (body color, missing appendages, external/gill fouling, black gills or gill choking, *etc.*), gut condition, and growth in terms of weight or length. Shrimp behaviour and feeding trends should also be monitored. The gut content colour is a good indicator of the probable health status and corrective action to be taken. A black/brown/ green gut content implies under feeding whereas a red or pink gut showed disease manifestation whereas a pale whitish gut showed gut infection. A normal gut will have a light or golden brown colour.

Probiotics, immunostimulants, bioremediating agents can be employed as prophylactic and disease treatment measures in grow out culture and hatchery system. Probiotics is a live microbial feed supplement which beneficially affects the shrimp

by improving its intestinal microbial balance. Most commonly used probiotics in shrimp culture are Yeast, *Bacillus*, *Vibrio alginolyticus*, *Lactobacillus acidophilus* etc. Immunostimulant is a chemical or drug which enhances the non specific defense mechanism of the animals thereby impart better generalized protection. Many substances such as β -glucan, chitin, lipopolysaccharide of gram negative bacteria, peptidoglycan of gram positive bacteria, lactoferrin, levamisole, and many nutritional factors and cytokine are used as immunostimulants. There is a serious concern on the use of antibiotics, and their use in shrimp farming should be avoided.

Better management practices during Disease Outbreak

During disease outbreak, the following actions should be taken:

- a. Check any abnormalities in water and soil condition and take immediate action to correct the problem.
- b. Remove dead animals and bury them away from the ponds.
- c. Emergency harvesting, if the mortality rate is increasing rapidly and shrimp are not feeding. It can be carried out preferably using cast netting to avoid discharge of infected water into the main water source.
- d. Bleaching of pond water for 5 – 7 days before releasing into to the drainage.
- e. Neighboring farmers should be kept well informed about shrimp disease problems, emergency harvesting and the time and date of water discharge.
- f. The pond water should be treated in an effluent treatment system (ETS) before discharging to a common water source.
- g. To avoid the cross contamination during periods of disease outbreak, surrounding farmers should try to avoid water exchange and should not use any equipment (nets, tanks, pumps, boat, etc) from affected farms.

Farm Record Maintenance: Records are necessary to identify problems in the pond environment and shrimp health and to rectify these problems at the earliest during the production cycle. Record keeping also helps the farmer to learn from past mistakes, thus reducing risk and costs of production in subsequent crops.

Better Practices for harvesting

Harvesting must be avoided during moulting period and agr.lime can be applied 3-4 days before harvesting.try to do harvesting in the early morning or evening. harvesting should be done with drag with minimum delay. After harvesting transport crates with crushed ice at 1:1 ratio for better preservation.Nutrient rich pond effluent must be treated before get discharged in to water source.

Impacts of BMP includes Social Environment, Reduced costs and improved Profits, Reduced risk to small-scale farmers, Increased co-operation and harmony among farmers , Better organized farmer groups, Reduced disease incidence, Reduced FCR and increased efficiency of resource use (feed, seed, energy, finance in particular), Reduced pollution , Reduced chemical and antibiotic use.

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NUTRITION AND HEALTH MANAGEMENT OF BRACKISHWATER FISH AND SHRIMP CULTURE

T. K. Ghoshal

Maintenance of fish and shrimp in good and healthy condition is very much essential to make the aquaculture more profitable and sustainable. Slow growth, poor feed conversion, low yields, increased disease incidence and mortality, consequently, low profitability are the results of poor fish health. Successful fish health management is considered as prevention of disease rather than treatment. Prevention of fish disease is accomplished through good water quality management, nutrition, and sanitation. Moreover, proper nutrition is one of the key of success of producing healthy fish that can fight against other environmental stress and pathogens. Without this foundation it is impossible to prevent outbreaks of opportunistic diseases. The fish is constantly bathed in potential pathogens, including bacteria, fungi, and parasites. Even use of sterilization technology (i.e., ultraviolet sterilizers, ozonation) does not eliminate all potential pathogens from the environment. Suboptimal water quality, poor nutrition, or immune system suppression generally associated with stressful conditions allow these potential pathogens to cause disease. Presently aquaculture is a fast growing agricultural sector with growth rate of 10% annually. As a result production losses due to diseases also have increased and indicate the need for precisely formulated, economical diets that adequately meet the needs of fish and shrimp grown in high- density production system. If there is no utilizable feed intake by the fish, there can be no growth and death eventually results. Under-nourished or malnourished animals cannot maintain health and growth, regardless of the quality of the environment. The goals of the aquaculture industry are to optimize growth and to produce high-quality fish. As in all farming, the outbreak of diseases in fish farming can be a major concern. The high susceptibility of fish to stress and the rapid spread of diseases in water have forced aquaculturists to concentrate their efforts on maintaining their fish in good health in order to achieve sustainable economic performances. Thus nutritional balance feed and proper feeding management in all aquaculture practice especially in brackishwater aquaculture, have to be required for healthy growing fishes and shrimps.

Vitamin C as a nutritional factor

Vitamin C is one of the most important vitamins which are very much essential and responsible for fish or shrimp growth and immunity. Since no vitamin C synthesis occurs in fish/ shrimp, they are dependent on a vitamin C supplementation via the feed. To optimise

stability of vitamin C during feed production and storage as well as bioavailability in the fish, the use of vitamin C in phosphorylated form is recommended. With the exception of perhaps two or three species, vitamin C biosynthesis does not occur in fish due to the lack of the last enzyme of the biosynthetic pathway: L-gulonolactone oxidase. Major signs of ascorbate deficiency include reduced growth, scoliosis, lordosis, internal and fin haemorrhage, distorted gill filaments, fin erosion, anorexia and increased mortality. The recommended inclusion levels of vitamin C are described as follows (Table 1.) on the basis of the combination of experimental results obtained in several controlled studies and field experience:

Table-1. General inclusion rates of vitamin C in feeds for aquaculture species are:

Species	Recommended vitamin C level (mg/kg feed)*
Carp	150-250
Tilapia	150-250
Seabass / Seabream	150-250
Salmon	150-250
Trout	150-250
Catfish	150-250
Eel	150-300
Shrimp	250-500

*** Vitamin C-activity in phosphorylated form**

Based on findings on the influence of vitamin C supplementation on immune response and on increasing field experience in this important area of fish nutrition, the following vitamin C levels are recommended whenever the immune system is challenged (stressing situations, e.g. handling and grading, vaccination, winter wounds, disease outbreaks and release of smolt into the sea) and after reduced intake during winter.

Other Vitamins

Vitamins are micro-nutrients required for normal growth, reproduction, health and maintenance of fish metabolism. The requirements of fish depend upon the intake of other nutrients, size of the fish, and environmental stresses. Four fat-soluble (Vit.- A,D,E & K) and nine water-soluble vitamins (Vitamin B₁, B₂, B₃, B₅, B₆, B₇, B₉, B₁₂,C) are known to be required by the fish or shrimp for their better nutrition and health. However, recommended dietary levels and deficiency signs (NRC 1981 and 1977) are summarized in Tables 2. In the early days of fish culture, the most common nutritional deficiencies were those associated with vitamins. Now a days most practical diets contain sufficient quantities of vitamins. In spite of the addition of excess amounts of vitamins to most fish diets, vitamin deficiency disorders still occur in fish culture. Vitamins are very susceptible to destruction by oxidation

in the presence of excessive moisture, heat, and trace minerals, particularly if rancid fat is present. Many of the vitamin deficiency signs are non-specific and it is difficult to analyze. Nutritional disorders caused by vitamin deficiencies can impair the utilization of other nutrients, weaken the health of the fish, and lead to disease. Therefore, need for supplementation of vitamins in is recommended for better growth, health and survival of fish and shrimp.

Table-2. Different vitamins and minerals Deficiency Signs in Finfish/ shrimp:

Signs of different deficiency in Finfish/ shrimp	Possible Nutrient Deficiencies
Anemia	Folic acid, inositol, niacin, pyridoxine, riboflavin, rancid, fat, vitamins B12, C, E&K
Anorexia (poor appetite)	Biotin, folic acid, inositol, niacin, pantothenic acid, pyridoxine, riboflavin, thiamine vitamins A, B12 and C
Atrophy, gills	Pantothenic acid
Atrophy, muscle	Biotin. Thiamine
Cataract	Methionine, riboflavin, thiamine, zinc
Cartilage abnormality	Vitamin C, tryptophan
Clotting blood	Vitamin K
Coloration, dark skin	Biotin, folic acid, pyridoxine, riboflavin
Deformation, bone	Phosphorus
Distended stomach	Inositol
Equilibrium loss.	Pyridoxine, thiamine
Exudated gills	Pantothenic acid
Fatty liver	Biotin, choline, fatty acids, inositol, vitamin E
Feed efficiency, poor	Biotin, calcium, choline, energy, fat, folic acid, inositol, niacin, protein, riboflavin
Fragility, fin	Folic acid
Hemorrhage, gill	Vitamin C
Hemorrhage, kidney	Choline, vitamins A and C
Hemorrhage, skin	Niacin, pantothenic acid, riboflavin, vitamins A and C
Necrosis, liver	Pantothenic acid
Swimming, erratic	Pyridoxine, pantothenic acid
Nerve disorder	Pyridoxine, thiamine

Minerals

Minerals are defined as inorganic elements which are needed by animals to maintain many of their metabolic process and provide material for major structural elements (e.g. skeleton). These are necessary for maintenance of osmotic pressure, acid-base balance, thus the regulation of pH of blood, haemolymph, urine, and other body fluids. They are also components of soft tissues, enzymes, some vitamins, hormones and respiratory pigments and

are essential for muscle contraction and transmission of nerve impulses. Though the requirement of minerals in fish or shrimp is minimum it plays a vital role in fish or shrimp nutrition and health. As fishes are living in water which is also containing different minerals, it is assumed that fishes can fulfill their minerals requirement from ambient media directly. However, marine fishes and shrimps have some advantage as marine water contain more minerals than freshwater. But Phosphorus is not abundant in seawater and has to be taken from the food ingested. After discovering different disorder like, the appearance of cataract in fish (Ketola, 1979), made the mineral nutrition of fish a topic of immediate concern to the aquaculture community. The minerals required for normal metabolism can be divided in to two groups, major and trace minerals. Major minerals are required in large quantities included calcium, phosphorous, magnesium, sodium, potassium, chlorine and sulphur. Trace minerals are those require in trace amount and this are iron, iodine, manganese, copper, cobalt, zinc, silicon, tin and chromium. The major micro nutrient such as calcium and phosphorus is found to present in skeleton, scale and shell of bony fishes and shrimps. Calcium is also important for Variety of other physiological process including metabolism nerve, shell development, muscular function and osmoregulation. Around twenty two minerals, both macro and micro, have been found essential to animals, fish and shrimp. Some minerals identified as essential for shrimps health, are: calcium, phosphorus, magnesium, potassium, iron, zinc, copper, iodine and selenium. Penaeids may need dietary sources of minerals for growth and health because of repeated moultings wherein minerals are lost. The availability of minerals to shrimp is dependent on the dietary source and form of the mineral that is ingested, amount stored in the body, interaction of other elements present in the gastrointestinal tract and body tissues and mineral interactions with other dietary ingredients or metabolites.

Dietary protein and aminoacids

Dietary protein is the major and most important component of formulated aquafeeds. Proteins compose approximately 70% dry weight of the organic material in fish tissue; therefore, protein content is one of the most important nutritional compounds of fish feeds. Usually, fish growth will be directly proportional to the protein level of its feed if the level is within the range of approximately 20 to 50% crude protein. Optimum dietary protein levels vary with fish species, stage of life, water temperature, food consumption, daily feed allowance, feeding frequency, quality of protein (amino acid composition), and quantity of non-protein energy. Fish do not have a true protein requirement but require a balanced combination of the 20 major essential and nonessential amino acids that make up proteins

(Table-3). Recent evidence shows that some amino acids and their metabolites are important regulators key of metabolic pathways that are necessary for maintenance, growth, feed intake, nutrient utilization, immunity, behavior, larval metamorphosis, reproduction, as well as resistance to environmental stressors and pathogenic organisms in various fishes. Fish utilize dietary proteins by digesting them into free amino acids, which are absorbed into the blood and distributed to tissues throughout the body where they are then reconstituted into new specific proteins of the fish tissues. Protein in fish tissues is formed from all 20 major amino acids. Fish can synthesize some of these amino acids in their body, but others cannot be synthesized and must be consumed. The 10 amino acids that fish cannot synthesize are the "essential amino acids" that must be provided in proper amounts in their diet. If any of the 10 "nonessential" amino acids are not consumed as such in the diet, fish can synthesize these in required quantities from other amino acids. It is not clear how effectively cultured fishes can utilize synthetic (crystalline) amino acids, such as free lysine and methionine, in their diets. Proteins are present in all animals and plants in varying amounts and compositions of amino acids. However, each protein varies in its digestibility and content of available amino acids to fish. Therefore, the amino acid composition and availability in feed ingredients may not be balanced and may be limiting relative to the requirements of a specific fish species. On the basis of their crucial roles in cell metabolism and physiology, we anticipate that dietary supplementation with specific amino acids may be beneficial for: (i) increasing the chemo-attractive property and nutritional value of aquafeeds with low fishmeal inclusion; (ii) optimizing efficiency of metabolic transformation in juvenile and subadult fishes; (iii) suppressing aggressive behaviors and cannibalism; (iv) increasing larval performance and survival; (v) mediating timing and efficiency of spawning; (vi) improving fillet taste and texture; and (vii) enhancing immunity and tolerance to environmental stresses. Functional amino acids hold great promise for development of balanced aquafeeds to enhance the efficiency and profitability of global aquaculture production. The preferred sources of feedstuffs for providing protein for fish feeds are fish meal and soybean meal, because these products are high in percent crude protein, contain high levels of all essential amino acids, are universally available in high quantities.

Table1. Nutritionally essential and nonessential amino acids for fish/shrimps and other aquatic animals

Essential AA	Nonessential AA	Conditionally essential AA
Arginine	Alanine	Cysteine
Histidine	Asparagine	Glutamine
Isoleucine	Aspartate	Hydroxyproline
Leucine	Glutamate	Proline
Lysine	Glycine	Taurine
Methionine	Serine	
Phenylalanine	Tyrosine	
Threonine		
Tryptophan		
Valine		

Dietary fatty acid

Fatty acids comprise major components of most lipid classes and are especially common in nearly all fresh, marine and brackish water fishes and shellfishes. The fatty acid composition of membranes determines physical properties such as fluidity and permeability. Hence the activity of membrane associated enzymes, membrane receptors, binding of mitogens, antigens and soluble mediators can all be affected by fatty acid composition. In addition the cyclooxygenase and lipoxygenase products of fatty acids (prostaglandins, thromboxanes and other eicosanoids) have important roles in immune modulation. Macrophages and other immune cells are the main producers of these eicosanoids. Most fatty acids >18 carbons in length are transferred into a consumer's tissues with little or no alteration from diet. Fish and mammals appear to be unable to synthesize fatty acids that are unsaturated in the w-3 or w-6 positions unless a suitable precursor is supplied in the diet. Thus, the lipid component of the diet must provide an adequate amount of essential fatty acids for growth as well as for required dietary fuel. Polyunsaturated fatty acids (PUFAs), fatty acids containing more than one double bond, are a prevalent substance in marine organisms and are typically the most abundant and highly conserved fatty acids in the marine environment. In recent years, fatty acids have been increasingly used as natural tracers of diet for a variety of marine organisms including invertebrates, fishes, shrimp, sea turtles and marine mammals. Studies using fatty acids as dietary indicators typically focus on PUFAs since they cannot be synthesized, are rarely modified by brackishwater fishes or shrimps, are essential for life processes and are therefore the optimal choice in delineating feeding pathways. The inability to synthesize PUFAs in fishes/shrimps are thought to be due to a relaxation of constraints on biochemical pathways, specifically elongase and desaturase enzymes (Hastings et al., 2001). Comparatively, most saturated and monounsaturated fatty

acids can be produced endogenously. The ability to modify monounsaturates into PUFAs, or modify PUFAs into longer and further desaturated fatty acids, however, has been lost or severely limited in marine fishes (Hastings et al., 2001). Modification of fatty acid profiles in the tissues of cultured fish species have been studied extensively. However, the purpose of such studies often involves the alteration of tissue composition to increase the growth, survival or marketability of cultured species. In previous studies attempting to quantify PUFA transfer in marine consumers has suggested that diet influences PUFAs in consumer tissues, they failed to examine the extent of dietary PUFA incorporation into consumer tissues. Therefore, even though studies are increasingly using fatty acids as biomarkers for investigating trophic interactions in marine ecosystems, basic experiments quantifying dietary PUFA transfer in estuarine and marine fishes have yet to be completed.

Toxin and antinutrients

Toxins which may be present in fish feeds include mycotoxins, residues of polychlorinated biphenyls, pesticides, herbicides and other agricultural and industrial chemicals. Mycotoxins are produced by many molds on plant products such as oilseed byproducts (soybean, cotton seed, and peanut meals) and grain byproducts. In particular, aflatoxin B₁ (at less than 1 ppb) in the diet will produce liver cancer in rainbow trout in one year, and at 8-20 ppb will induce grossly visible hepatomas in 4-6 months (Sinnhuber et al. 1977). Other toxins and antimetabolites in plant materials are protease inhibitors, hemagglutinins, glucosinolate, cyanogens, saponins, tannin and gossypol (NRC 1977 and 1981). These antinutrients are defined as substances which by themselves, or through their metabolic products arising in living systems, interfere with food utilisation and affect the health and production of animals (Makkar, 1993). However, these compounds can be destroyed or inactivated by proper processing (e.g. heating or chemical treatment). The antinutrients which are commonly present in most of the plant feed stuffs have been found to have negative consequences on growth and feed utilization of farmed fish, depending on fish species and inclusion level in their diets. Microbial toxins which are produced by microorganisms associated with feed contamination or spoilage cause bacterial toxicoses. Mycobacteriosis (fish tuberculosis) in Pacific salmon was related to the practice of feeding raw salmon offal to hatchery stocks (Ross et al. 1959). Converting to diets containing pasteurized salmon viscera effectively eliminated the primary source of infection (Hublou et al. 1959).

Dietary approaches and feeding management

A proper diet has long been recognized as a necessity in preserving an animal's health and maintaining its ability to resist diseases. Nutrients may alter immune responses by acting on immune cells directly or indirectly through metabolic, neurological, or endocrine pathways. Dietary modulation of the immune response has been accomplished in a variety of fish or shrimp species using various approaches, including manipulating levels of various nutrients, adding certain amino acids, vitamins, minerals and altering feeding strategies. Accomplishing with this objectives, water stable feed, optimum feed allowance (>80% to <100% satiation) and proper feeding methods and proper feed distribution techniques are very much desire. Pelleted feeds must be water stable for at least 10 minutes before it consumed by the fish/shrimp. Determining optimum feed allowances for each feeding time/day, is a simple concept but sometimes a difficult practice. The concept is to feed the fish at each feeding time carefully measured amounts of feed until the fish stop eating. The amount of feed consumed to that point is considered 100% satiation and the maximum end point of the feed allowance optimum range. Feed allowance adjustments should be made at any time overfeeding or underfeeding is obvious. The effect of overfeeding is one of the probable cause of water pollution and thereby increasing the chance of disease outbreak of fish and shrimp. Underfeeding is preferable to overfeeding but neither is desirable.

Further Reading

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NURSERY REARING AND GROW-OUT CULTURE OF BRACKISHWATER FINFISH

G. Biswas and J. K. Sundaray

Several species are available in brackishwater, but commercial farming is restricted to only few species due to some important factors, viz., economic importance, growth rate, culture compatibility, quality and adequate seed availability etc. It is useful to have knowledge on basic biology of any fish species before undertaking their handling and maintenance.

Biology of Cultivable Brackishwater Fishes

Biology of Asian seabass, *Lates calcarifer*

The Asian seabass, *L. calcarifer* is a much esteemed food fish and it belongs to the family Centropomidae. Seabass is carnivorous in nature. However, juveniles are omnivores. It is opportunistic predator and its diet changes with size. Stomach contents of smaller fish (1-10 cm) showed 20% phytoplankton and 80% fish, shrimp etc. In bigger fish (20 cm), the gut contained 100% animal prey (70% crustaceans and 30% fishes). It prefers pelagic fishes than benthic crustaceans. It also has cannibalistic habit.

Biology of Milkfish, *Chanos chanos*

Milkfish, *C. chanos* is the only member of the family Chanidae. It is an important fish from aquaculture view point and cultured in large scale in South East Asia. Milkfish is an herbivorous fish. But their larvae feed mainly on zooplankton. Juveniles and adults eat cyanobacteria, soft algae (Cyanophyta, *Lyngbya spp.* and diatoms) small benthic invertebrates, decayed organic matter and even pelagic fish eggs and larvae. The plant animal complex, namely *lab-lab* formed in shallow water is one of the preferred food items. It has fine gill rakers and long intestinal tract which help in retaining and digesting this kind of food ingested. They can be adapted to accept artificial diet very easily. It is a euryhaline species which can withstand sudden changes in salinity and can be grown in fresh, brackish and marine waters. Its salinity tolerance limit is 0 to 158 ppt. It can tolerate a temperature range of 15-40°C, but the optimum is between 20°C and 33°C. Milkfish is a marine species with catadromous migratory habit. Adults spend part of their lives in littoral waters and go to sea for breeding. In nature it grows to a maximum of 1.5 m. In well maintained culture ponds it grows to a marketable size of 300-400 g in 3-4 months.

Biology of Striped Grey Mullet, *Mugil cephalus*

The striped or jumping grey mullet, *M. cephalus* is the most important species of the grey mullets which belong to the family Mugilidae. *M. cephalus* is the fastest growing grey mullet among 77 species and enjoys a very cosmopolitan worldwide distribution. Fry of less than 30 mm feed principally on zooplankton. The juveniles feed preferably on diatoms and epiphytic cyanophyceae. Gut contents of adult consisted of sand, decayed organic matter, diatoms, dinoflagellates, foraminifera, algae and miscellaneous items like copepods and tintinnids. Qualitative composition of gut contents says that it is an iliophagous subsisting mainly on decayed organic matter. They feed in a head down position, moving its head from side to side. The movement sometimes is so vigorous that the whole body shakes, as a result a cloud of mud along with soft flocculant matters rich in microorganisms are sucked through its protrusible mouth. Although it can adapt itself to artificial diet, it has a preference for natural food.

Biology of Pearlsplit, *Etroplus suratensis*

The pearlsplit, *E. suratensis* belonging to the family Cichlidae is also referred to as green chromid and has high market value in the southern states of India, especially Kerala. Young ones feed almost on zooplankton, but from juvenile stage onwards, they are mainly herbivorous and detritivorous and feed on algal weeds, filamentous algae, detritus etc. However, miscellaneous food items such as insects, molluscs, crustaceans and sponges also form part of its food.

Biology of Spotted Scat, *Scatophagus argus*

The spotted scat, *S. argus* is a euryhaline teleost widely distributed in nearshore waters of Indo-Pacific region. It is popular aquarium fish and an important food fish in its available areas. The binomial nomenclature *Scatophagus argus* is translated from Greek as 'spotted faeces-eater', and was derived from the habit of scat to gather in harbours and feed on offal and other wastes dumped from ships. It is uncertain, however, if scats are true coprophages since their acceptance and/or preference for faecal matter has not been confirmed. Gut content analysis revealed that adult scats are primarily herbivorous. They accept green filamentous algae and brown seaweeds also. Worms, crustaceans and insects constitute part of their food items. The evidence of herbivorous food habit is supported by presence of their long coiled intestine approximately 3.5 times the body length.

Nursery Rearing of Seabass (Bhetki)

The availability of fish seed is considered to be one of the most important factors for the success of seabass culture in any water body. Till recently for the practice of seabass

culture in India, the fish farmers have to depend on collection of seabass fry or fingerlings from the wild and then stock them in ponds or cages. The availability of fish seed varies considerably from year to year and the uncertain supply of this input results in limited seabass culture. With the development of technology package for artificial propagation and year round mass scale production of seabass fry by CIBA, it is anticipated that the much awaited seed demand would be met up by interested entrepreneurs after adoption of this technology. Seabass larvae grown for 25-30 days in hatchery are reared further for a period of 30-45 day nursery phase till they attain suitable size for stocking in grow-out culture systems. Generally, seabass nursery is carried out in fertilized brackishwater ponds and in net cages placed inside a pond or in an open coastal area. Here different types of pond based rearing systems are described.

Nursery rearing methods

Seabass fry is highly carnivorous and voracious feeder and development of shooters drastically reduces the survival percentage through cannibalism. From a management standpoint, priority must be given to control the cannibalistic behaviour which is one of the most important causes of mortality in nursery. So during designing any nursery system due consideration must be given towards manageability of these causes of mortality. Two pond based nursery rearing systems are followed for seabass.

Nursery rearing in pond

Ponds used for nursery rearing should be not more than 2000 m² for easy management with most preferred size of 200-500 m² holding at least 70-80 cm water. Ponds should have the provision of inlet and outlet fitted with small mesh net. Ponds are prepared and fertilized to eradicate predators and grow zooplankton at least two weeks prior to stocking. When the pond water is with natural algae growth, freshly hatched *Artemia* nauplii are introduced. Usually, for 1 ha pond 1 kg cyst is required. Seabass fry acclimatized to pond condition is stocked @ 20-30 nos/m². At least 30% water is exchanged daily. Supplementary feeding is done with chopped, cooked fish/shrimp meat 3 times daily @ 100% body weight in the 1st week followed by gradually reduced to 80, 60, 40 and 20% during 2nd, 3rd, 4th and 5th week, respectively. Excess feeding should always be avoided to maintain the optimum pond environment. At the end of rearing around 40-50% survival is achieved with 3-5 g body weight.

Nursery rearing in net cages (hapas)

Seabass fry is reared in net cages or hapas fixed in ponds. Hapas are either rectangular or square shaped enclosures made of nylon thread webbing and are kept inverted with top portion open.

Pond preparation and hapa arrangement

The ponds where hapas are to be fixed are prepared well following weed eradication, liming etc. to maintain desirable water condition one week prior to start of rearing. Mosquito net hapas made with 1/8-inch nylon webbing of size 1x1x1.5 or 1x1x2 m³ are fixed with bamboo poles in both sides of a catwalk in the pond by keeping 20 cm free board. The catwalk is erected to facilitate feeding, hapa checking, grading and other management activities. Sometimes use of double-layered hapas are suggested where chance of breaching of hapa wall by crabs or any other animals is high to prevent escape of fry under rearing.

Stocking & Feeding

Healthy fry with 1 to 1.5 cm size are stocked @200-500 nos./m² after proper acclimatization during cool hours of day. Seeds are counted and distributed to each hapa. Uniform size fry are stocked. Initial biomass is calculated by taking samples and it is required for feed calculation. The fry are fed either with slow sinking formulated feed or minced meat 3 times daily during day hours. Formulated feed is provided @5-10% body weight daily throughout the rearing period, whereas for minced meat feeding the rate is similar to that of pond rearing system. Feed amount is adjusted weekly from sampling data of growth increment.

Grading of fry/ separation of shooters

To get higher survival it is needed to sort and size-grade shooters regularly to lower competition for space and food, thus controlling cannibalism. Shooters are the individuals which have a minimum size difference of approximately 33% from the rest of the stock and they are removed weekly by hand picking from each hapa. These shooters are reared separately.

Hapa management

Hapas provide almost natural condition for the growing fishes. There are chances of the mesh get clogged due to adherence of debris or weeds and thus restricting water movement and resulting in stagnation of water and accumulation of waste products and algal growth in the hapas. Hapas are cleaned at 2-3 day intervals to avoid this problem. At the same time the hapas should be checked for any damage by crabs.

Growth and survival

Growth and survival depend on the stocking density, feed and feeding, shooter separation, etc. If all management activities undertaken properly, in 4-6 weeks, the fry attain 50-70 mm size with 2-3 g body weight and survival rate upto 80% is achieved.

Advantages of hapa rearing

There are a number of advantages of hapa nursery rearing compared to other nursery rearing methods. These are, i) hapas can be easily managed and require less space and capital investment, ii) farmer can extend it to any scale depending on his necessity and capability, iii) it can be maintained in a corner of grow-out pond or near the grow-out cages itself, iv) since cages or hapas are in *in situ* condition, it provides natural environmental condition, v) water flow in the cage site washes away the metabolites and uneaten feed, vi) growth and survival are higher in hapa rearing, vii) total harvesting of seed is possible.

Causes of mortalities in nursery rearing

There are some reasons which are directly or indirectly involved in mortalities during nursery rearing.

Cannibalism

Seabass is highly cannibalistic especially in the early life stages. The cannibalistic behaviour of this species is definitely one of the major causes of high nursery mortality, it would appear to be dependent upon a number of factors. Cannibalistic rate increases with increasing stocking density, water transparency and decreasing light intensity, number of feeding per day, etc.

Stocking density

High stocking density is another common cause of nursery mortality especially in the absence of stock management measures. As stocking density increases, the percentage of mortality accordingly increases for the same culture period.

Differential growth

Differential or uneven growth of the same stock is a common phenomenon in seabass throughout the life period. It promotes competition among the individuals for feed, space and other essentials for survival. The resulting additive effects of stresses on the smaller and weaker fry make them dark to black colour, posing them much more susceptible to being preyed and diseases. This uneven growth may also be attributed to dietary and environmental factors.

Disease infection

Disease infection has been known to be responsible for mass mortalities of fry in seabass nursery operations. The causative agents for the diseases are generally referred to viruses, bacteria, fungi, protozoans and other harmful pathogens including helminths. In the case of seabass disease infection the culturists pay proper attention until it is in advanced stages when the symptoms are more easily discernible, but by that time treatment becomes ineffective. There are some treatments like immersion in formalin or other antimicrobial substances. Although these external treatments have certain positive effect on the sick fish, but from practical application point of view, they may not be suitable for mass scale administer under cage and pond conditions. So the only way to prevent these diseases is maintenance of optimum culture environment.

Grow-out Culture of Brackishwater Fishes

Historically, brackishwater fish culture appears to have developed in India, Indonesia, Philippines and some other countries of South and South-East Asia. In India, traditional brackishwater fish and shrimp farming known as 'Bhasabadha fisheries' in West Bengal and 'prawn filtration' in Kerala have been in vogue during last few centuries. These techniques were developed by skilled farmers of these regions for exploiting the naturally available fish and shrimp seeds. Recently, however, a system of fish and shrimp farming has been developed in specially designed brackishwater ponds. Today, brackishwater aquaculture also known as coastal aquaculture is mostly dependent on a single species, tiger shrimp, *Penaeus monodon*. Development of finfish culture in brackishwater has not been remarkable and it is sporadic. The present practices are based on the availability of natural fry and often several species are either purposely stocked or they gain entry with the tidal water, in spite of preventive measures. The most common species in brackishwater ponds are seabass, mullets, milkfish, pearlspot and catfish. Tilapia has also got entry to the coastal aquaculture. Here, the existing traditional and some improved methods of brackishwater fish culture are presented.

Traditional farming practices

Traditional method of aquaculture is practised in the extensive backwater systems of Kerala and Sunderbans mangrove swamps of West Bengal for a few centuries. Traditional fish/ shrimp farming is also done in a smaller scale in the coastal paddy fields in the states of Karnataka, Goa and Orissa. The traditional prawn culture technique locally known as 'prawn filtration' is practised in low lying coastal paddy fields called *Pokkali* fields in Kerala. Both seasonal and perennial fields exist. Here, alongwith shrimp species some fishes are also encountered in these fields. Different fish species occurred are mullets, pearlspot, seabass,

milkfish, threadfin bream, ten pounder, tilapia etc. Average yield from these fields is around 800-1000 kg/ha. Brackishwater tidal wetlands, namely mudflats, swamps, marshes, paddy fields etc., situated in low-lying areas of North and South 24 Parganas districts of West Bengal, in which fisheries are developed, are locally known as *bheries*. These are of irregular shaped and their sizes range from 2 to 267 ha with the average size of 15-37 ha spreading over low, medium and high salinity zones. The major sources of water supply for *bheries* fisheries are the estuaries of Saptamukhi, Thakuran and Matlah (under Hooghly-Matlah estuarine system) and other minor estuaries like Gosaba, Muriganga, Haribhanga, Kulti, Ichamati, Raimangal etc., with their tributaries and distributaries. There are two types of *bheries*, seasonal and perennial. In the seasonal *bheries* both fish/shrimp and paddy are raised either through simultaneous or rotational systems, whereas in the perennial *bheries* fish/shrimp are raised almost throughout the year. Polyculture systems are practiced in *bheries* with large number of fish and shrimp seeds brought in through the tidal water and partial stocking. Fish culture operations start from February and continue till April. Large scale fishing in the *bheries* is done between September and November. The important fishes occurred in the *bheries* include *L. calcarifer*, two species of mullets, *Liza tade*, *L. parsia*, cat fish, *Mystus gulio*, *Elops spp.*, *Megalops cyprinoides*, *Eleutheronema tetradactylum*, *Therapon jarbua*, *Glossogobius giuris* etc., while the shrimps trapped consist of *Penaeus monodon*, *Fenneropenaeus indicus*, *Metapenaeus monoceros*, *M. brevicornis*, prawns, *Macrobrachium resenberghii*, *M. rude*, *M. malcolmsonii*, *Palaemon styliferus* etc. and crab, *Scylla serrata*. Now-a-days, supplementary stocking of selected fish species, such as *L. calcarifer*, *L. tade*, *L. parsia*, *Mugil cephalus*, *Oreochromis mossambicus*, *O. niloticus* is followed but in most cases the stock is left at the mercy of nature and the predators. Supplementary feeding is not generally practiced as the entire production system is dependent on utilization of natural productivity of *bheries*. However, some framers use oil cake and rice bran as supplementary feed. In this system production ranges from 500-1000 kg/ha/year. At present with the regular occurrence of white spot disease in shrimp, *bheries* farmers are opting some eco-friendly and sustainable culture systems mainly comprising of finfishes. Selective stocking is done in the *bheries* depending on the availability of natural seeds. Fish species cultured here are seabass, mullets and tilapia. Carps are also being used in low saline areas. The traditional fish farming fields are known as *Khazan* lands in Goa. In *Khazan* fields fish/shrimp farming is rotational with paddy culture. The average production from the traditional culture in Goa is about 350 kg/ha/year. In Karnataka, paddy-cum-shrimp culture is practised

in low lying coastal *Khar* lands. It is similar to prawn filtration practice of Kerala. Yield is about 400 kg/ha of which shrimps constitute 55% and the rest of the catch by fishes.

In Orissa, traditional practice of 'trapping and holding' locally known as *Ghery* has been in vogue for a long time in low lying areas and paddy fields. The average yield in the fields is 600 kg/ha/year.

Improved culture practices

The age old traditional methods of brackishwater fishes and shrimps still exist in several countries because of their simplicity and cheapness. These are being gradually replaced by modern improved culture practices.

Culture of Asian seabass, *Lates calcarifer*

In brackishwater sector, with the failures faced in shrimp farming caused by the dreaded white spot disease in India, the aquafarmers are gradually turning to finfish culture and in this context *L. calcarifer* is the most lucrative species of choice. Similarly, in freshwater sector, as the market price for IMC and other carps has become almost static and the margin of return is reducing with the regular increase in production cost, the farmers are inclined towards culture of some high-value, high-demand fish like seabass. As seed is regarded as the main critical input, so in the areas where wild seeds are available, the farmers go for improved farming system of seabass culture. Farmers generally follow polyculture or monoculture methods depending upon the availability of feeding materials.

Culture methods

Improved culture techniques include pond culture (polyculture and monoculture) and enclosure culture (cage farming). The improved culture techniques for seabass farming are of recent origin and gaining rapid popularity among farmers.

Polyculture (Predatory-prey culture) System

Polyculture is a farming practice where two or more species of fishes are reared together. This method is a modification and improvement over the traditional method. In this improved polyculture method, the food required for seabass is produced in the pond itself and seeds are stocked thereafter. The pond for seabass polyculture is prepared first, following eradication of unwanted organisms and application of manures and fertilizers. Tilapia, *Oreochromis mossambicus* and *O. niloticus* which are omnivore and prolific breeders are the best suited candidates for polyculture with Bhetki as the primary crop. Tilapia fry as forage fishes are introduced first @15,000-20,000/ha, 1-2 months prior to seabass seed stocking. Tilapia are fed with cheap feed mixture consisting of rice bran and mustard oil cake at 1:1 ratio. The tilapia grows and breeds gradually. As a result, in the same pond adult fishes,

small fry, late fry, fingerlings, juveniles are found and served as food for seabass. In this pond, seabass seeds of 3-5 g size are stocked @ 8000-10,000/ha. To provide more food, at the time of water exchange, tidal water is directly pumped into the pond so that many small fishes and crustaceans enter the pond and serve as food for seabass. If there is reduction in forage fish noticed, it is supplemented with further introduction. This practice is more sustainable. The culture period lasts for 8-12 months. In this practice production upto 2 to 4 tons/ha is achieved. Owing to its euryhaline nature, it is also cultured in freshwater polyculture systems with Indian major carps, medium carps, barbs and tilapias. IMC, *Puntius gonionotus* and tilapia fry serve as desired food for seabass. After 8-10 months culture a total production of 1.5-2.5 tons/ha is obtained.

Monoculture

Monoculture of *L. calcarifer* in ponds is a highly developed aquaculture industry in Taiwan. In India, it is practiced in some pockets where cheap trash fish as feed is available in plenty. Seabass seeds are stocked @10,000-15,000/ha in well-prepared culture ponds. In this system the stock is totally raised on supplementary feed. In wild, seabass prefers live food. So the fish is weaned to accept dead trash fish. Stocked fishes are fed with minced flesh of cheaper trash fishes collected from landing centres. Seabass does not feed at pond bottom, so the chopped trash fish is broadcasted slowly twice a day and the sinking feed material is engulfed actively by them. Feed is provided *ad libitum* at not more than 100% of total biomass initially and then gradually decreased to 10% at the last phase of culture. In this method, after a culture period of 8-10 months seabass attain average size of 800 gm with a survival rate of about 60-70% and a production of 2.5 to 4 tons/ha is achieved. For the trash fish feeding, feed conversion ratio of 6-8 is obtained on wet weight basis.

Enclosure culture (Cage farming)

Open net cage culture of Bhetki was first established in Southeast Asia in the late 1970s and continued to expand rapidly. Cage culture technology is an eco-friendly culture method and can be adopted in open waters for which soil characteristics and basin topography do have any significant role. Bhetki fingerlings after grading and acclimatization are stocked in the grow-out cages. There are two types of cages, floating and stationary cages. Floating net cages are attached to wooden frames and kept afloat using plastic drums with anchors, whereas stationary cages are fixed enclosures which are fastened to wooden poles erected in the water body at the corners. In the cages, 10-15g fish are stocked @40-50 nos./m³ initially. When the stocked fingerlings attain of about 150-200g, they are thinned to 15-20 nos./m³ and transferred to different cages of required mesh size. Cage culture is

normally done in two phases- till attaining of 150-200g in 2-3 months and thereafter upto 600-800 g in 5 months. Feeding in cages is done with either extruded pellets @5-2% or trash fish @20-5% at decreasing rate with the progress of culture period. Frequent size grading and culling of larger individuals are performed for obtaining high production. Here production level of 15-20 kg/m³ can be obtained with normal management protocol. In India, cage culture of seabass has not yet gained the shape of commercial scale production. The reports whatever available are of experimental kind. Recently, RGCA has demonstrated a production of 12 ton/ha seabass from cage culture systems at their Aquaculture Demonstration Farm, Puducherry.

Culture of other non-predatory brackishwater fishes

It has been observed that monoculture of non-predatory brackishwater fishes is not as lucrative as mixed culture or polyculture systems. This is mainly due to low production in monoculture. Experimental monoculture of these fishes has been carried out since 1980s to till date to find out economically viable technology packages. However, farmers mainly practice the mixed or polyculture based on the natural seed availability. The combined culture of two or more species of fish and shrimp having compatible feeding habits is followed for increasing production from brackishwater farms. For polyculture in brackishwater, the selective fish and shrimp species should have the following characteristics:

- Hardiness and ability to tolerate considerable fluctuations in salinity and temperature
- Ability to grow fast in brackishwater ponds
- Ability to accept natural as well as supplementary feed
- High survival rate
- Good market demand

Based on the above criteria fish species like *Mugil cephalus* (Striped grey mullet), *Liza tade* (Tade grey mullet), *L. parsia* (Goldspot mullet), *Chanos chanos* (Milkfish), *Etroplus suratensis* (Pearlspot) and shrimp, *Penaeus monodon* are considered suitable for brackishwater fish culture:

Before stocking of seeds, pond is prepared well following eradication of pests and predatory fishes, removal of bottom mud and liming, fertilization etc. The ready ponds are stocked with seeds of fish species at 8000-15,000 nos./ha alongwith tiger shrimp seeds of 10,000-30,000 nos./ha. The stocking density varies with the quantum of seed availability. Natural pond productivity is maintained by fertilization. In addition, supplementary feed prepared from locally available ingredients can be used at 2-5% body weight. This kind of system can yield a total production of 1.5-2.5 ton/ha in 6-8 months.

Conclusion

Nursery rearing of seabass fry is an important phase to get suitable size stocking materials for grow-out farming, as stocking bigger size seeds directly increases survival and production. It is an important intermediate stage of rearing between hatchery and grow-out phase. Among the two pond based rearing systems discussed here, rearing in *hapas* is more advantageous over direct pond rearing in terms of management and output point of views. Farmers should take up this type of rearing before stocking in culture ponds. Development of eco-friendly and cost-effective culture technologies of finfish targeting small-scale farmers is the need of the hour. Some steps towards brackishwater aquaculture development are extension of culture to inland saline areas, bringing more areas under culture, species diversification from existing shrimp to fishes etc. Adequate availability of quality fish seeds will also help in expansion of culture.

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BACTERIAL AND FUNGAL DISEASES OF SHELLFISH, ITS DIAGNOSIS AND CONTROL STRATEGY

Sujeet Kumar, Shyne Anand, R. Ananda Raja and Asutosh D. Deo

Aquaculture represents the fastest growing food sector industry in the world. Disease poses the biggest challenge for this growth. Fish and shellfish encounter diseases from all bio-aggressors such as viruses, bacteria, parasites and fungi. In this chapter, some of the important bacterial and fungal diseases of fish and shellfish are discussed. The bacteria causing diseases of penaeid shrimp constitute part of the natural microbial flora of seawater. Thus, bacterial infections of shrimp are primarily stress related. Adverse environmental conditions or mechanical injuries are important predisposing factors. The most common shrimp pathogenic bacteria belong to the genus *Vibrio*. Other Gram-negative bacteria such as *Aeromonas* spp., *Pseudomonas* spp., and *Flavobacterium* spp. are also occasionally implicated in shrimp diseases.

Bacterial diseases

1. Vibriosis

Diseases caused by *Vibrio* sps. are of considerable importance to marine shrimp culture. *Vibrio* epidemics are a frequent problem in shrimp hatcheries, intensively raised farm and pond culture stocks especially at the juvenile stage. Increasing frequency of vibriosis infection had characteristic luminescence which aid in their rapid identification during disease outbreak.

Outbreak of vibriosis is linked with exposure to various stresses such as higher salinity, increased ammonia level in the culture environment, low dissolved oxygen, rise in temperature and higher stocking density.

Diagnosis of bacterial infection of larval shrimp can be made by microscopic demonstration of bacterial rods in haemolymph. Symptomatically the disease is associated with melanised cuticular lesions of the appendages and melanised nodules in the gills and other organs. In pond, disease is diagnosed by disoriented shrimp swimming weakly, gathering along the edges of the pond, cloudiness of the musculature, red discoloration of the appendages, dorsal flexure at the third abdominal segment with slight rigidity, and the gathering and feeding on the weakned shrimp by fish eating birds. Haemolymph from moribund shrimp will fail to clot or will clot very slowly. Bacteria can be isolated on Thiosulphate citrate bile salt agar (TCBS), Marine agar, Tryptic soya agar and nutrient agar

with 1.5-3% NaCl and should be confirmed by biochemical testing. Histopathological evaluation of moribund shrimp can also be done to see the presence of pathogens.

A variety of strategies have been applied to manage vibriosis in shrimp farming. For shrimp hatcheries, the prevention and control is the major management consideration. Some of the management practices that apply to control of vibriosis in shrimp hatcheries include batch culture of larvae and system disinfection between stockings, disinfection of intake water for the hatchery system, rigorous sanitation of live feeds and equipment disinfection before introduction into production tanks, lower larval stocking density, control of water temperature to avoid fluctuation, reduction of fecal contamination, form broodstock in spawning tank, clean water rinsing of nauplii prior to stocking into hatchery tanks, use of periodic water exchange to reduce nutrients and bacterial population, addition of antibiotics or other chemicals to water or feeds, attentive management to reduce cross transfer of water or other vectors between larval rearing tanks.

In the shrimp production hatchery, an outbreak of bacterial disease should be responded quickly for effective disease control. Furthermore, continuous 24 hour monitoring of larval behavior and condition and further feed, water or treatment manipulations until the larval population has recovered may be essential to minimize the disease loss. Application of antibiotics in shrimp hatchery for disease control is controversial. It is preferable to use probiotics as a preventive measure.

Management strategies that have been applied for control of vibriosis in nursery and growout culture of marine shrimp include use of medicated feed, fertilization of pond with 20 Kg/ha sucrose, partial harvest to reduce pond biomass, herbal treatment with application of neem leaves at the rate of 20 kg/ ha, disinfection of pond bottom between growout cycles by drying and removal of excessive organic sediments and application of quick lime. In extreme cases, immediate farmwide harvesting of affected ponds should be done to minimize the economic losses.

2. Fouling

Fouling is also referred as black gill disease. Microbial colonization of cuticle surface of penaeid shrimp and other crustaceans is a common occurrence in larvae, juveniles, and adult shrimp. Under natural conditions, the density of microorganisms attached to shrimp remains low without recognizable harm to the host. However, under some culture conditions, density of biofouling organisms can increase to levels on shrimp where physiological function is impaired.

The predominant bacterial fouling organism affecting shrimp culture is a filamentous bacterium, *Leucothrix mucor*. Other related genera include *Thiothrix* sp., *Flavobacterium*, *Flexibacter* sp., *Cytophaga* sp. Sometimes *Vibrio* and other rod shaped organism also colonise cuticle surface. Infestations are usually a heterogenous mixture of filamentous and non-filamentous bacteria, blue green algae and protozoa (*Zoothamnium*).

The disease occurs when there is abundant colonization of the gill lamellae resulting in compromised respiratory function. Disease also occurs after colonization in mouth parts or swimming appendages. In affected shrimp growth rate gets slow with sporadic but persistent mortality of shrimp. Low oxygen and other stressor can precipitate high mortality in shrimp with heavy gill fouling.

Diagnosis of bacterial biofouling on shrimp is done by microscopic examination of whole shrimp, gill/ mouth part appendages in wet-mount preparations. Infestation severity can be numerically graded to assist decisions on the need to implement control procedures.

Control of biofouling disease of shrimp is done by routine monitoring of gill and mouthparts fouling level, adjustment in feeding, and periodic chemical treatment with soluble copper compounds. Antibiotic bath treatment can be administered to control bacterial fouling of tank reared larval and post larval shrimp.

Fungal diseases

1. Larval mycosis

Larval mycosis is the disease of penaeid hatcheries. The fungus *Lagenidium* and *Sirolopidium* are the responsible for larval mycosis. *Lagenidium* infections often occur in nauplia and protozoal stages while the *Sirolopidium* disease is usually in the late protozoal to mysis stage. Juvenile or older shrimp are not susceptible to fatal infection by the phycomycetes fungi, presubaly because of thicker exoskeleton.

Diagnosis of larval mycosis is based on microscopic demonstration of characteristic fungal mycelia that fills the body cavity of dying or dead infected larvae. The fungus can be grown on standard fungal media such as Sabouraud dextrose agar.

Once a larval shrimp becomes infected by a phycomycete fungi, chemical treatment has no effect on the survival of individual animal. However, chemical treatment of culture water with trifuralin, an herbicide, effectively inhibits the motility of zoospore and thus widespread spread of infection is prevented. Trifuralin is unstable in seawater and frequent re-dosage should be applied to maintain the therapeutic level. Likewise, malachite green oxalate could be used for control.

In hatchery, spawned eggs should be rinsed with sterile seawater. These fungus are relatively resistant to chlorine and thus 500 ppm of chlorine concentration should be maintained for 24 hours to completely destroy the fungus.

Conclusion

Vibriosis is a big challenge in shrimp hatcheries. Comparatively, its prevalence is less in shrimp and finfish grow out culture. Still, sometimes it becomes a major challenge in grow out culture in presence of various biotic and abiotic stressors. Application of antibiotics is banned in aquaculture. Therefore, bacterial diseases are mainly controlled by adopting the preventive measures such as maintaining water quality, stocking of good seed, using good quality feed. Apart from this, application of probiotics and immunostimulants has also been proved useful.

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VIRAL DISEASES OF SHRIMP, ITS DIAGNOSIS AND CONTROL STRATEGY

R. Ananda Raja and Sujeet Kumar

The advances in intensive aquaculture have brought parallel acute concern for problems associated with diseases whether by slow continuous attrition or by sudden catastrophic epizootics leads to great economic loss. Generally, infectious diseases of shrimp and other aquatic animals are caused by virus, bacteria, fungi and parasites, and other diseases and abnormalities are due to environmental stresses, genetic factors and nutritional deficiencies. This chapter is aimed at highlighting the major viral diseases of shrimp. General diagnosis, treatment and control measures are discussed at the end of the chapter to avoid duplication in many diseases.

i) Monodon baculo virus (MBV) disease

MBV is associated with high mortalities in hatchery-reared larval, post-larval and early juvenile stages of *Penaeus monodon*, caused by Type A baculo dsDNA virus first reported in Taiwan during 1983. It infects other penaeids species also. It is transmitted by ingestion of free virus and occlusion bodies, and by cannibalism. It is also believed to be transmitted vertically from broodstock to offspring, but not yet been proven. The virus affects all life stages of shrimp. It is severe in postlarvae with over 90% cumulative mortalities and senescent in adults. Clinically, sudden onset of mortality in the early life stages reduced feeding and growth rates, and an increase in gill and surface fouling. Severely infected prawns may display a white hepatopancreas and midgut. Histopathology shows large, single or multiple, roughly spherical, eosinophilic, polyhedral intranuclear occlusion bodies (OB's) in hepatopancreatic cells when stained with malachite green.

ii) White spot disease (WSD)

It is an acute, highly contagious disease of shrimp, caused by a large, ovoid, bacilliform, non-occluded enveloped dsDNA virus belonging to the family *Nimaviridae* (Nima: In Latin, means thread) and genus *Whispovirus* consisting of thread-like polar extension. The virus infects all life stages of decapod crustaceans of marine and brackishwater sources. The virus infects ectodermal and mesodermal tissues, especially the cuticular epithelium and subcuticular connective tissues. Both vertical and horizontal transmission is reported. Clinically, the disease is characterized by lethargy, inappetence, crowded at pond margin, red to pink discoloration of the body, loose cuticle, swelling of branchiostegites, broken antennae, damaged appendages, and most conspicuous feature of

small to large white spots on the inner side of the carapace which spread all over the body in advanced infection. Cumulative mortalities in infected populations may reach 100% within 3 to 7 days of the onset of clinical signs. Histologically, degenerated cells are characterized by hypertrophied nuclei with marginated chromatin and eosinophilic to basophilic intranuclear inclusions.

iii) Infectious hypodermal and haematopoietic necrosis (IHHN)

The disease is caused by ssDNA virus of family *Parvoviridae*. In *P. vannamei* it causes the chronic disease called runt deformity syndrome (RDS) characterized by lower overall crop production, shrimp with increased size variability, and cuticular deformity. Infected shrimps have been observed to rise to the water surface, remain motionless for a few moments then roll over and sink to the bottom. This behaviour may be repeated until mortality occurs. In juvenile *P. stylirostris*, more than 90% mortality reported within several weeks of onset of infection. Gross signs of infection include white to buff mottling of the cuticle, opacity of striated muscle and melanised foci within the hypodermis. In the later stages of infection *P. stylirostris* and *P. monodon* may appear bluish in colour. Infected *P. vannamei* display deformed rostrums, cuticle and antennal flagella. IHHNV forms Cowdry Type A intranuclear inclusion bodies (IB's) associated with widespread cytopathological changes including hypertrophy of the nucleus and margination of the chromatin in cells of ectodermal (epidermis, gills, fore and hind gut, antennal gland and neurons) and mesodermal origin (haematopoietic tissue, haemocytes, striated muscle, heart, lymphoid organ and connective tissues).

iv) Hepatopancreatic parvovirus (HPV)

Hepatopancreatic parvo virus is an ssDNA virus. High levels of HPV infection occurs in early juvenile stages result into slow growth which finally stops to grow at approximately 6 cm in length. Horizontal transmission occurs by cannibalism and vertical transmission by infected broodstock. Microscopically, single prominent basophilic intranuclear inclusion bodies in the hypertrophied hepatopancreatic epithelial cells are seen.

v) Infectious myonecrosis (IMN)

Infectious myonecrosis (IMN) is a recently identified viral disease caused by dsRNA infectious myonecrosis virus (IMNV), a putative totivirus. IMNV particles are icosahedral in shape and 40 nm in diameter. It causes mortalities in juvenile and subadult pond-reared stocks of *L. vannamei* and the mortality range from 40 to 70%. Outbreaks of the disease seems to be associated with certain types of environment and physical stresses (i.e. extremes in salinity and temperature, collection by cast net, etc.), and possibly with the use of low

quality feeds. Experimental infection is observed in tiger shrimp, *P. monodon* and blue shrimp, *P. stylirostris*. IMN affected shrimp presents focal to extensive white necrotic areas in the striated (skeletal) muscle, especially of the distal abdominal segments and tail fan, which can become necrotic and reddened in some affected shrimp. By histopathology, shrimp with acute phase disease presents lesions with coagulative necrosis of skeletal muscle. In shrimp recovering from acute disease or those in the more chronic phase of the disease, the myonecrosis appears to progress from coagulative to liquefactive necrosis accompanied with hemocytic infiltration and fibrosis. Significant lymphoid organ spheroid formation is typically present, and ectopic lymphoid organ spheroids are often found in the hemocoel and loose connective tissues, especially in the heart lumen and adjacent to antennal gland tubules. In some histological preparations, perinuclear pale basophilic to dark basophilic inclusion bodies are evident in muscle cells, connective tissue cells, hemocytes, and in cells that comprise lymphoid organ spheroids.

vi) Yellowhead disease (YHD)

Yellow-head disease is first noted by Limsuwan in cultured *P. monodon* adults in central Thailand during 1991. It appears to be widespread in cultured stocks of *P. monodon* in South-East Asia and India. It is caused by ssRNA virus of genus *Okavirus*, family *Roniviridae* of the order *Nidovirales*. Contaminated water, cannibalism of weak or moribund shrimp, animate vector, net and other equipment transmit the disease. Vertical transmission has not been reported. Shrimps with YHD display yellow colouration of the dorsal cephalothorax caused by the underlying yellow hepatopancreas showing through the translucent carapace. Within the ponds, infected animals, usually between 5 and 15 g, begin consuming feed at an abnormally high rate for several days then cease feeding entirely. One day after cessation of feeding, moribund shrimps may be seen swimming slowly near the edges of the pond. By the third day, mass mortality occurs and the entire crop is typically lost. Histologically, moribund shrimps suffering YHD usually have extensive abnormalities in the lymphoid organ. These include foci of necrotic cells which resemble degenerate tubules with occluded lumens and contain cells with hypertrophied nuclei, pyknotic nuclei, large vacuoles and cytoplasmic, basophilic, Feulgen-positive inclusions. Similar inclusions may also be found in the interstitial tissues of the hepatopancreas, connective tissues underlying the midgut, cardiac tissues, haematopoietic tissues, haemocytes and gill tissues.

vii) Loose shell syndrome (LSS)

LSS has been reported in India from farmed black tiger shrimp *P. monodon* since 1998 and is recognized as a major disease problem next to WSD causing significant

economic loss. It is caused by a filterable infectious agent. It causes low-level progressive mortalities unlike the rapid mortalities associated with viral pathogens such as WSSV and YHV. The clinical signs include a flaccid spongy abdomen due to muscular dystrophy, space between the exoskeleton and muscle, and shrunken hepatopancreas with reduced feed conversion efficiency and poor meat quality. Histologically, LSS-affected shrimp shows shrinkage of extensor and flexor muscles with occasional hemocytic infiltration, inflammation of hepatopancreatic tubules with enlargement of intertubular spaces and hemocytic infiltration, and low levels of lipid reserves in the R cells. In advanced stages, many tubules are in highly necrotic condition with a sloughed epithelium, reflecting the dysfunction of the digestive gland.

viii) Monodon slow growth syndrome (MSGS)

MSGS is a condition of *P. monodon* cultivation ponds characterized by abnormally slow growth and coefficients of size variation greater than 35 per cent. It is assumed to be caused by Laem-Singh virus (LSNV). It is a positive-sense ssRNA virus measuring 25 to 30 nm, first identified in Laem Singh Province in Thailand during 2003. The virus is observed in the lymphoid organ, heart and other tissues of the affected shrimps. The major lesion observed is retinopathy. Retinopathy comprises abnormally enlarged haemolymphatic vessels, haemocytic infiltration and rupture of the membrane that separates the fasciculated zone from the overlying row of reticular cells. By transmission electron microscopy (TEM) and *In situ* hybridization (ISH), LSNV is detected in the fasciculated zone and in onion bodies of the organ of Bellonci in the affected shrimp. It suggests that retinopathy associated with LSNV may be linked causally to stunting of *P. monodon* in MSGS.

ix) Taura syndrome (TS) / Red tailed disease

It is caused by a nonenveloped, linear, ssRNA virus called Taura syndrome virus belonging to the family *Dicistroviridae*. TS is widely distributed in the shrimp-farming regions of the Americas and South-East Asia. Both horizontal and vertical transmission is reported. During the preacute/acute phase of infection, shrimps appear pale red while their tail fans become bright red. They are soft shelled, lethargic and anorexic. Those with severe infections die during moult and cumulative mortalities may reach 80-95%. Recovering, chronically infected shrimps generally display multifocal, melanised cuticular lesions and may also have soft cuticles and red body colouration with normal feeding. Microscopically, Feulgen-negative inclusion bodies, which may first appear eosinophilic then change to basophilic observed in the cytoplasm of cells in areas of necrosis.

Diagnosis

The Correct diagnosis is obviously a critical step in any control program. The available diagnostic methods that may be selected for diagnosis of the shrimp diseases or detection of their etiological agents are based on:

- 1) Anamnesis.
- 2) Gross and clinical signs.
- 3) Direct bright-field, phase-contrast or dark-field microscopy with whole stained or unstained tissue wet-mounts, tissue squashes, and impression smears; and wet-mounts of faecal strands.
- 4) Post mortem findings.
- 5) Histology of fixed specimens.
- 6) Bioassays of suspect or subclinical carriers using a highly susceptible host (life stage or species) as the indicator for the presence of the pathogen.
- 7) Transmission or Scanning Electron Microscopy (TEM or SEM).
- 8) Antibody-based tests for pathogen detection using immune sera polyclonal antibodies (PAbs) or monoclonal antibodies (MAbs).
- 9) Molecular methods.

Prevention and control

The disease prevention and control strategy is the best practice for successful hatchery and grow out culture practices in shrimp industry. Quarantine measures should strictly be adopted to import broodstock to avoid entry of existing or emerging pathogen. Extensive cannibalistic nature of shrimp warrants removal of the moribund shrimp from the pond to avoid 100 per cent mortality during the outbreak of any infectious diseases. With good management practices and less stress to the animals, there are many reports of successful harvest even in the presence of dreadful disease causing pathogen WSSV. So, some of the salient points to be followed in disease prevention strategy are discussed as follows.

1. Pond preparation

Pond preparation such as removal of bottom sludge, ploughing of soil and use of lime are essential to significantly reduce the risks of shrimp disease outbreaks. The sludge must be disposed away from the pond site to avoid seep back into waterways, ponds, or cause other environmental problems. Ploughing is done to expose the black soil layer underneath bottom soil to sunlight and atmospheric oxygen for oxidative process. After ploughing, dry the pond is dried for 5 to 7 days and repeated the procedure till no more black soil is seen. Compaction of the pond bottom using heavy rollers is required to avoid turbid water during culture period.

Liming during pond preparation optimizes pH and alkalinity conditions of soil and water. Quick lime should be used only if the soil pH is low i.e. pH <5. If it is applied on soils of pH >5, then it may increase the water pH after filling and this high water pH condition may remain for a prolonged period even after stocking, which is not desirable. If the soil pH is >5, then shell lime or agricultural lime dolomite should be applied.

2. Water intake and preparation

Management practices during water intake and preparation of water before stocking can reduce the risk factors associated with disease outbreak. Water filtration (mesh of 60 holes/sq inch) and disinfection (bleaching powder @ 60 ppm) reduces the risk of disease outbreak through reduced introduction of carriers to the pond. Inorganic fertilizers like urea or superphosphate at 30-50 Kg/ha in 3-4 dosages must be used to boost the plankton production. Fertilizer should be first applied at least 10 days before the planned stocking date so as to obtain a good plankton bloom with green water color for stocking. If the water color remains unstable, fertilizer application should be continued even after stocking. This would avoid a risk of plankton bloom collapsing suddenly. The use of water reservoirs is normally good to combat disease outbreak. The organic juice containing yeast, *Saccharomyces cerevisiae* is found to be more effective in curbing the disease incidence by controlling the *Vibrio* load and improving the health status of the animals.

3. Seed selection and stocking process

Zero water exchange shrimp culture practice is very well proven to reduce the disease incidence. The whole success of grow out culture depends on the quality of the seed stocked. The seeds should be disease free from the disease free broodstock. Specific pathogen free (SPF) seeds are advisable where the closed biosecured system of grow out culture is practiced. As the stocking density is directly related to the disease outbreak, the stocking density should be 6 nos./sq.m for farms within the coastal regulation zone (CRZ) and 10 nos./sq.m outside the CRZ as per permitted limit by the coastal aquaculture authority (CAA) for sustainability of the industry whole. Longer transport time (>6 hours) of the seed from hatchery or nursery to the pond also increases the likelihood of a subsequent disease outbreak. Grossly and microscopically examined, and PCR tested PL should be used for stocking to achieve the good grow out culture. Seed acclimatization is very much important to avoid mass mortality at the time of stocking.

4. Water quality management

A number of risk factors that are significantly related to shrimp disease outbreaks and shrimp production can be addressed through routine water quality management like water

exchange from reservoirs, aeration, liming and fertilization. Periodical monitoring and management of salinity, pH, dissolved oxygen and microbial load will lead to successful grow out culture. The water depth at the shallowest part of the pond should be at least 80 cm.

5. Pond bottom management

The occurrence of black and toxic bottom sediments has been shown to adversely affect shrimp health and lead to disease outbreak or poor production. More attention must be given to pond bottom management where the farmers stock with higher density. The pond bottom soil should be checked on weekly basis, especially at the feeding area or trench. The occurrence of black soil, benthic algae and bad smell should be recorded. If the soil is black and smelly, water exchange should be carried out and feed reduced (using a feed tray to monitor requirements). During water exchange, the feeding area and where black soil occurs should be mildly and carefully agitated to dislodge the soil from the pond bottom. This will facilitate its drainage from the pond. Special attention has to be given to management of feed to reduce wastage and avoid deterioration of pond bottoms.

6. Feed management

Good feeding practice is essential to maintain water and soil quality and a healthy environment within the ponds. Pellet feed should be given according to a fixed schedule which depends on the body weight of the shrimp. Excessive feeding is dangerous than under feeding. So, feeding should be monitored with check trays to get optimum profit.

7. Routine health monitoring

The table explains about the important aspect of the gross examination for the gut contents and probable causes of disease. Moreover, one should check for abnormal behaviours and other gross lesions.

Gut content color	Probable food item	Probable cause(s)
Light or golden brown	Manufactured feed	Normal
Black, dark brown	Benthic detritus, sediment	Under-feeding; inadequate feeding frequency
Green	Benthic algae	Under-feeding
Red, pinkish	Cannibalized body parts from dead shrimp	Disease event in pond
Pale, whitish	None (disease condition)	Gregarines, or some other disease condition

8. Record keeping

Maintaining the accountability of the activities carried out through out the culture operation is very much essential for accruing the good and bad about the practices adopted.

9. Treatments and use of chemicals

The study showed that many chemicals were used against shrimp disease without any beneficial effect. Farms with higher stocking density tend to use more chemicals, both in terms of numbers and the quantities applied. The study showed that antibiotics and a range of probiotics had no significant effect on the risk of shrimp disease outbreaks. Use of lime, fertilizers in low input systems, and disinfectants in farms using higher stocking densities had some protective effect against shrimp disease. Feed additive including vitamin and mineral premix as well as some bacterial products had some beneficial effect on shrimp production in higher stocking density farms. There is a concern to residues in harvested shrimps which are hazardous for human health. Only calcium hypochlorite (bleach) should be used as disinfectant. Other chemicals such as lime compounds, fertilizer, zeolite and related compounds that do not lead to residues in the harvested shrimps can be used to manage the pond condition and shrimp health. It has been clearly shown that shrimps of good quality can be grown with very minimal chemical usage and without the usage of antibiotics. Minimal usage of chemicals and no usage of banned chemicals lead to reduced cost of production and easy marketing of the harvested shrimp in domestic and export markets. There is a serious concern on the use of antibiotics, and their use in shrimp farming should be avoided.

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DISEASES OF BRACKISHWATER FINFISH, ITS DIAGNOSIS AND CONTROL STRATEGY

Sujeet Kumar, K.P. Jithendran, R. Ananda Raja, Shyne Anand

Asian seabass (*Lates calcarifer* Bloch) is an important food fish in Asia. Its importance is rising with the success of captive breeding and standardization of larval rearing technology. This fish species is found to be susceptible to various pathogens of parasitic, bacterial and viral origin. Viral nervous necrosis (VNN) and Iridovirus infections are the major viral fish diseases which is gaining importance worldwide. In this chapter, some of the important diseases associated with fish culture are discussed.

1. Viral diseases of fish

1.1 Viral Nervous Necrosis

Introduction: Viral nervous necrosis (VNN) is a devastating disease of marine fish species cultured worldwide. It affects more than 40 wild and cultured marine fish species, especially the larval and juvenile stages which records high mortality. Based on characteristic lesions, VNN is also known as viral encephalopathy and retinopathy (VER).

Etiology: The disease is caused by piscine nodavirus of the genus betanodavirus, family Nodaviridae. The virus contains single stranded, bipartite positive sense RNA genome.

Affected species: In India, betanodavirus infection has been observed in cultured and wild population of brackishwater / marine fish species such as *Lates calcarifer*, *Mugil cephalus*, *Chanos chanos*, *Epinephelus tauvina*, *Sardinella longiceps*, *Amblygaster clupeioides*, *Thriposocles dussumieri*, *Leiognathus splendens*, *Upeneus sulphureus*, *Mystus gulio*, etc.

Mode of transmission: Nodaviruses are regarded as pathogens of marine fishes. However, natural development of disease has also been reported from low saline and freshwater environments. Betanodaviruses are quite resistant to environmental conditions, which make it possible to get translocated by commercial activities via influent water, juvenile fish, utensils, vehicles, etc. Translocation of species for stocking purpose from one location to another may be another way but it is yet to be validated. Latent infection among wild fishes also serves as source of infection. Apart from all these means of horizontal transmission, vertical transmission is highly suspected to takes place from infected spawners to fry.

Symptoms and lesions: The major clinical signs of VNN are characterized by cellular necrosis and vacuolation in the central nervous system (brain and spinal cord) and retina accompanied with common behavioural changes such as lack of appetite, erratic, spiral or belly-up swimming and dark coloration of body. The severity of disease is the maximum in

juveniles with higher rate of mortality. In seabass, the earliest onset of clinical signs of the disease is during 18-21 day post hatch. Recently, the disease is observed in fishes irrespective of their age. Instances of asymptomatic / sub-clinical infection among wild fishes may possibly act as potential carriers.

Diagnosis: The first step in combating an infectious disease is detection and identification of the causative agent using a reliable diagnostic method followed by better surveillance techniques. Viral nervous necrosis can be diagnosed by demonstrating characteristic lesions in the brain and / or retina by light microscopy. Detection of virions by electron microscopy, viral antigens or antibodies by serological methods such as indirect fluorescent antibody test, and enzyme linked immunosorbent assay or detection of viral nucleotides by molecular techniques such as RT-PCR and nested PCR and by tissue culture.

The RT-PCR is the most sensitive test and has become the main diagnostic method for fish nodaviruses. The capsid protein gene sequence is the target sequence for this test. More recently, it was reported that nested PCR was 10-100 folds more sensitive than the RT-PCR and permitted diagnosis by using blood, sperm, as well as nervous and ovarian tissues.

Clinically, the affected animals show spiral or looping swim pattern, swim bladder hyperinflation and later wasting. Vacuolation is seen in the grey matter of brain and retina of eye. Necrosis is observed in the spinal cord, brain and retina while intracytoplasmic inclusion in nervous cells.

Prevention and control: Against VNN, no vaccine or treatment method is available. Broodstock fish acts as the most important source of the virus to their larvae by vertical transmission. Therefore, after RT-PCR screening, virus-carrying broodstock should be eliminated. Fertilized eggs should be disinfected with disinfectants such as iodine or ozone. Strict hygiene within the hatchery should also be maintained by regular disinfecting the hatchery facility and farm materials with chlorine. Separate rearing facility of larvae / juveniles from brooder should be maintained with each batch of larvae / juveniles in the separate tanks supplied with sterilized (UV or ozone) seawater.

1.2 Iridovirus infection

Introduction: Iridovirus infection is a significant cause of mortality in farmed red sea bream (*Pagrus major*) and more than 30 other species of cultured marine fish belonging mainly to the orders Perciformes and Pleuronectiformes. The infection has also been detected in Asian seabass. It affects all the stages of fish but the susceptibility of juveniles is generally higher than adults.

Etiology: The disease is caused by double stranded DNA virus of genera Lymphocystivirus and Ranavirus. *Ranavirus* causes systemic disease in infected fish and are associated with high morbidity and mortality.

Mode of transmission: The principal mode of transmission of Iridovirus infection is horizontal via the water. The vertical mode of transmission has not yet been established.

Diagnosis: Outbreaks of the disease have been mostly reported in the summer season at water temperatures of 25°C and above. Depending on host fish species, fish age, water temperature, and other culture conditions, mortality rates range between 0 and 100%. Affected fish become lethargic, exhibit severe anaemia, petechiae of the gills, and enlargement of the spleen.

The sample should be collected from moribund fish. Gill and visceral organs such as spleen, heart, kidney, liver and intestine should be collected. However, the spleen and / or kidney tissues are the most appropriate organ for pathogen detection by IFAT. The fish sample should be stored at 4°C for use within 24 hours or –80°C for longer periods.

The disease is characterised by the appearance of abnormally enlarged cells stained deeply with Giemsa solution in the histopathological observations of the spleen, heart, kidney, intestine and gill of infected fish. These enlarged cells react to an anti-RSIV MAb by the antibody based antigen detection (Indirect fluorescent antibody test - IFAT) test. Electron microscopy confirms the presence of virions (200–240 nm in diameter) in these cells.

PCR is able to detect Iridovirus infection with high degree of sensitivity in short time. Recently real time PCR has been developed which shown improved rapidity, sensitivity, reproducibility, and the reduced risk of carry over contamination over normal PCR. An antibody based enzyme-linked immunosorbent assay (ELISA) has also been developed to detect iridovirus infection.

Prevention and control: A formalin-killed vaccine is available against Red Sea Bream Iridovirus infection (RSIVD) and is effective and commercially available. A number of general husbandry practices are used to reduce RSIVD associated losses. These includes introducing pathogen free fish, implementing hygienic practices on farms and avoiding practices that can decrease water quality and / or increase stress, such as overcrowding and overfeeding.

2. Bacterial diseases in fish

Bacterial diseases occur as a result of the complex interactions between pathogen, fish and environment. Environmental stresses can affect the homeostatic mechanism of fish thus reducing their resistance to disease-causing organisms. Fish reared in intensive culture

systems are exposed to extreme environmental fluctuations and they may be more sensitive to disease than wild populations.

2.1. Vibriosis

Different species of *Vibrio* such as *Vibrio anguillarum*, *V. ordalli*, *V. salmonicida* and *V. vulnificus* biotype 2 are commonly affects the fish. The typical symptom of vibriosis are hemorrhages at the mandible, eyes, isthmus, breast, bases and rays of dorsal, pectoral, pelvic, anal and caudal fins. Echymosis and petechiae are observed on the body surface and ulceration on the skin and muscle. The affected fish are annorectic. Internally, there is congestion and haemorrhages in the liver, spleen and kidney, frequently accompanied by the presence of necrotic lesions. The gut and particularly the rectum may be distended and filled with a clear viscous fluid. The body is completely covered by a thick layer of mucus. Microscopically, extensive lamellar hyperplasty, honey-comb vacuolization & fatty degeneration in hepatocytes, and aggregation of melano-macrophage centres (MMC) in kidneys and spleen are observed. The degree of the lesions varies with the severity of the disease. Fingerlings die more rapidly than adults.

2.2. Photobacteriosis

Photobacteriosis is also described as pasteurellosis. It is caused by the halophilic bacterium *Photobacterium damsela subsp. piscida* (formerly called as *Pasteurella piscida*). The disease is characterized by the presence of white nodules in the internal viscera, particularly, spleen and kidney, so called as pseudo-tuberculosis. Severe mortalities occur usually when water temperatures are above 18-20°C. Below this temperature, fish can harbour the pathogen as subclinical infection for long periods.

The presumptive identification of the pathogen is based on standard biochemical tests. Several commercial vaccines against *Ph. damsela subsp. piscida* are available on the market, but their efficacy is dependent of the fish species, fish size, vaccine formulation and use of immunostimulants. As the majority of the pasteurellosis outbreaks occur from larval stages to fingerlings of 10–30 g, a vaccination programme which comprises a first dip immunization at the larval stage and a booster vaccination when fish reach a size of about 1–2 g is recommended to avoid high economic losses caused by this disease.

2.3. Columnaris disease

Columnaris disease is a disease of juvenile seabass in low saline condition. It is caused by *Flexibacter columnaris*, a gram negative, aerobic bacillus. Clinically, the condition may be chronic, acute or peracute. Grossly, saddle-shaped lesions in the mid-body position

near the dorsal fin of the fish and bilaterally symmetrical lesion as a fuzzy, pale yellow white plaque, with dark margins, observed in the affected fish.

2.4. *Aeromonas* infection

Aeromonas sp. is a common water-borne bacterium normally present in fishes. Whenever fishes are exposed to environmental stress or injury, it causes serious outbreaks of haemorrhagic disease with high mortality. Grossly, hemorrhages on the fin and tail and erosion of tail and fin can be clearly seen in severe cases. Temperature, pH, high CO₂, O₂ depletion, free NH₃, over-crowding and low salinity are considered as predisposing causes for *Aeromonas* infection.

Treatment of bacterial diseases in fish

Quality of water including pH and temperature greatly affects treatment results. Therefore, water quality needs to be carefully checked. A few fishes are first treated and checked for how they react before treating the entire group. The total volume of water in the pond must be known to treat it accurately. To determine the volume of a pond, the surface unit area of water is multiplied by the average depth of the pond. Then, the approved and calculated therapeutics is evenly administered all over the pond.

Bacterial diseases are treated best by injection or use of food additives or antibiotics. Terramycin @ 2.5–3.0 g/ 100 lb body weight/ day for 10 to 12 days with a withdrawal period of 21 days, Sulfamerazine, Nitrofurans and Furazin @ 10 g/ 100 lb BW/ day for 10 days, Erythromycin @ 4.5 g/ 100 lb BW/ day for two weeks are used for the treatment based upon requirements. Potassium permanganate (KMnO₄) is used at the rate of 2–3 ppm as a wide-spectrum treatment in ponds against bacterial infections.

3. Fungal diseases of fish

3.1 Epizootic Ulcerative Syndrome (EUS)

The disease is reported in freshwater as well as brackishwater fishes throughout India. It is caused by non septate fungus *Aphanomyces invadans* and is invariably associated with secondary infection by gram negative bacteria and rhabdoviruses. The transmission of disease occurs through zoospores which are transmitted through water, direct contact between fishes, and transport of infected fishes into new area.

The disease is diagnosed by sudden high mortality rate in wild and farmed fish with different types of fishes gets affected and show abnormal behaviour. Hemorrhagic ulcers are seen on head which often extend to skull leading to exposure of brain. Confirmatory diagnosis of disease is done by detection of non-septate, branching fungal hyphae in the

periphery of the lesion. Invasive fungus extends through muscle tissue to spinal cord, kidney, and peritoneum.

No vaccine is available for the control of disease. Controlled feeding regime should be operated during the outbreak of disease as a means of preventive measures. The affected fish should be destroyed and all farm appliances should be disinfected. The affected pond should be dried followed by liming application before restocking.

4. Parasitic diseases of fish

The major groups of parasites in brackishwater fishes are (i) protozoans and (ii) metazoans. Based on the location on the host fish it can be also categorised under ecto- or endoparasites.

4.1 Protozoan parasites

Protozoans are single celled microscopic organisms with specialized structures for movement, food gathering and attachment. They are either external or internal parasites. The following are the major protozoan parasites in fishes.

4.1.1 Dinoflagellates

Amyloodinium is a common dinoflagellate found in fishes particularly young fishes. These are microscopic external parasites with flagella for movement. It is usually attached with gill filaments or body surface of the affected fish. Predisposing factors are high level of organic matter in water and higher stocking density of fish.

Symptoms and lesions: The gill and skin of affected fish shows signs of necrosis and destruction. Darkening of body surface is observed. Affected fish often found on the surface or near the source of aeration. Mass mortality of fish is observed in severely affected pond, if not treated in time.

Treatment: Short bath treatment with 200 ppm formalin for 1 hr. and copper sulphate bath with 0.5 ppm for 3-5 days with aeration and daily water replenishment.

4.1.2 Ciliates

Ciliates are microscopic external parasites with cilia for movement. The most common ciliates are *Cryptocaryon* and *Trichodina*.

4.1.2a *Cryptocaryon*

Cryptocaryon is a pear shaped (0.3-0.5 mm in size) parasite. The parasites are found on the external body surface of fish. High stocking density and low water temperature serve as predisposing factor for infection.

Symptoms and lesions: It affects mainly gills and body surface. The affected part has increased mucus production. The white spots are observed over the body surface. Wound

often has the complication of secondary bacterial infection. Affected fish are found to rub body against submerged objects. Respiratory distress with mass mortality is seen in untreated cases.

Treatment: Copper sulphate bath with 25 ppm or / and formalin bath with 0.5 ppm for 5-7 days with aeration and daily replenishment of water.

4.1.2b *Trichodina*

Trichodina has a circular body with 100 µm diameter with cilia around the perimeter. It infects mainly gills, body surface and fins. High level of organic matter in water or poor water exchange serves as predisposing factors.

Symptoms and lesions: Affected fish has pale gills with irritation on body and hence rubs body against objects. Excessive mucus production is observed on gills and body surface. This leads to respiratory distress by clogging of gills by mucus.

Treatment: Short bath treatment with 200 ppm formalin for 30-60 minutes with strong aeration or extended bath treatment with 25 ppm formalin for 1-2 days with good aeration and daily replacement of water.

4.1.3 Myxozoans

Myxozoans are microscopic internal parasites. The parasites either infect tissue of various organs or live freely in cavities like gall bladder, abdominal cavity, etc.. Poor water quality, high stocking density, feeding with infected trash fish and lack of quarantine measures facilitate infection. Common examples of myxozoans are *Myxobolus*, *Myxidium*, *Kudoa*, *Ceratomyxa* etc.

Symptoms and lesions: Clinical symptoms are not apparently visible. However, white or black cysts may be seen on body surface, gills, fins and internal organs. The parasites invade all major organs and forms cysts or freely floating mass called pansporoblast. They destroy gills and all major target organs of the fish.

Treatment: No effective treatment is available. Preventive steps should be taken by having efficient water exchange, avoiding feeding of trash fish and by quarantine measures.

4.1.4 Microsporidian

Microsporidians are intracellular internal parasites. It forms spore and resembles myxozoans. The oval shaped spores have average size of 6 µm. Poor water quality and poor nutrition is the predisposing factor.

Symptoms and lesions: No visible symptoms of infection are observed. Cysts are observed in various internal organs like intestinal wall, ovary, fat tissue etc. These cysts are brown or black in color and are of various size and shape called xenoma.

Treatment: No treatment is available. The disease can be prevented by good water exchange.

4.2 Metazoan parasites

Metazoans parasites of fishes include mainly helminths of different classes, arthropods dominated by parasitic crustaceans and some annelids such as leeches. Among helminths, trematodes or flatworms are of significance in cultured fishes though nematodes and cestodes are found rarely as massive infections. Trematodes constitute mainly external parasites with adhesive structures for attachment to the host. These are two types: 1. Monogeneans and 2. Digeneans.

4.2.1 Monogeneans

Monogeneans are mostly external parasites with specialized posterior attachment organs. It needs single host for completing its life cycle and multiply very fast in confined water bodies. They include skin and gill flukes.

4.2.1a Skin flukes

Skin flukes are 2-6 mm long and the most common forms are *Benedinea* sp., *Dactylogyrus* sp. The infection sets in when there is high stocking density, poor water exchange and overlapping of cultured fish.

Symptoms and lesions: The disease mainly affects body surface, fins, eyes, and sometimes gills. Mass mortality is observed among affected fish due to rapid multiplication of the parasites. The parasites damage the host tissue leading to secondary bacterial infection. Fish becomes lethargic with excessive mucus production on gills and body surface with opaque eyes and skin lesions. Affected fish rubs body against submerged objects. Eye infection sometimes leads to blindness.

Treatment: Short bath with 100 ppm formalin or freshwater for 10-30 minutes or 150 ppm hydrogen peroxide for 10-30 minutes with strong aeration.

4.2.1b Gill flukes

Gill flukes are ectoparasites. The common gill fluke are *Diplectanum* sp., *Gyrodactylus* sp. High density and poor sanitation serve as predisposing factors.

Symptoms and lesions: Mass mortality with respiratory problems is observed in severely affected fish. Other symptoms are pale gills, low consumption of feed, erratic swimming behaviour and mucus production on gills.

Treatment: Short bath treatment with freshwater for 10-30 minutes or 200 ppm hydrogen peroxide for 60 minutes or 100-200 ppm formalin with strong aeration are recommended.

4.2.2 Digeneans

Digeneans as a group are mostly intestinal parasites and involves more than one host for completing their life cycle. These are of variable size based on the species. No treatment method is suggested, although anthelmintic drugs are used. Digeneans complete their life cycle in a molluscan host, therefore, elimination of molluscs from the culture facility should stop the transmission cycle of the parasite.

4.2.3 Nematodes

Nematodes or roundworms are large intestinal parasites, with un-segmented body and mostly 1-2 cm length. Infection is prevented by maintaining hygienic conditions.

4.2.4 Crustaceans

Parasitic crustaceans include mainly copepods such as *Caligus* sp., *Ergasilus* sp., *Lernanthropus* sp. etc. and are commonly found on the gills, skin and fins. *Caligus* sp. is oval shaped parasite with about 3 mm length and 1.6 mm width. The parasite has four pair of legs and a pair of suckers over the frontal edge of the body. Poor water quality serves as predisposing factors.

Symptoms and lesions: Affected fish had sluggish behaviour, shows anorexia and become weak due to heavy infestation. Gills and skin have erosions and secondary bacterial infection leading to high mortality, if not treated.

Treatment: Short bath treatment with 100 ppm hydrogen peroxide for 30 minutes or 200-250 ppm formalin for 1 hr. with good aeration. Mild infection can be controlled by simple freshwater bath for 10-15 minutes.

Lernanthropus sp., *Ergasilus* sp. and parasitic isopods such as *Cymothoa* sp. are relatively larger than *Caligus* and commonly found on the gills and fins. Dichlorvos @ 1ppm is effective against these parasites.

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PARASITIC DISEASES OF SHRIMP

R. Ananda Raja, K.P. Jithendran and Sujeet Kumar

Viral, bacterial, fungal and parasitic diseases are major causes of production and economic losses in the aquaculture industry. With the rapid increase of production, the sector experienced parallel increase in quantum of diseases. In last two decades many disease and its causative agents have been identified which lead to search for innovative means of control. This chapter describes the major diseases attributed to parasites of finfish and shell fish and their diagnosis, treatment and control. Apart from virus and bacteria, other pathogens such as fungus and parasites are responsible for causing important diseases such as larval mycosis, protozoan fouling, cotton shrimp disease and bopyrid infestation in shrimp.

i) Protozoan infection

a) Endocommensal / Invasive protozoa

Order: *Microsporida*

It affects muscle and causes “**milk or cotton shrimp disease**”. Shrimps are infected by ingestion of spores. The affected shrimp show cooked- muscle appearance and the exoskeleton appears bluish black, and white tumor-like swellings may be found on gills and subcuticle. The spores extrude a filament that penetrates the gut wall and deposits an infective unit which enters nucleus of the gut cells where schizonts are developed. The schizonts divide and develop to form spores which are located in muscles and other organs of the shrimp. The infected shrimp is infective to fish species and advice versa but not to the shrimp itself. The major genus are *Pleistophora* sp. (>8 spores / envelope), *Thelohania* sp. and *Agmasoma* sp. (8 spores / envelope).

Order: *Haplosporida*

It affects the digestive glands of shrimp but the incidence is rare.

Order: *Gregarina*

It affects the digestive tracts and other tissues of shrimp. Shrimps are infected by ingestion of spores. The developed sporozoites attach to the gut wall, grows in to trophozoites and form gametocysts. Gametocysts undergo multiple divisions to produce gymnosporozoites which are released outside and be an infective stage for invertebrates such as clams, snails or marine worms. Spores are developed in the intermediate hosts and released in mucous strings which become infective to shrimps. The major genus involved are *Nematopsis* sp., *Cephalolobus* sp. and *Paraophrioidina* sp.

Body invaders

There are several protozoa invade the body and feed on tissues of weakened and diseased shrimps. They are *Parauronema* sp., *Leptomonas* sp and *Paranophrys* sp. and amoeba.

b) Ectocommensal protozoa

They are found on the gill and body surface of the shrimp. They are *Zoothamnium* sp., *Epistylis* sp., *Acineta* sp., *Lagenophrys* sp., *Ephelota* sp., and *Apostome* ciliates. Special staining technique like silver impregnation staining is used for identification of *Apostome* ciliates. They cause “**protozoan fouling**” and “**Fuzzy mat-like appearance**” due to ciliate fouling. The affected shrimps show restless and difficulty in locomotion and respiration.

ii) Metazoan parasites / Helminthiasis

Metazoan parasites in shrimp are categorized as Trematodes, Cestodes, and Nematodes. Immature forms and adult worms are found in the different parts of the body in shrimps.

(a) Trematodes (flukes)

The cercarial forms of the flukes are infective to the shrimps. The cercaria penetrates the shrimp and encysted in the form of metacercarial forms in tissues which are infective to the first intermediate host (fish). The metacercaria develops in to adult and release eggs. Eggs are hatched out and miracidia released which penetrate second invertebrate host, snail and develops in to sporocysts. Cercarias develop inside the sporocysts in second intermediate host and released in to water which become infective stage for shrimp. Eg. *Opecoeloides fimbriatus*, *Microphallidae* sp. and *Echinostomatidae* sp.

(b) Cestodes (tape worms)

Shrimp ingest copepods or other crustaceans with larval form of tape worm which develop in to advanced larval stage in shrimp. The advanced larval stage enters the first intermediate host (sting ray) by ingestion of the infested shrimp and develops in to adult and release eggs. The eggs are eaten by copepods, second intermediate host and develop in to larvae which are infective to shrimps. Eg. *Prochristianella penaei*, *Parachristianella* sp., *Renibulbus* sp., Pear shaped worms, and Cyclophyllidean group.

(c) Nematodes (round worms)

Shrimp ingest copepods or other crustaceans with larvae which develop in to advanced larval stage in shrimp. The advanced larval stage enters the first intermediate host (toad fish) by ingestion of the infested shrimp and develops in to adult and release eggs. The eggs are eaten by copepods, second intermediate host and develop in to larvae which are

infective to shrimps. Eg. *Spirocumallanus pereirai*, *Leptolaimus* sp., *Ascaropsis* sp., and *Hysterothylactum reliquens*.

iii) Other infestations

Single cell plant diatoms (on larvae), Over growth of algae of mixed variety, barnacle, leeches, colonial hydriod *Obelia bicuspidata*, insects eggs, Isopods -*Aega* sp are occasionally observed among wild and poorly farmed shrimp populations. Bopyrid parasitic infestation caused by *Epipenaeon spp.* belonging to family *Bopyridae* which are lodged in the brachial cavity leads to impaired respiration and reproductive failures.

Diagnosis

The Correct diagnosis is obviously a critical step in any control program.

1. Rapid diagnosis by history, clinical signs and lesions.
2. Isolation and identification of the causative agents.
3. Histopathology and histochemistry.
4. Electron microscopy.
5. Restriction Endonuclease Analysis.
6. Serological methods.
7. Molecular methods.

Treatment

a) External parasites

1. Formalin 15–25ppm as a pond treatment or Dip treatment of affected animals with 250ppm for one hour at temperature below 15.6°C and 100ppm for one hour at higher temperature.
2. Combination of 25ppm formalin and 0.1ppm malachite green against cryptocaryon.
3. Malachite green as a dip treatment at a concentration of 1:15,000 to control fungus on both fish and eggs.
4. Acetic acid at a 1:5 concentration for 1–2 min and acriflavin at 3–5ppm to control external parasites.
5. Organo phosphorus compounds, Dylox (Dimethyl phosphonate) at the concentration of 0.25ppm to control anchor worm, crustaceans, gill and body flukes but is not effective against protozoans.
6. Cypermethrin 10% w/v– 10-15ml/1000 sq. m. water spread area.

b) Internal parasites

1. Anthelmintics Di-N-Butyl tin oxide mixed in the food at the rate of 1% and fed at 3% of BW for three days.

Prevention and control

1. Dry the pond before starting the culture.
2. Use sieve at water inlet, bleach before stocking to weed out wild shrimp & fishes and intermediate hosts.
3. Maintain the good water quality through out the culture.
4. Use only disease-free stock, select a good genetic strain of broodstock and develop resistant stocks.
5. Supply of adequate balanced nutrition.
6. Regulate population density, periodical size grading and proper disposal of dead animals.
7. Handle the animals with good care and control other animals.
8. Proper chemical prophylaxis and vaccine development for immunological protection.
9. Regulations to prevent transfer of pathogens from one host population to another, nationally or internationally.
10. Proper destruction and disposal of infected animals.
11. Sanitation and disinfection of hatchery and equipments.

Conclusion

The impact of parasites on aquaculture is well documented and economically important for the local and international markets. Prerequisite for establishing effective control measures in fish and shrimp cultures is a quick and exact diagnosis of the causative agent, and the knowledge on the complex host–parasite–environment relationships. The correct parasite identification is of great significance, and may decide on the survival or death of the infested fish. Further attention should be paid towards the parasitic infestation in view of the change in climatic conditions.

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COMMONLY REPORTED DISEASES IN FRESHWATER AQUACULTURE AND THEIR CONTROL

Ashutosh D Deo, R. Ananda Raja and Sujeet Kumar

Intensification of aquaculture has helped in achieving the goal of self-sufficiency in fish production to a greater extent. At the same time, it has also propagated artificial stress to overcrowding, nutrients and metabolites accumulation, dissolved oxygen and other water quality problems. All such potential stressors have magnified the risk of outbreak of diseases and thus an integrated health management approach is needed involving control of pathogenic organisms, decreasing host susceptibility and correction of impaired environmental conditions, etc. Such a radical management system of fish health is now becoming a decisive factor for the success of aquaculture.

Aquatic organisms are susceptible to many communicable and non-communicable diseases. Manifestation of infectious diseases in susceptible fish species is the combined action of potential pathogen and the stress caused by improper environmental factors. Potential pathogens are virus, bacteria, fungus, protozoan and metazoan parasites etc. Many fish may carry small numbers of pathogenic organisms either harbouring chronic, low grade infections or serving as carriers while in some cases it may account for sudden massive outbreaks of acute disease that sometimes occur when such fishes are subjected to stress. The most significant among them are:

Name of the disease	Species affected	Causative agent	Symptoms	Control measures
Eye disease	Mainly Catla, lesser extent other carps and air breathing fish (<i>Channa</i> spp.)	Bacteria	Epidemic disease affects eyes; cornea vascularised & becomes opaque; subsequently eye ball gets putrefied, leading to death	Treat affected ponds with 0.1 ppm KMnO ₄ , followed by 300 ppm lime, maintain dissolve oxygen level
Columnaris or Cotton wool disease	Catla, Rohu, Mrigal, Silver carp and Grass carp	Bacteria	Raised white grayish plaques, often with reddish peripheral zone leading to haemorrhagic spots on body	Dip treatment with 500 ppm KMnO ₄ , Improve water quality
Dropsy	Juveniles and	Bacteria	Body scales	Disinfect affected ponds

	adults of Catla, Rohu and Mrigal		stretch out resembling pine cone; inflammation; ulceration; exophthalmia; abdominal distension	with 1 ppm KMnO ₄ ; dip treatment of 5 ppm KMnO ₄ for 2 minutes
Ulcer	Catla, Rohu, Mrigal, Silver carp, Grass carp, Mangur and Singhi	Bacteria	Ulcerations; exophthalmia; abdominal distension	Destroy badly infected fish; disinfect affected ponds with 0.5 ppm solution of KMnO ₄ ; add sulphadiazine (100 mg/kg) or Terramycin (75-80 mg/kg) to feed for 10-12 days
Fin and Tail rot	Catla, Rohu, Mrigal, Silver carp, Grass carp, Magur and Singhi	Bacteria	Whitish margin on fins and caudal region reaches up to base, lacerations, cell death	Dip treatment with 500 ppm KMnO ₄ , Improve water quality, reduce organic load.
Saprolegniosis	Eggs, Fry and Fingerlings of Catla, Rohu, Mrigal, Silver carp, Grass carp, <i>Channa</i> spp., <i>Puntius</i> spp	Fungus	Mould grows like cotton wool on body, penetrating into the muscle; morbid muscle rot	3-4 % NaCl or KMnO ₄ at 160 mg/litre bath for five days; 1-2 mg/litre malachite green bath for 30 minutes to 1 hour; add formalin at 20 ml/litre to affected ponds, Maintain water quality and stocking density.
Branchiomycosis (gill rot)	Fry and Fingerlings of Catla, Rohu, Mrigal and Silver carp	Fungus	Fungus grows out through gill blood vessels and causes death of surrounding tissues; yellow-brown discolouration & disintegration of gill tissues	Addition of quick lime (50-100 kg/ha) to affected ponds; in case of limited infections, use 3-5 % NaCl bath for 5-10 minutes, or 5 ppm KMnO ₄ bath for 5-10 minutes, Reduce organic load, maintain hygienic condition
Ichthyophthiriasis	Fry and Fingerlings of Catla, Rohu and Mrigal	Parasite (protozoan)	Skin, fin rays & operculum covered with white spores; sick fish keep rubbing against hard substratum	Dip in 1:5 000 formalin solution for 1 hour for 7-10 days or in 2 % NaCl for 7-10 days; affected ponds should be disinfected with quicklime at 200 kg/ha
Trichodiniasis	Fry,	Parasite	Invasion of	2-3 % NaCl bath for 5-10

	Fingerlings and adults of Catla, Rohu , Mrigal, Common carp, Silver carp, Grass carp	(protozoan)	parasites in skin & gill region, creamish coating due to excessive mucus secretion, hampered respiration.	minutes or 4 ppm KMnO ₄ bath for 5-10 minutes; treat affected ponds with 25 ppm formalin, Maintain good water quality.
White gill spot disease	Catla, Rohu and Mrigal	Parasite (protozoan)	Weakness; emaciation; raising of scales along their posterior margins; scale loss; perforation of scales; loss of chromatophores	Reduce density; add yeast to feed (1 g/kg); 2-3 % NaCl bath
Dactylogyrosis and Gyrodactylosis	Fry, fingerlings of Catla, Rohu and Mrigal	Parasites (monogenean trematodes)	Gill, fin & skin affected; excessive mucus secretion, growth reduction.	3-5 % NaCl dip treatment for 5-10 minutes; 100 ppm formalin bath; treat affected ponds with 25 ppm formalin or 4 ppm KMnO ₄
Black spot or Diplostomiasis	Fry, fingerlings of Catla, Rohu and Mrigal	Parasites (digenean trematode)	Black nodules due to metacercarial cysts in the host body; infects eye and causes blindness	Remove resident molluscan population
Argulosis Ergasilosis Lernaeosis	Catla, Rohu , Mrigal, Silver carp and Grass carp	Parasite (crustacean)	Parasites visible to naked eye attached to head ,gills, buccal cavity operculum & fin rays; haemorrhagic spots found in chronic cases,	Drain and dry ponds showing severe Argulus infection; short duration 5 ppm KMnO ₄ dip; treatment with 'Butox' three times at 35 ml/ha/m at weekly intervals
Epizootic Ulcerative Syndrome	Catla, Rohu , Mrigal, Silver carp ,Grass carp, <i>Channa</i> sp., <i>Puntius</i> sp, Magur, Singhi, <i>Mystus</i> spp, Many other fishes	Unknown Necessary causative agent: fungus <i>Aphanomyces invadans</i>	Most dreadful diseases, severe in winter, Large red or grey shallow ulcers with necrotic areas on skin; fungus extends deep into the musculature; lesions of acute dermatitis and	200 kg/ha quicklime or 0.1 ppm CIFAX

		ulcers	
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Common diseases of freshwater prawn and their control

Most commonly cultured shellfish in fresh water is *Macrobrachium rosenbergii*, the scampi (Giant Freshwater Prawns) and *M. malcomsonii* (Indian River Prawn). Considering vast fresh water natural resources in the various states of India and high export potential, the giant fresh water prawn, *M. rosenbergii*, enjoys immense potential for culture in India. Scampi can be cultivated for export through monoculture in existing as well as new ponds or with compatible freshwater fishes in existing ponds.

With the rapid development in hatchery technology particularly in the artificial seawater, the grow out culture of scampi in inland states is gaining pace due to which outbreak of diseases are very common. The situation becomes worst when the good husbandry and environmental management is neglected for a very long time. Here are some of the most commonly occurring diseases of freshwater prawn encountered in hatchery and grow out systems.

Diseases occurring in Hatchery systems

1. **Bacterial Necrosis:** High mortality in larval stages IV and V, older stages are more resistant. In severe case mortality goes up to 100% in 48 hours. The affected larvae turn bluish green in color with empty stomach. The antennae and newly formed appendages have brown spots. As stress is the pre disposing factors, removal of stress can prevent this disease. The occurrence of disease can be minimized by good water quality and proper feed management. The use of antibiotic like Bipencillin–streptomycin in the ambient water at the dose of 2ppm should be considered as the last option to treat this disease.
2. **Mid cycle larval disease (MCD):** This is responsible for large scale mortality in hatchery. The larvae show weak movement, reduced feed consumption and settling at the bottom. Most of the times moribund individuals are eaten by the healthier larvae. Clinical signs are similar to bacterial necrosis. The disease is more severe in early larval stages IV to IX. This disease is due to mixed bacterial infection and the numbers of PLs are reduced to 1-2 PL/L from 10-15Pl/L. Proper sanitation and good nutrition have been proven effective in eliminating the disease.
3. **Luminescence disease:** This disease is very common in hatcheries. The causative agents are *Vibrio* species bacteria mainly a *V. harveyi*. The infected larvae glow in the dark/night. Larvae swims slowly, show opacity, aggregation and finally die. Mortality

may reach up to 100 percent. Treating infected larvae with calcium hypochlorite (20-30ppm) or formalin (50ppm) have been found significant; if situation does not improve whole stock may be discarded.

4. **Exuvia entrapment disease:** This occurs in the final stages of larvae and early post larvae. Affected larvae are unable to free appendages, eyes or rostrum from the exuvia in which they become entrapped. Some animals even shed the exuvia, but die soon after molting. Mortality varies from 20 to 30 percent. It is believed that the disease is due to poor water quality or nutritional inadequacy. Addition of lecithin in the larval diet reduces the molting related problems. The use of algae in larval rearing tanks and feeding late stage larvae with lecithin rich feed are suggested to control the disease. Poor water quality is reported to be the pre disposing factor of the disease.

5. **Protozoan disease:** This is very common parasite infection in hatcheries but less common in grow out systems. This is caused by the protozoa *Zoothamnium* sp., *Epistylis* sp, *Vorticella* sp, *Acineta* sp. and heavily infested larvae turn opaque. Heavy infestations can obstruct molting, suppress the growth and can cause death. Improve water quality and give static bath in formalin.



Fig: Protozoan infections in prawn.

6. **Fouling by epibiont:** This disease is caused by epibionts, filamentous bacteria and many non-filamentous bacteria, algae *etc.* common to the aquatic environments. This disease is most common in larvae and post larval stages. This disease has been reported from grow out system also. Poor water quality is the most suitable cause for the development of epibionts. Treating infected larvae with formalin (50-150ppm) removes epibionts.

Diseases occurring in grow out systems

1. **Viral disease/ white tail disease (WTD)/ white muscle disease (WMD):** This disease is mainly reported from *M. rosenbergii* which is due to nodavirus (MrNV) and extra small virus (XSV). MrNV infected post larvae and adult show milky to opaque abdomen. The discoloration appears in telson region gradually progress towards the head. Eventually all muscles in the abdomen and cephalothorax become white. Severely affected tissues of juveniles are striated muscles of the abdomen and

cephalothorax and intratubular connective tissues of hepatopancreas. Mortality rate reaching up to 95 percent. This is the most severe problem at present in the fresh water prawn industry/culture system. No treatment has been reported. The brood stock or seed tested positive for white tail disease must not be used for culture. Usually sanitation and control protocols for viral infections are recommended.

2. **Brown/ black spot disease:** This disease is characterized by ulcerative to raised focal or multifocal lesions. The disease is usually found focally on gills, carapace, appendages, uropod's, telson and body cuticle. In severe cases infection may spread to the epithelium, muscle and viscera, resulting in mortalities. For control of the disease maintain good husbandry practices and nutritional requirements.
3. **Black gill disease:** This disease is caused by poor water quality which is represented by accumulation of nitrogenous waste product. Increasing levels of ammonia and nitrite in growout ponds results in growth suppression and mortality. Sometimes such diseases have been reported due to iron precipitation from acid sulphate soil. Improvement of water quality gives better result.
4. **Isopod infestation:** Isopod infestations are most common in eastern and north eastern states of India. Isopod parasite affects the gill cavity and causes deformities in the gill filaments. Infected specimens show poor growth rate. As the parasite settles in the gill cavity, market value of prawn goes down. No specific treatment has been reported.



Fig: Prawn with Isopod infestation

Health management

Fish health management measures which have to be followed for preventing and controlling disease outbreaks involves the following major steps aiming towards attacking all the three interlinked causative factors - pathogen, host susceptibility and the environmental stress. These are:

- Maintain environmental parameters within limits.
- Provide nutritionally balanced feed.
- Avoid overcrowding.
- Stock disease free fry/ fingerlings/PL /Juveniles.

- Apply lime at regular interval.
- Disinfect farm appliances regularly.
- Use immunostimulants/probiotics/bioremediators

Precautionary considerations

- Drugs and chemicals are often used as a stop-gap arrangement to correct errors in management. it should not be used to prop up poor culture programme. A sound husbandry practice is the best approach to disease control.
- Indiscriminate feeding of low levels of antibiotics will remove only those bacteria most sensitive to the drug and can lead to the development of drug resistant strains. Drug resistant bacteria can transmit this resistance to other bacterial species that have never been exposed to the drug. Therefore treatment with antibiotics should be only at prime need.
- Monitor D.O. levels before and during treatment. Fishes are stressed during treatment and their D.O. need is increased.
- A small group of fish should always be treated first before treating the whole lot with a new compound or formulation or using a product for the first time.
- Calculations for dilutions of the drug should always be rechecked.
- Once treatment has been started, rigidity should be maintained to dose and treatment schedule even when mortality has been stopped.
- If antibiotics are used to treat diseases give sufficient withdrawal period.

Conclusion

Disease prevention and control mainly depends on the identification of predisposing factors, correct diagnosis, preventive measures and treatment which can be achieved through the use of realistic stocking densities, preventing the entry of pathogen, improving the unkempt condition and by providing the balanced nutrition. The dictum prevention is better than cure should be the objective of aquaculture. Quarantine of newly introduced fish in the farm should be followed holistically. The regular observation of fish while feeding them is also essential to maintain good health of fishes. When faced with a severe outbreak of disease, the culturist should consult the local fish disease experts or the institutions where the examination of samples of infected fish and water can be done.

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PRINCIPLE OF MICROBIAL IDENTIFICATION: A TRADITIONAL APPROACH

Sujeet Kumar and R. Ananda Raja

Identification of bacteria by traditional means involves a number of tests which includes (a) morphological, (b) cultural, (c) biochemical and (d) serological, and (e) phage typing. Serological tests and phage typing are mainly used in epidemiological analysis and rarely used in bacterial identification. In all cases it is absolutely essential that the tests should be carried out on pure cultures.

Generally for this purpose, first bacteria are grown on the non-selective medium, and examined for their colonial morphology, and their chromogenic effect on the medium like color. Cultural characteristics on simple media are often too variable to be of diagnostic value. However, a few groups of bacteria produce characteristic morphology which may allow the experienced microbiologist to make an educated guess. Some of the cultural characteristics useful for bacterial identification are presented in Table 1.

Cowan and Steel (1983) suggests three alternative approaches to dealing with unknown organisms. These are:

- (a) A blunderbuss approach in which every test is done and the results are compared with Bergeys manual.
- (b) A probability approach which involves an informed assessment of the microorganisms likely to be present in a given situation. For example, in brackishwater we suspect presence of *Vibrio*. In sewage we suspect the presence of *E. Coli*, faecal *Streptococci* and *Clostridium perfringens*. Tests would be carried out to indicate the presence or absence of these groups.
- (c) The progressive or step by step approach, which is the one most commonly used.

Table 1. Cultural characteristics of bacteria grown on solid media

Sl. No.	Colony character	Description
1	Shape	Circular, irregular, radiated, rhizoidal etc.
2	Size	Size of colony in mm
3	Surface	Smooth, contoured, rough, ridged, striated, dull, Glistening
4	Edge	Entire, undulate, lobate, crenated, fimbriate, effuse, spreading
5	Opacity	Translucent, transparent, opaque
6	Colour	Different colours due to production of pigments

Other than cultural characteristics, traditional methods mainly rely on staining and morphology in the initial phase and then biochemical testing at a later stage. Morphological characteristics like cell shape (rod, cocci, coccobacilli, and comma), sporulation, flagellation, capsule etc. are important aid in preliminary grouping of the bacteria. Some of the morphological characteristics useful for distinguishing bacteria are presented in Table 2.

The Gram stain should be carried out on fresh cultures grown overnight, as old cultures frequently give a Gram-negative response. This stain indicates differences in cell wall chemistry. The crystal violet and iodine form a complex which cannot be removed from Gram-positive cells using ethanol, and cells thus appear purple. Gram-negative cells, however, are decolourised and when counterstained appear pink/red. When handling unknown cultures, it is sound practice to stain cells whose Gram reaction is known at the same time.

Table 2. Morphological characteristics of bacteria

Sl. No.	Parameter	Description
1	Shape	Cocci, oval, short rod, long rod, filamentous, comma, spiral
2	Size	Length and breadth in μm
3	Arrangement	Single, pairs, chains, in four (tetrads), in groups, grape like clusters, irregular
4	Flagella	Polar, monotrichous, amphitrichous, peritrichous
5	Spores	Spherical, oval, elliptical, single or multiple, terminal, subterminal or central
6	Capsule	Present or absent
7	Staining	Gram positive or negative

After getting the background information of cultural characters, and microscopical examination by staining techniques some routine biochemical tests such as motility test, oxidase, catalase, oxidation-fermentation test etc. should be conducted. These three category of tests in many cases enable a well versed microbiologist to identify a bacteria upto genus level. The routine biochemical tests are presented in table 3.

Under morphological characteristics a number of points should be carefully noted down such as appearance of the cells, like whether they are cocci or rods, and whether the cells are arranged singly, in filaments, in groups, or in chains. In a few groups of bacteria, it may be difficult to decide whether the cells are cocci or rods. This problem may be related to the stage of the growth cycle from which the cells were taken. Further the size of bacterial cells should be measured after calibration.

In the spore stain, cells are being examined for the presence of resistant spores. The test works on the basis that the dye, malachite green, is driven into the spores by the heating process and cannot be removed by washing with distilled water. Washing does remove the dye from the vegetative portion of the cell, which can then be counter-stained with safranin. Spores therefore stain green, and the vegetative portion of the cell stains red. If spores are present, then the organism is either from the genus *Bacillus* or *Clostridium*. These can easily be distinguished as *Bacillus* is aerobic and *Clostridium* is anaerobic. The position and size of the spores should be noted as these may be of diagnostic value, i.e. are they central, subterminal or terminal, and are they the same diameter as the cell or do they cause the cell to bulge?

Table 3. Routine biochemical tests for identification of bacteria

Sl. No.	Test	Basis	Inference
1	Gram's staining	Cell wall peptidoglycan helps in retaining dye	Gram + bacteria show violet colour and Gram –bacteria Red/pink colour
2	Motility test (Hanging drop method)	Presence of flagella	Zig-zag motility indicates presence of polar flagella
3	Oxidase test	Presence of cytochrome oxidase	Development of blue colour shows positive reaction
4	Catalase test	Catalase enzyme which breakdown toxic hydrogen peroxide	Production of gas bubble (Effervescence) indicate positive reaction
5	Carbohydrate fermentation test	Utilisation of sugar in presence and absence of oxygen	Acid production in open tube indicate oxidative metabolism while in paraffin covered tube fermentative metabolism
6	Indole test	Production of indole by degradation of tryptophan	Development of pink colour indicates positive reaction
7	Voges proscauer's test	Detect acetoin or acetyl methyl carbinol an intermediate product of glucose metabolism	Production of red/crimson colour indicates production of acetoin
8	Sensitivity to O/129	Vibrio are sensitive to O/129	Differentiate <i>Vibrio</i> from closely related bacteria like <i>Pseudomonas</i> , <i>Aeromonas</i> etc

Motility must also be checked. This may be done in one of several ways, for example by direct observation using the hanging drop method, by flagellar staining, or by a cultural method. If the first two methods are attempted, then it is essential that a young liquid culture is used. The temperature of incubation is also important as a number of organisms are motile at 30°C but not at 37°C, even though this may be the optimum temperature for growth. The hanging drop method and flagellar staining both require some experience to produce reliable results. The cultural method is the most reliable in inexperienced hands. This involves the preparation of a bottle of a sloppy agar medium with a glass tube in the middle which is open at both ends (Craigie tube). The central tube is inoculated with bacteria using a straight wire, incubated overnight, and examined for growth outside the tube. If growth is confined to the centre of the tube then the organism is non-motile, but if growth is found throughout the medium, then the cells must have been motile.

Both the oxidase and catalase tests are very rapid, producing results in 2-3 minutes. The other primary tests include the ability to produce acid from glucose, and the Hugh and Leifson oxidation/fermentation (O/F) test.

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PRINCIPLE OF MICROBIAL IDENTIFICATION – A MODERN APPROACH

Sujeet Kumar and R. Ananda Raja

Modern approach of bacterial identification relies upon molecular tools. These are rapid and distinguish closely related species and strains. In the present chapter few commonly used molecular approaches have been described.

G+C content of bacterial genome: This includes the melting temperature of DNA (G+C content of DNA) and similarities in DNA sequences. The G+C content of DNA is about 40% in eukaryotic species. But in bacteria it varies from species to species. The G+C content of some bacteria belonging to genus *Mycoplasma* and *Clostridium* are as low as 21%. While in *Micrococcus* it is as high as 70%. The G+C content is determined by hydrolysis of DNA, by melting curve analysis and by analysing the bases by High Performance Liquid Chromatography (HPLC).

Nucleic acid Hybridisation: The similarities between two genomes can be compared by nucleic acid hybridisation. In this procedure the genome of two bacteria is heated to melt it to a single strand. Then DNA of two different bacteria is allowed to hybridise. If the DNA is homologous then hybridisation will take place. DNA-DNA hybridisation is used to study the relationship between closely related organisms. For studying the distantly related organisms, RNA-DNA hybridisation can be used. In this process the 16S rRNA of one organism is hybridised with DNA of another organism.

Nucleic acid sequencing: Recent advances in sequencing of part or full bacterial genome have opened up new horizon for bacterial taxonomy. The genome of almost 1000 bacterial and archeal organisms have been fully sequenced. The nucleic acid sequence of 16S ribosomal RNA gene is considered as gold standard for identification of microbes. This is because of its molecular clock properties and availability of large database. Moreover 16S ribosomal RNA is universal in nature and present in all the bacteria. The length of 16S rRNA gene is almost 1550 bp, which is very much suitable for taxonomic analysis. The 16sr RNA of bacteria consists of three region viz. conserved region, semi-conserved region and variable region. The huge database of sequencing of 16S rRNA of wide range of bacteria is available at National Centre of Biotechnology Information (NCBI). After sequencing of 16s rRNA gene of any bacteria, the sequences can be compared with database resources available with NCBI by BLAST (Basic Local Alignment Search Tool). Based on sequence comparison, the phylogenetic tree or phylogram can be formed. Several softwares are available for

transforming sequence data into phylogenetic tree e.g. Clustal X, Clustal W2, Bioedit etc. Apart from NCBI, DDBJ (Data Bank of Japan) and EMBL (European Molecular Biology Laboratory) are some others DNA database.

Protein profiles and Amino acid sequencing: Analysis of proteins is done by Sodium dodecyl sulphate-Polyacrylamide gel electrophoresis (SDS-PAGE). Proteins of different molecular weight migrate differently on SDS-PAGE. After electrophoresis, each organism yields unique pattern of proteins bands which is visualised after staining of SDS-PAGE by Coomassie blue. Amino acid sequence of some protein is also very important in bacterial identification and taxonomy.

DNA bar coding: In recent years, this is also an important method for identification of species. In DNA bar coding, a short section of DNA sequence is used to identify the species. These are conserved sequence for some gene and are specific for particular species.

RNA based methods: RNA is comparatively small molecule than DNA thus containing a low genetic information as compared to DNA. Among RNA molecule, mostly 5S rRNA and tRNA molecules have been targeted for sequencing. RNA sequencing is done by alkaline hydrolysis following end labelling. 16S rRNA molecule of bacteria is generally more conserved than the part of the genome. The major drawbacks of RNA sequencing are that this is more tedious than sequencing of DNA and RNA molecule is not stable.

Molecular typing of bacteria

Molecular fingerprinting of bacteria is important in epidemiological study, disease monitoring, source tracking, population genetics and in food safety related investigations. The common genotyping methods include Randomly Amplified Polymorphic DNA (RAPD), Amplified Fragment Length Polymorphism (AFLP), Restriction Fragment Length Polymorphism (RFLP), Pulsed Field Gel Electrophoresis (PFGE), gene probes, ribotyping, insertional sequences analysis, multilocus sequence typing and repetitive deoxyribonucleic acid sequences analysis.

Randomly amplified polymorphic DNA (RAPD): The RAPD technique is a PCR based method. It uses short arbitrary primers usually of 10 random bases which anneal to multiple random target sequences. The RAPD-PCR reaction requires low stringency annealing conditions, which results in the amplification of randomly sized DNA fragments. The multiple band pattern generated by the RAPD-PCR is followed by dendrogram analysis to generate fingerprint profiles for the test organism. The RAPD method has been used for fingerprinting of enteropathogenic *Escherichia coli*, *Salmonella*, *Shigella*, *Vibrio*, *Aeromonas* and *Listeria* from food and water samples.

Amplified fragment length polymorphism (AFLP): It involves digestion of genomic DNA by two restriction enzymes, followed by binding at 5' and 3' end by ribonucleotide primers in stringent annealing condition, PCR amplification and thereafter gel analysis. Thus, AFLP selectively amplifies the restriction fragment from the digest of total genomic DNA using the PCR. This fingerprinting technique has several advantages over other DNA fingerprinting techniques. The most important are the capacity to inspect the entire genome for polymorphism and its reproducibility. It has the potential to become a universal fingerprinting system because it can be applied to any DNA sample including DNA of bacteria, human, animal and plant.

Restriction fragment length polymorphisms (RFLP): Chromosomal DNA restriction analysis was the first of the chromosomal DNA-based typing schemes. It is regarded as the most sensitive method for strain identification and discrimination. Because of the high specificity of the restriction enzymes and the stability of chromosomal DNA, a reproducible pattern of fragment is obtained after the complete digestion of the chromosomal DNA. The procedure involves isolation of DNA, digestion of DNA with restriction endonucleases, size fractionation of the resulting DNA fragments by electrophoresis gel matrix to membrane, preparation of radiolabelled and chemiluminescent probes, and hybridization of membrane bound DNA.

Ribotyping: Ribotyping is the powerful molecular fingerprinting tool to determine the genetic variation in bacteria based on rRNA gene. Typically each ribosomal operon consists of the three genes encoding the structural rRNA molecules, 16S, 23S and 5S co-transcribed as a polycistronic operon. The copy numbers, overall ribosomal operon sizes, nucleotide sequence, and secondary structure of the three rRNA genes are highly conserved within a bacterial species due to their fundamental role in polypeptide synthesis. Knowledge of intra-species conservation of the 16S rRNA gene sequence and basic 16S-23S-5S ribosomal operon structure led Grimont and Grimont to develop ribotyping technique for bacterial classification. The conventional ribotyping technique is based on restriction endonuclease cleavage of total genomic DNA followed by electrophoretic separation, southern blot transfer, and hybridisation of transferred DNA fragments with a radiolabelled ribosomal operon probe. Following autoradiography, only those bands containing a portion of the ribosomal operon are visualised. The no. of fragments generated by ribotyping is the reflection of multiplicity of the rRNA operons present in a bacterial species. The copy no. of rRNA operons have been found to range from 1 in *Chlamydia trachomatis* to 15 in *Photobacterium profundum*.

Pulsed field gel electrophoresis (PFGE): All linear double-stranded DNA molecules that are larger than a certain size (~ 40 kb) migrate through agarose gels at the same rate. Above this critical length, the locity of DNA molecules becomes almost independent of their size and depends chiefly on the strength of the electric field. Pulsed field gel electrophoresis technique, switches electric field periodically between two different directions with pulse times ranging from 0.1 to 1000 seconds. This allows the separation of DNA molecules upto ~ 5 Mb in length.

With each reorientation of the electric field relative to the gel, smaller sized DNA will begin moving in the new direction more quickly than the larger DNA. The larger DNA lags behind, thus, providing a separation from the smaller DNA. This provides an important tool to separate larger DNA molecules of interest. Large DNA molecules are easily sheared and difficult to pipette due to its high viscosity. This problem is solved by embedding the bacteria in agarose plugs and then treating the plugs with enzymes to digest away the cell wall and proteins, thus leaving the naked DNA undamaged in the agarose. The plugs then are cut to size, treated with restriction enzymes, if necessary, loaded in the sample well, and sealed into place with agarose. The field strength has a profound effect on the pulsed field separations, in general, 4-6 volts/cm is required for resolving DNA upto 2000 Kb in 1-2 days.

Denatured gradient gel electrophoresis (DGGE): The general principle of DGGE is separation of individual rRNA genes based on differences in their chemical stability. Polyacrylamide gels consisting of a linear denaturing gradient formed by urea and formamide are employed for DGGE. It is a useful technique for monitoring dynamic changes in mixed bacterial populations over time. The rRNA gene sequences from bacterial species in a mixed culture are first amplified using conserved bacterial primers of a hyper-variable region of the rRNA gene, producing amplicons of the same length but with differing sequences that are specific to a given species. DGGE allows the separation of these amplicons, producing a molecular fingerprint of the bacterial species.

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APPLICATION OF MOLECULAR TOOLS FOR IDENTIFICATION OF BACTERIA RELEVANT TO AQUACULTURE

R. Ananda Raja, Sujeet Kumar and Ashutosh D Deo

An explosive development and application of molecular tools for identifying microbes and analyzing their activity has gained its great importance in aquaculture. For identifying microbes and analyzing their activity, handful of techniques are available based on the nucleic acids. The nucleic acid- based tools are more frequently used because of the high throughput potential. These techniques include PCR amplification, *ex situ*, or *in situ* hybridization with DNA, RNA, or even peptide nucleic acid probes. The 16S rDNA-based methodologies are robust and superior to traditional methods based on phenotypic approaches, which are often unreliable and lack the resolving power to analyze the microbial composition and activity of bacterial populations. In addition, panoply of approaches that are based on DNA sequences other than rDNAs have been applied frequently for strain identification. DNA arrays can be used in comparative genomics or genome- wide expression profiling. Overviews of these approaches are presented in this chapter.

16S Ribosomal RNA (16S rRNA)

16S rRNA technique is most commonly used for genus to species level identification. Hybridizations with ribosomal RNA (rRNA)-targeted probes have provided a unique insight into the structure and spatiotemporal dynamics of complex microbial communities. Nucleic acid probes can be designed to specifically target taxonomic groups at different levels of specificity by virtue of variable evolutionary conservation of the rRNA molecules. Appropriate availability of large databases or the online resources offer powerful platforms for a rapid identification of bacteria, and hence, they are used as reliable and rapid diagnostic tools.

Nucleic acid probing

Nucleic acid probing is based on two major techniques such as dot-blot hybridization (FISH) and whole-cell *in situ* hybridization. Dot-blot hybridization is an *ex situ* technique in which total RNA is extracted from the sample and is immobilized on a membrane together with a series of RNAs of reference strains. Subsequently, the membrane is hybridized with a radioactively labeled probe, and after stringent washing, the amount of target rRNA is quantified. Because cellular rRNA content is dependent on the physiological activity of the cells, no direct measure of the cell counts can be obtained. In contrast to dot-blot hybridization, fluorescent *in situ* hybridization (FISH) is applied to morphologically intact cells and thus provides a quantitative measure of the target organism. The listed probes can all be used for dot- blot hybridizations, but for application in FISH, specific validation is

required because some regions of the rRNA are not accessible because of their secondary structure and protection in the ribosome. Hence, the number of validated FISH probes is much smaller than that of the probes suitable for *ex situ* analysis. Many bacteria, however, even well-known species, do not grow well under laboratory conditions. FISH can be used to detect directly the presence of the suspect on small samples, to compare the genomes of two biological species, to deduce evolutionary relationships, and to visualize the distribution of specific species within the biofilm.

Genotypic typing

Genotyping is done for identification and classification of microbiota at or near the strain level. The most powerful genetic-based molecular methods are ribotyping, randomly amplified polymorphic DNA (RAPD), pulsed-field gel electrophoresis (PFGE), and amplified fragment length polymorphism (AFLP). Basically, these methods rely on the detection of DNA polymorphisms between species or strains and differ in their dynamic range of taxonomic discriminatory power, reproducibility, ease of interpretation, and standardization.

Ribotyping

Ribotyping is a variation of the conventional restriction fragment length polymorphism (RFLP) analysis. RFLP analysis of bacterial DNA involves the digestion of genomic DNA with rare-cutting restriction enzymes to yield a few relatively large fragments. Ribotyping combines Southern hybridization of the DNA fingerprints, generated from the electrophoretic analysis of genomic DNA digests, with rDNA-targeted probing. The probes used in ribotyping vary from partial sequences of the rDNA genes or the intergenic spacer regions to the whole rDNA operon. Ribotyping has been used to characterize strains of bacteria. However, ribotyping provides high discriminatory power in aquaculture microbiology at the species and subspecies level rather than on the strain level.

Randomly amplified polymorphic DNA (RAPD) or Arbitrary amplification

RAPD has been widely reported as a rapid, sensitive, and inexpensive method for genetic typing of different strains of bacteria. This PCR-based technique makes use of arbitrary primers that are able to bind under low stringency to a number of partially or perfectly complementary sequences of unknown location in the genome of an organism. Several factors such as annealing temperature, DNA template purity and concentration, and primer combinations have been reported to influence the reproducibility and discriminatory power of the RAPD fingerprints.

Pulse field gel electrophoresis (PFGE)

By RFLP, the bacterial genomic DNA is digested with rare-cutting restriction enzymes to get a relatively large fragment. The restriction fragments are then size-

fractionated using PFGE that allows separation of large genomic fragments. The generated DNA fingerprint obtained depends on the specificity of the restriction enzyme used and the sequence of the bacterial genome and is therefore characteristic of a particular species or strain of bacteria. This fingerprint represents the complete genome. It can detect specific changes like DNA deletion, insertions, or rearrangements within a particular strain over time. Its high discriminatory power has been reported for the differentiation between strains of important probiotic bacteria. A new approach combining RFLP with DNA fragment sizing by flow cytometry has been reported recently for bacterial strain identification. DNA fragment sizing by flow cytometry is found to be faster and more sensitive than PFGE. PFGE is shown to be more discriminatory in typing closely related strains than either ribotyping or RAPD analysis.

Amplified restriction length polymorphism (AFLP)

It combines the power of RFLP with the flexibility of PCR- based methods by ligating primer-recognition sequences (adaptors) to the digested DNA. Total genomic DNA is digested using two restriction enzymes, one with an average cutting frequency and another with higher cutting frequency. Double-stranded nucleotide adaptors are usually ligated to the DNA fragments serving as primer binding sites for PCR amplification. The use of PCR primers complementary to the adaptor and the restriction site sequence yields strain-specific amplification patterns.

Enterobacterial Repetitive Intergenic Consensus (ERIC) – PCR

ERIC elements comprise repetitive sequences in bacterial genomes not restricted only to enterobacterial genomes, which are targeted to produce unique fingerprint for target bacteria based on the relative abundance of the repeats within the bacterial genome. The discriminatory power of ERIC-PCR have been claimed to be at par or more than the RAPD-PCR for *Aeromonas* spp.

BOX-PCR

Like ERIC elements, BOX elements too are repetitive sequences of 154 bp length found within the bacterial genome. PCR amplifying these repeats can be useful for fingerprinting and typing of bacteria in epidemiological studies.

Denaturing gradient gel electrophoresis (DGGE)

DGGE offers a unique and comprehensive tool for the characterization of bacterial communities. With DGGE, double- stranded DNA is denatured in a linearly increasing denaturing gradient of urea and formamide at elevated temperatures. As a result a mixture of amplified PCR products will form a banding pattern after staining that reflects the different melting behavior of the various sequences. The DGGE-generated patterns make it possible to

monitor shifts in the structure of microbial communities over time and/or following different treatments. Subsequent identification of specific bacterial groups or species present in the sample can be achieved either by cloning and sequencing of the eluted bands or by hybridization of the profile using phylogenetic probes. One major limitation of DGGE fingerprinting is its low sensitivity in detecting rare members of the community. However, with group- or species-specific primers, the sensitivity of detecting less-frequent bacteria has been significantly improved.

Other PCR approaches

Other PCR-based approaches such as amplified ribosomal DNA restriction analysis (ARDRA), repetitive extragenic palindromic PCR (Rep-PCR), and triplicate arbitrary primed PCR (TAP-PCR) have been used for molecular typing, and have been shown to offer a high discriminatory power for the identification and differentiation of microbiota.

DNA microarray

High-throughput methods for determining the composition of microbiota can be achieved with DNA microarrays. Identification can be accomplished with designed diagnostic arrays within a matter of hours without prior cultivation and knowledge. Detection-type arrays generally contain oligonucleotides targeting a set of sequences, usually the 16S rRNA gene. The probes have to be designed so that they will hybridize with similar efficiencies to a target group of sequences. Essentially, the oligonucleotides are designed by in silico prediction using sequences from the databases and printed or arrayed onto slides. The target DNA or rRNA of the bacteria from the sample is purified and prepared to incorporate a fluorescent label, fragmented, and hybridized to the arrays. For a very complex microbial ecosystem highly specialized arrays may be used to target specific microbes.

Conclusion

The smorgasbord of molecular techniques innovatively experimented upon, painstakingly optimized, rigorously evaluated and ready to use were available in aquaculture. Although these advanced various throughput techniques have been successfully applied to the identification and taxonomic classification of a number of bacteria, the outcome can be highly variable between laboratories. Furthermore, a basic limitation in genotypic typing procedures is that the organism to be typed must be isolated because DNA from other sources disturbs the DNA fingerprints. Considering the cost/time-effectiveness and ambiguities that are still inherent to some of these techniques, 16S rRNA sequence-based methods offer a viable option for the rapid and reliable identification of bacterial populations.

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POLYMERASE CHAIN REACTION IN DISEASE DIAGNOSIS

R. Ananda Raja and Sujeet Kumar

Polymerase Chain Reaction (PCR), a technique very often used in molecular biology. It is a means of selectively amplifying a particular segment of DNA. The basic principles of replicating a piece of DNA using primers were described by **H. G. Khorana** in 1971 but the progress was limited by primer synthesis and polymerase purification issues. The DNA was properly amplified by **Kary Mullis** during 1983 who was awarded **Nobel Prize** in Chemistry in 1993 for his contribution in PCR. PCR derives its name from one of its key components, a DNA polymerase used to amplify a piece of DNA by *in vitro* enzymatic replication. As PCR progresses, the DNA thus generated is itself used as template for replication which is further exponentially amplified. Recent past years, PCR has become indispensable technique in medical and biological research. This chapter discusses about the various steps and compounds involved in PCR and its application in disease diagnosis and monitoring in aquaculture.

Components and reagents

1. Buffer solution

It provides a suitable chemical environment for optimum activity and stability of the DNA polymerase.

2. Mg^{2+} concentration

The Mg^{2+} ion forms complex with dNTPs, primers and DNA template. It plays major role in the yield of PCR product. So the optimal concentration of $MgCl_2$ has to be selected for each reaction. Too few Mg^{2+} ions result in a low yield of PCR product, and too many increase the yield of non-specific products, and promote misincorporation and multiple bands to appear. Lower Mg^{2+} concentrations are desirable when fidelity of DNA synthesis is critical. The concentration of $MgCl_2$ should be selected empirically, starting from 1 mM and increasing in 0.1 mM steps, until a sufficient yield of PCR product is obtained. If the DNA samples contain EDTA or other chelators, the $MgCl_2$ concentration in the reaction mixture should be raised proportionally.

3. dNTPs

The concentration of each dNTP in the reaction mixture is usually 200 μM . Inaccuracy in the concentration of even a single dNTP dramatically increases the misincorporation level. So it is very important to have equal concentrations of each dNTP

(dATP, dCTP, dGTP, dTTP). When maximum fidelity of the PCR process is crucial, the final dNTP concentration should be 10-50 μ M.

4. & 5. Primers

PCR primers are usually 15-30 nucleotides in length. Longer primers provide higher specificity. The GC content in primers should be 40 – 60% and distributed uniformly throughout of the primer. To avoid nonspecific priming, more than three G or C nucleotides at the 3'-end of the primer should be avoided. The primers should have neither self-complementarity nor inter-complementarity in order to avoid primer-dimer and hairpin formation. The melting temperature of flanking primers should not differ by more than 5°C, so the GC content and length must be chosen accordingly. The annealing temperature should be approximately 5°C lower than the melting temperature of primers.

The melting temperature (T_m) of the primers is calculated using the following formula:

$$T_m = \left[\left(\text{Percentage of GC content in primers} \times 0.41 \right) + 62.3 \right] - \left[\frac{500}{\text{Primer length}} \right] - 5$$

6. Autoclaved Distilled Water

Autoclaved distilled water is used to make up the desired volume for the reaction.

7. DNA template

It contains the DNA region (target) in question to be amplified.

8. DNA Polymerase

In 1957, **Arthur Kornberg** identified the first DNA polymerase and was awarded the **Nobel Prize** in 1959 for the same. In 1969, **Thomas Brock** reported the isolation of a new species of thermophilic bacterium, *Thermus aquaticus* from which Taq DNA polymerase was isolated in 1976. The enzyme, Taq DNA polymerase is world widely used in PCR reaction. It should be stored at -20°C in a non-frost free freezer, typically in 50% glycerol. The tubes should never be allowed to reach room temperature and gloves should be worn when handling to avoid contamination. Before opening a new tube of enzyme, it is spun briefly as there is often enzyme in the cap. When pipetting enzyme from a stock tube, the end of the tip is just plunged far enough into the enzyme to get what is needed to avoid excessive adherence of enzyme to the peripheral tips. Enzyme should never be added to unbuffered water to avoid its denaturation. Usually 0.5-1.0 U of *Taq* DNA polymerase is used in 50 μ l of reaction mix. Higher *Taq* DNA polymerase concentrations may cause synthesis of nonspecific products.

The Cycling Reaction

There are three major steps in a PCR, which are repeated for 30 or 40 cycles depending on the product size. This is done on an automated cycler, which can heat and cool the tubes with the reaction mixture in a very short time.

1. Initial Denaturation

The initial denaturation should be performed over an interval of 1-3 min at 95°C. This interval may be extended up to 10 min for GC-rich templates. If the initial denaturation is no longer than 3 min at 95°C, *Taq* DNA polymerase can be added into the initial reaction mixture. If longer initial denaturation or a higher temperature is necessary, *Taq* DNA polymerase should be added only after the initial denaturation, as the stability of the enzyme dramatically decreases at temperatures over 95°C.

2. Denaturation

During the denaturation, the double strand melts open to single stranded DNA which is important in PCR reaction. Incomplete denaturation of DNA results in the inefficient utilization of template in the first amplification cycle leads to a poor yield of PCR product. Usually denaturation for 0.5-2 min. at 94-95°C is sufficient. If the amplified DNA has a very high GC content, denaturation time may be increased up to 3-4 minutes. Alternatively, additives such as glycerol (up to 10-15 vol.%), DMSO (up to 10%) or formamide (up to 5%) may be used to facilitate DNA denaturation. In the presence of such additives, the annealing temperature should be adjusted experimentally, since the melting temperature of the primer-template DNA duplex decreases significantly when these additives are used. The amount of enzyme in the reaction mix should be increased since DMSO and formamide, at the suggested concentrations, inhibit *Taq* DNA polymerase by approximately 50%. Alternatively, a common way to decrease the melting temperature of the PCR product is to substitute dGTP with 7-deaza-dGTP in the reaction mix.

3. Annealing

Usually the optimal annealing temperature is 5°C lower than the melting temperature of primer-template DNA duplex. Incubation for 0.5-2 min is usually sufficient. The annealing temperature should be optimized by increasing it stepwise by 1-2°C to avoid nonspecific PCR products. The primers are jiggling around, caused by the Brownian motion and ionic bonds are constantly formed and broken between the single stranded primer and the single stranded template. The more stable bonds last a little bit longer (primers that fit exactly) and on that little piece of double stranded DNA (template and primer), the polymerase can attach and starts copying the template. Once there are a few bases built in, the ionic bond is so strong between the template and the primer which does not break anymore.

4. Extension

Usually the extending step is performed at 70-75°C. The rate of DNA synthesis by *Taq* DNA polymerase is highest at this temperature. Recommended extending time is 1 min. for the synthesis of PCR fragments up to 2 kb and may be further increased by 1 min. for each 1000 bp. Primers that are on positions with no exact match get loose again (because of

the higher temperature) and don't give an extension of the fragment. The polymerase adds complementary dNTP's to the template from 5' to 3', reading the template from 3' to 5'. Because both strands are copied during PCR, there is an exponential increase of the number of copies of the gene. But, polymerization is not strictly doubling the DNA at each cycle in the early phase. The number of PCR cycles depends on the amount of template DNA in the reaction mix and on the expected yield of the PCR product. Usually 25-35 cycles are sufficient.

5. Final Extension

After the last cycle, the samples are usually incubated at 72°C for 5-15 min to fill-in the protruding ends of newly synthesized PCR products. Also, during this step, the terminal transferase activity of *Taq* DNA polymerase adds extra A nucleotides to the 3'-ends of PCR products.

The PCR can be standardized and optimized either by changing the denaturing time, annealing time and temperature, Mg²⁺ ion concentration, extension time and temperature, and or the individual quantity ingredients in reaction mixture. The final product is visualized in transillumination to know the amplification of the product of the interest.

Application of PCR

1. PCR is used in diagnostic application in pathology for the detection of infectious agents and the discrimination of non-pathogenic from pathogenic strains by virtue of specific sequence.
2. Different PCR-based methods are used in genetic fingerprinting to identify the extremely small amounts of target of interest.
3. Used to identify genetic and evolutionary relationships between living organisms.
4. Used to identify beneficial and pathogenic microbiota at strain level by techniques such as ribotyping, randomly amplified polymorphic DNA (RAPD), pulsed-field gel electrophoresis (PFGE) of rare-cutting restriction fragments, and amplified fragment length polymorphism (AFLP).
5. Larger quantities of DNA are required for Southern or northern hybridization and DNA cloning. PCR augments these techniques with high amounts of pure DNA, enabling analysis of DNA samples even from very small amounts of starting material.
6. DNA sequencing.
7. Recombinant DNA technologies involving the insertion of a DNA sequence into a plasmid or the genetic material of another organism.
8. PCR may also be used in the analysis of ancient DNA that is thousands of years old.

Conclusion

PCR has become inevitable for any molecular biologist to understand the basics of PCR since it has gained its importance in modern molecular biology over a period of last two decades. The same technique has become vital in diagnosis of various diseases, developing ds RNA therapy & vaccines to shrimp and fish diseases, developing transgenic fishes, and so on.

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CELL CULTURE AND ITS APPLICATION IN DISEASE DIAGNOSIS IN AQUACULTURE

P. K. Nanda and P. Swain

Cell culture designates a wide range of methods which have a single objective i.e to isolate a cell or group of cells from tissue or organ and place them in an environment designed to approximate the essential features of the conditions within the body, and thus to maintain and study them in actively metabolizing state. Cultured fish cells, either as short term or as established cell lines, from a broad range of tissues and organs, are now being used for many applications in fundamental and applied biological research. Moreover, cultured cells offer several advantages over studies with whole animals as it is more reproducible and produces less toxic waste. Results are obtained more rapidly than with intact animals and with less cost. Further, it allows cellular phenomenon to be studied in a controlled and in some cases in a completely defined environment. In other words, much cellular behaviour can only be studied in cell culture. Above all, cell culture systems are attractive, as they reduce both the number of experimental animals required and the variation in data caused by inherent differences between individual animals.

Types of cell culture

Two types of cultures can be done to study fish cells *in vitro*: primary cultures and cell lines. The maintenance of growth of cells dissociated from the parental tissue using the mechanical or enzymatic methods, in culture medium using suitable glass or plastic containers is called primary cell culture. Primary cultures are initiated directly from the cells, tissues or organs of fish and typically last only a few days, although exceptions exist. By convention, the primary culture ends and the cell line begins upon splitting or subcultivation of the primary culture into new culture vessels.

The two are interrelated because cell lines are developed from primary cultures. The basic difference is that both of them differ in their life span.

Different methods of cell culture

Two different ways are generally employed to obtain cells from tissues: explants from which cells must migrate; or dissociation techniques to produce primary culture. Thus, a primary culture is that stage of the culture following isolation of the cells from tissues or organs by mechanical or enzymatic means.

Explants

The use of explants is the simplest method in which tissues are removed using sterile techniques which involves finely chopping the tissue into fragments no larger than 1-2 mm³. These fragments are left to adhere to a tissue culture vessel. After the fragments have adhered, further nutrients and other growth factors can be gently added to cover them completely. Under conducive conditions, cells outgrow from the explant eventually forming a culture.

Mechanical Disaggregation

Another common method of cell isolation is mechanical dispersion. By forcing the tissue through a sieve or syringe individual or small clumps of cells are released from the bulk of the tissue. However this procedure causes a great deal of mechanical damage and produces lower yields than any other method. Furthermore it is only recommended for softer tissues *e.g.* brain, spleen.

Enzymatic Disaggregation

The most common method of isolation is through enzymatic disaggregation. Tissues are minced and suspended in a digestion mixture, usually trypsin, but other proteolytic enzymes may be used (*e.g.* collagenase, dispase or pronase). The purpose is that the enzyme digests connective tissue and separation of cells is facilitated. Cells are harvested from by neutralizing the enzyme with media containing serum, then pelleting the cells by centrifugation and re-suspending in fresh media. But all the enzymes are not suitable for every tissue, as enzymatic treatment can weaken the cell membranes. Therefore, enzymatic dissociation method should only be done after standardizing the concentration; dose, duration and temperature otherwise it in turn can affect cell yield, viability and efficiency of attachment to the substrate.

Nutrients and other factors influencing cell culture

Media

Factors such as media, serum, temperature, attachment factors and other additives should also be considered as part of the primary culture procedure. Among these, to achieve the success in cultivation of fish cells *in vitro*, media and its composition plays a crucial role, as it provides all nutritional requirements of the cell. The basic component of the media is a balanced salt solution whose function is to provide essential inorganic ions, correct osmolality and pH. Further, the media to be used for cell culture must be sterile and isotonic with the cell cytoplasm and contain a pH value of 7.0-7.3 i.e it must be adequately buffered.

To control pH, the use of buffering solutions or CO₂ incubators in culture flasks is common in routine tissue culture

Although, cells will remain alive in balanced salt solution for several hours, many more ingredients i.e amino acids and serum must be included for proliferation of cells to occur. To achieve this, a complete medium for growth of fish cells are used which has two parts viz. a basal medium and a supplement. Complete basal media contains essential amino acids, vitamins and salts in varying amounts and combinations. The media developed for mammalian cell culture is mostly used to culture fish cells. Most common basal media that have been successfully used for fish culture include Dulbecco's modified Eagle's medium (DMEM), Glasgow MEM, Hank's MEM (H-MEM) and Leibovitz's L-15 medium (L-15) etc.

Serum

The most common supplement to basic media is serum. Serum contains growth factors, adhesion factors and is also a source of minerals, lipids and hormones. Serum also acts as a pH buffer. Although the precise contribution made by the serum is not fully understood, some components of the serum exert a beneficial effect by promoting attachment and spreading of the cells and by stimulating cell division. Although sera from different animal species are available, fetal bovine serum (FBS) and newborn calf serum (NBCS) are most widely used in cell culture works.

Attachment factors

Most of the cells obtained from tissues and organs are anchorage-dependent, i.e they need a substratum to attach and grow. Although glass was the original substrate because of its optical properties and surface charge, it has been replaced by synthetic plastic (usually polystyrene). As polystyrene is hydrophobic and does not provide a suitable surface for cell attachment, it necessitates considering the use of attachment factors in the flasks to produce a charged, wettable surface. Presently there are numerous attachment factors such as collagen, gelatin, fibronectin, laminin, poly-D-lysine and ready-coated flasks available to purchase. Many of these can be commonly used as substrate to enhance attachment and improve growth of the cells.

All these factors play an important role in the initial stages of primary culture and in the attachment, survival, and finally growth of the primary culture.

Secondary cell cultures

When a primary culture is sub-cultured, it is known as secondary culture. Subculture (or passage) refers to the transfer of cells from one culture vessel to another culture vessel. Subculturing or splitting cells is required to periodically provide fresh nutrients and growing

space for continuously growing cell lines. The process involves removing the growth media, washing the plate, disassociating the adhered cells, usually enzymatically. Such cultures may be called secondary cultures.

Cell Line

A cell line or cell strain may be finite or continuous depending upon whether it has limited culture life span or it is immortal in culture. On the basis of the life span of culture, the cell lines are categorized into two types:

Finite cell Lines

The cell lines, which have a limited life span and go through a limited number of cell generations (usually 20-80 population doublings) are known as finite cell lines. These cell lines exhibit the property of contact inhibition, density limitation and anchorage dependence. The growth rate is slow and doubling time is around 24-96 hours.

Continuous Cell Lines

Cell lines transformed under laboratory conditions or *in vitro* culture conditions give rise to continuous cell lines. The cell lines show the property of ploidy (aneuploidy or heteroploidy), absence of contact inhibition and anchorage independence. They grow in monolayer or suspension form. The growth rate is rapid and doubling time is 12-24 hours.

Disease constraints in the development of aquaculture

Fish and shellfish farming constitute an important resource for healthy food, and are a very important source of prosperity in many countries, and India is no exception to this. Therefore, it is expected that any long-term rise in fish production will depend on the future progress of aquaculture. However, fish, being obligate aquatic organism, are directly exposed to various biotic and abiotic factors because of their intimate relationship with the ambient water environment. This brings in a number of problems, including diseases of diverse etiologies. Often disease outbreaks are closely linked to environmental deterioration and stress associated with intensification of culture practices. Stress factors such as inadequate physico-chemical and microbiological quality of culture water, inferior nutritional status and high stocking densities can cause infection by opportunistic pathogens, leading to mortality. Further, unprecedented movement of fish and brood stock between region and localities has created fish and shrimp population with inherent pathogens, making the populations more prone to serious diseases, when they are under stress. Because of these reasons, fish in culture are susceptible to a wide range of bacterial, viral, parasitic and fungal infections thus making a significant impact on the quality and volume of the fish produced throughout the world and

India is no exception to this. The economic loss due to disease related problems is estimated to be of approximately 8 billion dollar annually worldwide. Further, the emergence of new pathogens, particularly intracellular bacteria and viruses, not only possess serious risks to the health of aquatic animals, but also to fish farmers and the consumers of aquaculture products. Addressing aquatic animal health issues through various means has, therefore, become an urgent requirement for the sustaining growth of aquaculture.

Aetiology of aquatic animal diseases

Diseases affecting aquaculture species can be of various aetiologies, including: (a) anatomical and/or functional traumas, including tumors and external injuries; (b) nutritional or unfit feeding (c) toxicological diseases, which in aquaculture are not only due to water quality, but also with food residues, particularly fungal toxins and with inappropriate drug (disinfectants, antibiotics . . .) dosages and (d) infectious agents such as viruses, bacteria, and parasites.

Application of cell culture methods to protect aquatic animal health

Cultured fish cells, either as short term or as established cell lines, from a broad range of tissues and organs may help to acquire knowledge for health protection and disease management in aquaculture.

Nutritional diseases

Lipids are an indispensable component of fish feeding, and dietary deficiencies in essential fatty acids have been related to lordosis and muscular dystrophy depending on the fish species. The application of cell culture techniques to study the relationships between nutrition requirements and disease in aquaculture could be done by studies on the nutritional requirements for polyunsaturated fatty acids (PUFA) between different farmed fish species. Several *in vitro* methods such as development of several cell lines, hepatocyte and astrocyte primary cell cultures from different fish species, can be used to study *in vitro* the elongation and desaturation of different PUFA.

Toxicological diseases

Aquatic animals have provided models for many toxicological studies, and it is well known that many aquatic invertebrate species, including farmed molluscs such as mussels, are good bioindicators of environmental pollution. Concerning fish health, there are numerous studies that have demonstrated the influence of different aquatic inorganic and organic pollutants on disease susceptibility in farmed fish. For toxicology in particular, cell cultures offer several advantages over studies with whole animals as dosing of cell cultures is easier and interpreting the results of mechanistic studies is often more definitive because cell

cultures lack *in vivo* complexities such as bioaccumulation and depuration. Researchers are now employing cell cultures methods and *in vitro* assays to understand the underlying mechanisms of the immunomodulation effects of heavy metals and fungal toxins, and their relationships with fish health. For example, Aflatoxin B1 is a mycotoxin that has carcinogenic and immunomodulatory effects in many vertebrates, and the rainbow trout is particularly susceptible to the carcinogenic effects of aflatoxin B1, either after exposure in feed or in water. The mechanisms underlying aflatoxin B1-induced hepatic carcinogenesis in trout, which involves the biotransformation of aflatoxin B1 by hepatic enzymes of the cytochrome P450 (CYP) family, have been investigated *in vitro* using cultures of isolated hepatocytes, in which the expression of hepatic CYP and the occurrence of DNA mutations can be analyzed. The toxicity of certain compounds of polycyclic aromatic hydrocarbons (PAH) can be studied *in vitro* using fish cell cultures. The expression of detoxifying enzymes, the detection of DNA breakage and the study of cell viability can also be analyzed *in vitro* using cell culture. In this aspect, primary cultured cells are considered to be more sensitive, with high metabolic capacity as compared to cell lines and are increasingly being utilized as tools in fundamental and applied research to analyze the effects of heavy metals thermal stress, xenobiotics, monitoring of aquatic pollution etc.

Diagnosis of infectious diseases

Viruses are known to be important pathogens to many species of aqua cultured fish and often cause mass mortality. These are obligatory intracellular pathogens and their isolation and propagation are totally dependent on the availability of a live host. Further, most viruses are host-specific and tissue-specific, and they can only be isolated and propagated using cell cultures established from tissues of the same host species. Therefore, if a virus is inoculated into an insensitive cell culture, the virus will not be able to replicate and a negative result will be obtained. On the contrary, when small amounts of virus are present in a specimen such as an early viral infection, a positive result may be obtained only when the most sensitive cell culture system is used. Besides, the use of cell cultures is a must to study the biology and pathogenic mechanisms of viruses affecting aquaculture. Thus, appropriate cell culture systems are necessary for the cultivation and isolation of pathogenic viruses from important cultivable fish species and can be an important component to the fish health assurance program of the aquaculture industry.

As cell culture methodologies are essential tools for the diagnosis of viral diseases in farmed fish, these along with allied *in vitro* assays can be applied for prevention of infectious diseases in aquaculture. Such techniques are mainly based on the use of fish cell lines

sensitive to different viruses, in which viral infection is primarily demonstrated by the cytopathic (due to cell lysis) or syncytial (produced by cell-cell fusion) effects. Furthermore, viral *in vitro* infection of susceptible cell lines facilitates several serological techniques for surveillance and diagnosis of viral diseases, including the detection of fish antibodies against the pathogenic viruses by means of the viral neutralization test. Cell culture has also been used also for the detection of some intracellular bacterial fish pathogens, as in *Rickettsiae* spp. and *Renibacterium salmoninarum*. For instance, such microorganisms may take longer than a month to obtain colonies of *R. salmoninarum* in bacteriological cultures, but these bacteria can be grown within one week in the EPC cell line from carp. The pathogen grows intracellularly in the EPC cells, and such culture methods also allow for the expression of specific virulence factors of *R. salmoninarum* and for the maintenance of the *in vivo* virulence of the isolates.

Concluding remarks

Cell culture and related *in vitro* assays have provided useful knowledge for disease diagnosis/control in aquaculture. But the potential of cell culture techniques and associated methodologies to provide tools and strategies for disease control in aquaculture has not been yet achieved in India, mainly due to the poor standardization of fish and shell fish cell culture methods. Therefore, unlike the international scenario, the researchers in the tropics, particularly in India, lack specific fish cell lines and so the number of reports on fish viruses. Further, there is a large number of fish species that are being used as research models and many of such species live in different aquatic environments and show disparate physiological requirements. On the other hand, some of the current uses of cell culture methods for disease diagnosis in aquaculture are being replaced by faster and less-laborious molecular biology methods. Still then, there is still a large spectrum of applications in which cell cultures would be indispensable. Of particular interest are those needed to study the pathogenic mechanisms of intracellular pathogens, and to isolate and characterize viral pathogens. As disease problems are one of the major stumbling blocks in the development of aquaculture, it is expected that future strategies to control disease in fish and shellfish will impel the progress of cell culture methods for rational health management in aquaculture.

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CELLULAR IMMUNOLOGICAL TECHNIQUES IN FISH

S. N. Joardar and R. Ananda Raja

Neutrophil functional assay

As a measure of non-specific immune responses, oxidative radical production by neutrophils is assayed by the NBT reduction test. Briefly, equal volume (0.1 ml) of the blood and 0.2% filtered NBT (nitroblue tetrazolium) solution is to be added and incubated for 30 min in room temperature. Then 0.05 ml from this mixture is to be added of the coulor formed is to be noted at 540 nm in a spectrophotometer.

Isolation of head kidney (HK) cell

The head kidney leucocytes is to be removed, cut into smaller pieces, teased to obtain cells and pushed through the stainless steel mesh with PBS (pH 7.4) containing 100 µg/ml streptomycin, 2 mg/ml gentamycin, and 100IU/ml penicillin. The cell suspension is layered onto Histopaque ® at the ratio of 1:3 (1 part of histopaque and 3 part of cell suspension) and centrifuged for 30 min at 400g. Following centrifugation, the white blood cell interface layer is collected using a micropipette, transferred into clean sterile test tube and washed thrice with PBS. Viable cells are to be enumerated by the try pan blue exclusion method using Neubauer's counting chamber.

Macrophage functional assay

Phagocytosis is examined using macrophage of fish. The number of cells is adjusted to 10^7 cells m/l in RPMI-1640 medium counting 10% foetal calf serum (FCS). One millilitre of cell suspensions allowed to adhere to a cover slip for 1h and non-adherent medium at 10^8 cfu/ml and added on to the cover slip and incubated for 2h at 28°C. The cell on the cover slip fixed with 95% methyl alcohol and stain with Giemsa. The numbers of phagocytic cells per 200 cells are counted. The mean numbers of Staphylococcus sp. phagocytosed/100 macrophages for each dietary group are calculated. Phagocytic activy (PA) and phagocytic index (PI) are determined using the following formula:

$$PI = \frac{\text{Number of engulfed bacteria}}{\text{Phagocytic cells}}$$

$$PA = \frac{\text{Number of phogocytosing cells}}{\text{Total number of phagocytes}} \times 100 \text{ (expressed as \%)}$$

***In vitro* nitrite production assay**

The production of reactive nitrogen intermediates is assayed using Griess reaction that quantifies the nitrite content of the macrophage supernatants, as NO is an unstable 2×10^6 viable cells/ml by diluting with RPMI – 1640 growth medium containing 2 mM L-glutamine, 24 mM Hepes buffer, 100 IU/ml penicillin, 100 µg/ml streptomycin and 10% V/V foetal calf serum. Cell suspension (100 µl/ well) is dispensed in a 96 well flat bottom tissue culture plate. The final volume of the wells is to be made up to 200 µl with LPS (Sigma, USA) at a concentration of 10 µg/ml and incubated at 28°C for 96h containing 5% CO₂ tension.

After incubation, 50 µl aliquots of supernatants from individual wells are removed and added 100 µl of Griess reagents containing 1% sulfanilamide, 0.1% N-1 for 10 min. the optical density is to be measured in a plate reader at 570 nm and quantified by comparison to NaNO₂ as a standard.

Lymphoproliferation assay

The number of head kidney leucocytes (5×10^5 cells/ml) is to be adjusted in RPMI – 1640 and 100 µl of cell suspension is seeded into each well of 96-well tissue culture plate. The final volume of the wells are made upto 200 µl with Con-A at a concentration of 10 µg/ml and incubated at 28°C for 48h containing 5% CO₂ tension.

The colorimetric 3 – [4, 5 – dimethylthiazool- 2yl] – 2, 5 – di phenyltetrazolium bromide (MTT) assay is used to determine the proliferation of HK leucocyte formazan crystals are dissolved by the addition to well of 150 µl of DMSO followed by 25 µl of glycine buffer (0.1 M glycine, 0.1 M NaCl, pH 10.5), mixed thoroughly with micropipette and incubated at room temperature for 10 min. the formazan development is to be read at 595 nm using a plat reader.

Stimulation index is to be calculated with the following formula:

$$SI = \frac{\text{Mean optical density of leucocyte wells with test mitogen at time x}}{\text{Mean optical density of negative control wells at time x}} - 1$$

Enzyme Linked Immunosorbent Assay (ELISA)

ELISA is most extensively used serological test for diagnosis of several viral diseases (as also bacterial, fungal and mycoplasmal diseases) of human, animals and fish. This is one of the most sensitive and specific tests next only to the virus neutralization test. When antibody is covalently linked to an enzyme and the enzyme-linked antibody is allowed to react with immobilized antigen, the antigen antibody reaction can be visualized by change in

color of the substrate due to enzyme action. Immobilization of antigen can be done either microtitre plate coated with polystyrene or in nitrocellulose membrane (NCM). The commonly used enzymes are horse radish peroxidases (HRPO), alkaline phosphates (AP) etc. The substrate for HRPO is 3-3 diaminobenzidine-di-hydrochloride (DAB), which gives brown color. The DAB is generally used in dot ELISA whereas OPD is in plate ELISA.), the substrate for alkaline phosphates is generally para- nitrophenyl phosphate, which gives yellow color reaction.

Advantage of ELISA

1. ELISA is highly versatile, rapid, relatively economical, highly specific, sensitive and reliable test for viral disease diagnosis
2. This can be applied as both qualitative as well as quantitative assays.
3. Can detect minimum quantity of either antigen or antibody without extensive purification.

Materials

- i. ELISA plate.
- ii. Antigen and known hyper-immune serum.
- iii. 0.05M carbonate-bicarbonate buffer (pH 9.6), phosphate buffer saline (pH 7.2), PBS+Tween-20(0.5%), citrate buffer (pH 5.0),
- iv. Bovine serum albumin (BSA)-3% solution in PBST.
- v. HARPO conjugated anti-species antibody.
- vi. OPD solution.
- vii. H₂O₂ (30%)
- viii. 3N H₂SO₄
- ix. Micropipettes (single & multichannel)
- x. ELISA reader.

Method

- I. Add 100µl of diluted viral as (known) in carbonate-bicarbonate buffer to each well of the ELISA plate except A₁, A₂, B₁ and B₂ wells (they serve as Ag control).
- II. Incubate the plate overnight at 4°C
- III. Wash the plate thrice PBST to remove unbound excess Ag.
- IV. Add 200µl of blocking solution (3% BSA in PBST) to all the wells and incubate the plate at room temperature for 2 hour.
- V. Wash the plate thrice with PBST.

- VI. Prepare two- fold dilution of serum (antibody) in PBS in a separate microtitre plate or Perspex plate.
- VII. Add 100µl of each dilution of serum of the corresponding wells of the ELISA plate except G₁₁, G₁₂, H₁₁ and H₁₂ (they serve as Ab control).
- VIII. Incubate the plate at 37°C for 2 hours.
- IX. Wash the plate thrice with PBST.
- X. Add 100µl of HARPO conjugated anti-species antibody in PBS all the wells.
- XI. Incubate the plate at 37°C for 2 hours
- XII. Wash the plate thrice with PBST.
- XIII. Add 100µl of substrate solution to all the wells and keep in dark at room temperature for 15-20 minutes.
- XIV. Stop the reaction by adding 50 µl of 3N H₂SO₄ solution in each well.

Observation

Record the OD values at 492 nm wavelengths in an ELISA plate reader.

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ANTIMICROBIALS IN AQUACULTURE

R. Ananda Raja and Sujeet Kumar

As the world's population continues to climb exponentially, the demand for producing food production remains challenging. New innovative sustainable technologies are need of the hour in agriculture, animal husbandry and fishery sector as a whole. As the new technologies emerge for increasing the food production, the parallel increase of problems associated mainly with the infectious diseases is inevitable. Despite many advances in vaccines, biosecurity, and health management, we have by no means eliminated infectious diseases from our animal populations, nor will this be accomplished in the foreseeable future. So, antibiotics are enormously important for the humane and efficient production of food animals. Antibiotics may be used for therapeutic, prophylactic, feed additive, soil and water treatment, or growth promoter purposes. Clearly, there will be a steady demand for safe and effective antibiotics to treat bacterial infections in terrestrial and aquatic animals. These benefits are somewhat offset by the antibiotic resistance risks posed by their use in animals. At present, the usage of antibiotics in aquaculture poses great threat by interaction with other environmental contaminants. Antibiotics may enter the environment by means of leaching from faecal materials and uneaten feeds. More research is needed to better understand the processes and pathways of antibiotics and their degradation products in the sediment and residue accumulated in the bottom of the ponds. This chapter provides an overview of what we have learned about antibiotic resistance as an issue in aquaculture and where that knowledge could lead us in the future.

Health hazards

Non-judicious use of antibiotics in human and animal health perspective leads to development of antibiotic resistance. It provides great concern by means of therapeutic failure or the need to use more costly, toxic, or expensive drugs. It can also be a problem when it increases the frequency, duration, severity of infection or increase the burden of illness. The total 40 per cent of the world living biomass is occupied by microbes. So any disturbance to the dominant population by human intervention will lead to irreparable damage to the natural ecosystem and we may experience the effect in a worst hit manner. So, care must be taken not to permanently end with new unsolvable problems in the prevailing ecosystem for short time profit.

Antibiotic resistance

Antibiotic resistance and shift in microbial community are the major concerns in usage of antibiotics. Some bacteria are intrinsically resistant to certain antibiotics. More concern is the ability of bacteria to acquire resistance attributes through genetic mutation or transfer of genetic elements from other bacteria of the same or even completely different species and genera. Depending on the drug and bacterial species, the rate at which resistance emerges in populations varies considerably. The sublethal concentrations of antibiotics in feed (Feed additive or Growth promoters) administered to large numbers of animals over long periods of time lead to the development of resistance. Resistant bacteria can spread among and between ecological niches of animals and humans. The mechanisms of spread are undoubtedly complex may be local, regional, national, or international. The vast majority of such spread is undetected and its extent is unknown. So, it is essential to make an extensive study on understanding the minimum inhibitory concentration (MIC) of each antibiotic towards each microbiota. The study should be made periodically by local, regional, national, and international level in association with the residual concentration of different antibiotics in the environmental samples.

Ban on application of antibiotics in aquaculture

Creation of awareness through a comprehensive programme is vital for achieving the objectives of regulating the use of antibiotics, drugs, chemicals, etc in aquaculture. Aquaculture authority in India imposed ban on the list of 20 antibiotics, drugs and pharmacologically active substances and permitted certain level of residue for four antibiotics and antimicrobials in the finished products. Maximum Permissible Residual Levels for Fish and Fishery Products approved by Coastal Aquaculture Authority (CAA) in India is as follows.

Sl. No.	Antibiotics and other Pharmacologically Active Substances	Maximum Permissible Residual Levels (in ppm)
1.	Chloramphenicol	Nil
2.	Nitrofurans including: Furaltadone, Furazolidone, Furylfulamide, Nifuratel, Nifuroxime, Nifurprazine, Nitrofurantoin, Nitrofurazone	Nil
3.	Neomycin	Nil
4.	Nalidixic acid	Nil
5.	Sulphamethoxazole	Nil

6.	Aristolochia spp. and preparations thereof	Nil
7.	Chloroform	Nil
8.	Chlorpromazine	Nil
9.	Colchicine	Nil
10.	Dapsone	Nil
11.	Dimetridazole	Nil
12.	Metronidazole	Nil
13.	Ronidazole	Nil
14.	Ipronidazole	Nil
15.	Other nitroimidazoles	Nil
16.	Clenbuterol	Nil
17.	Diethylstilbestrol (DES)	Nil
18.	Sulfonamide drugs (except approved Sulfadimethoxine, Sulfabromomethazine and Sulfaethoxypyridazine)	Nil
19.	Fluroquinolones	Nil
20.	Glycopeptides	Nil
21.	Tetracycline	0.1
22.	Oxytetracycline	0.1
23.	Trimethoprim	0.05
24.	Oxolinic acid	0.3

Alternatives to antibiotics

Age-segregation, all-in all-out management, biosecurity, sanitation, and vaccination are just a few examples of no antibiotic practices that may be used in the prevention and control of many infectious diseases of livestock. But, none of these measures is sufficient to completely eliminate the need for antibiotics on most farms because there are major technical and practical barriers to the elimination of all endemic bacterial infections from farms. Further, many research findings reveal that there is a scope with plants based extracts to combat some infectious viral and bacterial diseases.

Conclusion

Antibiotics are critically important for animal health and welfare. The supply of new antibiotics is limited and antibiotic resistance threatens their long-term usefulness. Therefore, we have an obligation to use them prudently, in the interests of both public health and aquatic animal health. We must view them as one of many tools to manage health and production in aquatic animals, and we must continue to collect and use data to improve the scientific basis for prudent antimicrobial use. The programs like antibiotic resistance monitoring and surveillance programs are important. Moreover strong, credible, science-based government regulatory programs for drug licensing, safety evaluation, control of distribution, sale, and use are essential components in aquatic animal health for antibiotic resistance risk management.

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APPLICATION OF PROBIOTICS AND PREBIOTICS FOR DISEASE MANAGEMENT IN AQUACULTURE

Debasis De and R. Ananda Raja

Aquaculture is one of the fastest growing industries in the world. The need for enhanced disease resistance, feed efficiency and growth performance of cultured species is substantial. If growth performance and feed efficiency are increased in commercial aquaculture, then the cost of production are likely to be reduced. Also if more fish are able to resist disease and survive until they are marketable size, the subsequent cost of medication and overall production cost would be remarkably reduced. Dietary supplementation of different feed additives, a substance that is added to a basic feed, usually in small quantities, for the purpose of fortifying it with certain nutrients, stimulants or medicines other than as a direct source of nutrient, have been found to be beneficial for improving immune status, feed efficiency and growth performance of crustaceans and finfishes. Application of different types of feed additives e.g. probiotics and prebiotics in aquaculture are discussed below.

Probiotics

The term “probiotics” was coined by Parker (1974) to describe “organisms and substances which contribute to intestinal microbial balance”. Probiotics are living organisms which when introduced through host feed have a positive effect on host health. Some reside in the digestive tracts of the individuals while others have an external origin. They affect the host animal by improving its intestinal microbial balance.

Properties of probiotics

The good probiotics should possess the following properties:

- a) It should be resistant to pH and bile acids
- b) It should have the ability to attach to the gut epithelial lining.
- c) It should be non-pathogenic.
- d) It should provide a beneficiary effect to the host animal.
- e) It should possess a high viability.
- f) It should be stable on storage and in the field.
- g) It should survive the gut environment and should have the potentiality to colonize in gut.
- h) It should be cultivable on a large scale.

Even if any new strains are used for probiotic development then they should possess all these aforementioned characteristics.

Mode of Action of Probiotics

Probiotic agents exert a beneficial effect via a wide array of actions. These include,

- i. Competition for adhesion sites and resistance to colonization.
- ii. Competition for essential nutrients.
- iii. Production of antagonistic compounds against pathogens.
- iv. Enhancement of the immune response and diseases resistance.
- v. Improve enzyme activity feed digestibility and feed utilization, fish health and performance.

Enhancement of the immune response

The immune systems of fish and higher vertebrates are similar and both have two integral components: 1) the innate, natural or nonspecific defense system formed by a series of cellular and humoral components, and 2) the adaptive, acquired or specific immune system characterized by the humoral immune response through the production of antibodies and by the cellular immune response, which is mediated by T-lymphocytes, capable of reacting specifically with antigens.

Probiotics for Aquaculture

Probiotics that currently used in aquaculture industry include a wide range of taxa – from *Lactobacillus*, *Bifidobacterium*, *Pediococcus*, *Streptococcus* and *Carnobacterium* spp. to *Bacillus*, *Flavobacterium*, *Cytophaga*, *Pseudomonas*, *Alteromonas*, *Aeromonas*, *Enterococcus*, *Nitrosomonas*, *Nitrobacter*, and *Vibrio* spp., yeast (*Saccharomyces*, *Debaryomyces*) etc.

Aquatic probiotics are marketed in two forms: 1) Dry forms: the dry probiotics that come in packets can be given with feed or applied to water. They have many benefits, such as safety, easy using, longer shelf life and etc. 2) Liquid forms: the hatcheries generally use liquid forms which are live and ready to act. These liquid forms are directly added to hatchery tanks or blended with farm feed. The liquid forms can be applied any time of the day in indoor hatchery tanks, while it should be applied either in the morning or in the evening in outdoor tanks. Liquid forms give positive results in lesser time when compared to the dry and spore form bacteria, though they are lower in density.

Prebiotics

Prebiotics are the indigestible components present in the diet which are metabolized by specific microorganisms which prove to be helpful for growth and health of the host. Prebiotics shift the microbial community to one dominated by beneficial bacteria, such as *Lactobacillus* spp. and *Bifidobacterium* spp.

Characteristics of prebiotics

1. It should stimulate the beneficial gut microbes.
2. It should reduce harmful microbial load.
3. It should be effective at lower concentration.
4. It should regulate viscosity of gut.
5. It should be non-carcinogenic
6. It should be derived from dietary polysaccharides.
7. It should have low calorific value.
8. It should not possess any residual effects.
9. It should exert anti-adhesive properties against harmful gut microbes.
10. It should be easy to incorporate in the feed or ration.

Commonly used prebiotics

Most efficient prebiotics which find their application in the field of aquaculture are fructooligosaccharide (FOS), transgalactooligosaccharide (TOS), inulin, glucooligosaccharide, xylooligosaccharide, isomaltooligosaccharide, soybeanoligosaccharide, polydextrose, lactosucrose. Natural sources of prebiotics in vertebrates include chicory, onion, garlic, leek, tomato, honey etc.

Modes of action of Prebiotics

FOS influence enteric microflora by ‘feeding the good bacteria’, which competitively excludes the colonization of pathogens and thus improving animal health and growth performance. The nutritional properties of prebiotics are related directly to the physiological changes they induce in the host. Bacterial metabolites are probably the main effectors of most observed effects. The most important metabolites are the short-chain fatty acids (SCFA) as acetate, propionate and butyrate. Prebiotic consumption can double the pool of SCFA in the GI tract. These SCFA acidify the GI environment, which is beneficial for the development of probiotic bacteria, and detrimental to the growth of harmful species. On the other hand unlike FOS, MOS is not used as a substrate in microbial fermentation, but it still exerts a significant growth-promoting effect by enhancing the animal’s resistance to enteric pathogens. MOS inhibits colonization of enteric pathogens by blocking bacterial adhesion to gut lining. Adhesion of pathogens to the epithelium surface of the gut (colonization) is believed to be the first critical stage leading to infection. Many potential pathogens as *E. coli*, *Salmonella* and *Vibrio* spp. as well as commensally bacteria attach to the mucosal surface of the intestine and that attachment is the key to their proliferation. They have mannose specific lectins on the cell surface that recognize specific sugars and allow the cell to attach to that sugar.

According to updated prebiotic concept not all dietary carbohydrates are prebiotics. Prebiotics (FOS, MOS, GOS, and XOS) are functional substrates that can only be consumed by a limited number of beneficial bacteria (lactobacilli and bifidobacteria). Among the group of beneficial bacteria present in the GI tract are those that most utilize prebiotic oligosaccharides being considered as the only microorganisms able to beneficially affect the host's health.

Effects on fish health and productivity

The use of probiotics and prebiotics for disease prevention and improved nutrition in aquaculture was relatively recent, but during the last decade, it is becoming increasingly popular due to an increasing demand for environment-friendly aquaculture. Since the first use of probiotics in aquaculture, a growing number of studies have demonstrated their beneficial effects in fish diet. Several new studies have reported that probiotics are effective in improving immunity, health status, feed efficiency and growth performance of fish species. As biological control agents probiotics provide nutritional benefits and protection against pathogens in the GI ecosystem. They are vital in modulating interactions with the environment, the development of beneficial immune responses, and in supporting the health. The supplementation of beneficial probiotic bacteria in the fish diet is important for enzyme activity, feed digestion, feed utilization and growth performance. By the above reasons, the use of microbial probiotics in finfish aquaculture is now widely accepted.

Conclusion

The current most popular targets for prebiotics use are lactobacilli and bifidobacteria. This is largely based on their success in the probiotics area. However, as our knowledge of the gut flora diversity improves, it may become apparent that other microorganism should be fortified through the use of prebiotics. With our increasing understanding of the bacterial diversity in the gut microflora effect of prebiotics on other target bacteria must be investigated. Prebiotics will then become unique tools to create gut microflora with controlled composition that will then be correlated with specific physiological conditions. In summary, probiotics and prebiotics have been investigated and reported to have numerous beneficial effects in terrestrial animals such as, increased disease resistance and improved nutrient availability. Similar type of effects can also be evidenced in fishes to increase the sustainability and profitability of aquaculture production. So, more research regarding usage and effect of different immunostimulants, probiotics and prebiotics is warranted.

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VACCINE IN AQUACULTURE

Sujeet Kumar and R. Ananda Raja

Vaccines are antigen preparation derived from a specific pathogen and stimulate the immune system in such a way to generate resistance to same pathogen from subsequent infection. Specificity and memory of the adaptive immune system are two key elements exploited in vaccination. An effective vaccine must be safe, immunogenic and protective.

Type of Vaccine

Success of any vaccination program depends upon types of vaccine available, its efficacy, safety, duration of immunity as well as its final cost. Currently, in aquaculture four types of vaccines are available.

1. Killed Vaccine
2. Live Vaccine (Attenuated vaccine)
3. Subunit Vaccine
4. DNA Vaccine

1. Killed Vaccine: This is the most commonly used vaccines in aquaculture. Heat or formalin is commonly employed for inactivation of bacteria, while, formaldehyde, glutaraldehyde, binary ethylenimine and β -propiolactone are used as inactivating agent for viral vaccines.

Advantages

- a. Very effective in inducing humoral antibody response so effective against most of the bacterial pathogens.
- b. The vaccine is safe as the microbes are in killed form and can't revert to virulence.
- c. Can be stored at room temperature.

Disadvantage

- a. Poor cell mediated immune response so not very effective against intracellular pathogens such as virus.
- b. Use of adjuvants and sometimes booster vaccination.
- c. Use of inactivating agents which may alter the form of a critical antigen, and therefore reduce vaccine effectiveness.

Successful commercial killed vaccines are available against Vibriosis (*Vibrio anguillarum*, *V. Ordalli*), Enteric red mouth disease for salmonids (*Yersinea ruckeri*), Furunculosis for salmon (*Aeromonas salmonicida*) and spring viremia of carp.

2. Live Vaccine: Live vaccines or attenuated vaccines are mutated strains of an infectious agent that have a reduced or no ability to cause disease. Such type of vaccine is highly desired for intracellular pathogens such as virus.

Advantages

- a. Long lasting immunity even in small doses.
- b. Strong cellular immunity thus very effective in protection against intracellular pathogen such as viruses.

Disadvantages

- a. Capacity of vaccine strains to revert to a virulent form and potential to cause disease in immuno suppressed host. By methods of genetic engineering now it is possible to irreversibly attenuate the microbes by removing virulent genes.
- b. Risk of transmission to non-farmed fish in the surrounding water for which the vaccine may be virulent.
- c. Cold chain maintenance for storage.

Live vaccine is available against Bacterial kidney disease using related bacterium *Arthrobacter davidneili*, Edwardsilosis (ESC) using RE-32 strain of *Edwardsiella ictaluri* and aroA and purA deficient *E. Ictaluri* strain.

3. Subunit Vaccine: Subunit vaccine includes immune response against the purified protein, synthetic peptides and recombinant protein (recombinant vaccine). For the subunit recombinant vaccines, the gene(s) encoding a particular antigen, which must be immunogenic, are cloned and subsequently introduced into a permissive host, e.g. bacterium, yeast or insect cells which then synthesize the recombinant antigen.

Advantages

- a. Well-defined, non-infectious and inexpensive to produce in large quantities.
- b. Differentiation of vaccinated from potential carrier by immunological methods is possible because fish is not exposed to the entire array of antigens that a pathogen expresses.
- c. Useful for pathogens which are difficult to bulk culture, such as viruses.

Disadvantages

- a. Need of adjuvant and vaccine delivery system
- b. Larger dose and need for booster immunization
- c. Shorter duration of immunity
- d. Poor cell mediated immune responses

Licensed recombinant vaccine for fish is available against infectious pancreatic necrosis (IPN) using VP2 gene of the virus. Glycoprotein of infectious hematopoietic necrosis (IHN)

and viral hemorrhagic septicemia (VHS) virus have shown good results in experimental trials.

4. DNA Vaccines: Genetic immunisation using naked DNA is most recent approach in vaccine design. This technique involves injection of naked DNA directly into the skeletal muscle of the fish where it is expressed extra chromosomally. Gene gun, electroporation, encapsulation of DNA in liposomes or polylactide-L-glycolide (PLG) microparticles enhances the cell uptake of DNA vaccine. Incorporation of CpG motifs or cytokines to plasmid expressing DNA vaccine creates a cytokine microenvironment conducive for developing an adaptive immune response.

Advantages

- a. Easy to manufacture: DNA vaccine is comprised of a plasmid with origin of replication, a selectable marker and the gene of interest under a strong promoter. This single platform makes DNA vaccines very attractive from the prospective of manufacturing.
- b. Safety: DNA vaccines are considered safe since it lack extraneous materials and are non-infectious. The chance of integration into host genome is also less as proved by many experiments.
- c. Multicomponent vaccines: Combining many plasmids encoding different genes of interest or introduction of two different genes in a single plasmid is also possible. This paves the way for vaccination against many diseases at single time in single stroke.
- d. Less vaccine dose: Fish rhabdoviruses such as Infectious hematopoietic necrosis virus (IHNV), Viral hemorrhagic septicemia virus (VHSV) a single IM injection of a 1.0 µg with no adjuvant or boosters, was found sufficient to provide a high level of protection (Kurath, 2008).
- e. Induction of both humoral and cellular immune response.
- f. Potent inducer of immunity in neonates in absence as well as presence of maternal antibodies.

Disadvantages

- a. Sub-optimal immunity: DNA vaccine has poor transfection efficiency leading to suboptimal induction of immunity.
- b. Licensing and commercialization of DNA vaccines is difficult partly due to public perception who confuse it with genetically modified organisms, and stringent law.
- c. May prove hazardous due to widespread release into aquatic environment which lack physical and physiological barrier.

Recently, a DNA vaccine was licensed to immunize fish against infectious hematopoietic necrosis virus for commercial use in Canada (Babiuk, 2008).

Vaccine Delivery System

Three different methods of administration are commonly used to vaccinate fish, namely injection, immersion and oral.

Injection Administration

Injection of antigens (intraperitoneal or intramuscular) is an effective way of provoking antibody response in fish by maximum retention of vaccine in the body. However, procedure is labor intensive and stressful to fish and suitable only for larger fishes like broodstock and not fit for fish below 15 grams.

Immersion Administration

Immersion of fish in antigen solutions has emerged as potent commercial process. Prior immersion of fish into hyperosmotic salt solution enhances the antigen uptake. The method is less labor intensive and less stressful to fish compared to injection method. It can easily be used to vaccinate small fish, while larger fish can also be vaccinated by spraying.

Oral Administration

Oral administration is “the ideal method” for administering vaccines to fish where the vaccine is incorporated into fish feed. It is least labour-intensive, avoid handling stress and can be used to vaccinate large numbers of fish of all sizes. The major limitation is lower levels of protection which is due to degradation of antigen by the gastric fluid and inefficient transport of antigen across the gut wall. Microencapsulation and bioencapsulation using live artemia has shown good results.

Fish Vaccination Research

Warm water aquaculture in Asia has problems with several bacterial diseases such as motile Aeromonad septicaemia or hemorrhagic septicemia, Vibriosis, Columnaris, Edwardsiellosis etc.

Hemorrhagic septicaemia is caused by motile species of *Aeromonas* such as *Aeromonas hydrophila*, *A. sobria*, *A. caviae* etc. Currently no commercial vaccine is available against hemorrhagic septicemia. Furunculosis in salmon and turbot is caused by non motile *Aeromonas salmonicida*. Oil adjuvant vaccine is commercially available against Furunculosis which provides lifelong protection.

Vibriosis is caused by *Vibrio anguillarum*. A widely acceptable formalin inactivated whole cell vaccine is available against Vibriosis which gives very good protection by immersion methods.

Columnaris disease is caused by *Flexibacter columnari* which affects both warm and cold water fishes. Recently a live vaccine based upon rifampicin resistant *Flavobacterium columnare* strain has been patented for commercial use.

Edwardsiellosis caused by *Edwardsiella tarda*, seriously affects carp, tilapia, mullet, catfish and eel culture. Till now no commercial vaccine is available for *E. tarda* due to variability in serotypes. Edwardsiellosis caused by *E. ictaluri* affect channel cat fish culture. Recently two live vaccines for *E. ictaluri* based upon RE-32 strain and aroA and purA deficient strain has been licensed for commercial use in channel cat fish.

White spot syndrome virus (WSSV) and other viral diseases became serious threat for shrimp industry worldwide. It is generally thought that shrimp lack the immunoglobulin based adaptive immune system. However many experiment exhibited the presence of 'quasi-immune response' in shrimp. Recombinant protein and DNA vaccines taking viral structural protein, VP28 has shown promising results on experimental trial.

In Indian context challenges lies to develop effective vaccine for cultivable fresh water fishes against *Aeromonas hydrophilla* and for cultivable marine shrimp against *Vibrio* and white spot disease virus.

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GENE KNOCK DOWN TECHNOLOGY: A NOVEL THERAPEUTIC APPROACH

R. Ananda Raja and Tobias G. Poehlmann

Gene knock down technology is a novel therapeutic approach with the RNAi-based strategies to silence gene expression in animal and plant cells. Short interfering RNAs (siRNAs), typically 21-bp dsRNA molecules with perfect complementarities to the target mRNA in one strand are used to induce dsRNA-mediated postranscriptional gene silencing (PTGS) i.e, RNA interference (RNAi) both in plants and animals. The siRNA antisense strand (AS) containing the thermodynamically less stable 5'end is preferentially incorporated as the guiding strand of the RNA induced silencing complex (RISC), while the sense strand (SS) of the siRNA duplex is cleaved by Argonaute 2 (Ago2), the RNAi machinery and liberated. This generates the activated RISC containing the guiding AS, which binds the complementary target mRNAs and trigger their degradation by RNase activities associated with RISC.

Overview to cell signaling pathways

It is a complex system of communication that governs basic cellular activities and coordinates cell actions. The basis of development, tissue repair, and immunity as well as normal tissue homeostasis mainly depends on the ability of cells to perceive and correctly respond to their microenvironments. Any errors in cellular information processing are responsible for diseases. By understanding cell signaling, diseases may be treated effectively. Cells communicate with each other via direct contact (juxtacrine signaling), over short distances (paracrine signaling), or over large distances and/or scales (endocrine signaling). The major cell signaling pathways are,

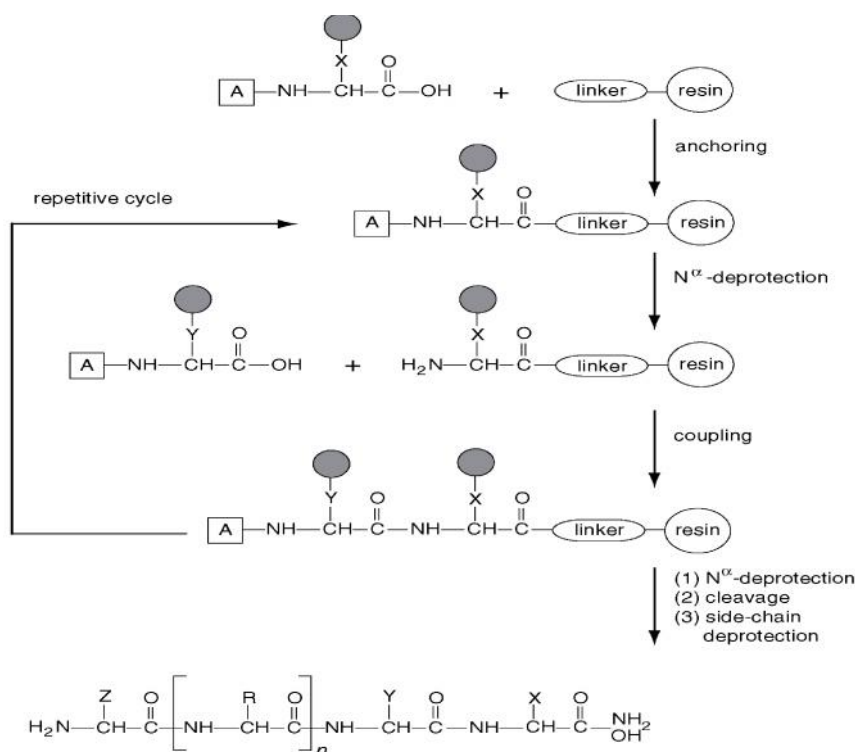
1. G protein coupled receptors (GPCRs), also known as seven-transmembrane domain receptors (7TM receptors), heptahelical receptors, serpentine receptor, and G protein-linked receptors (GPLR), comprise a large protein family of transmembrane receptors found only in eukaryotes which are involved in many diseases. There are two principal signal transduction pathways involving the G protein-coupled receptors: the cAMP signal pathway and the phosphatidylinositol signal pathway.
2. Receptor tyrosine kinases (RTKs) are the high-affinity cell surface receptors for many polypeptide growth factors, cytokines and hormones. They have been shown to be key regulators of normal cellular processes.

3. The notch signaling pathway is a highly conserved cell signaling system present in most of the multicellular organisms. It promotes proliferative signaling during neurogenesis.
4. The JAK-STAT signaling pathway transmits information from chemical signals outside the cell, through the cell membrane, and into gene promoters on the DNA in the cell nucleus, which causes DNA transcription and activity in the cell. The JAK-STAT system is a major signaling alternative to the second messenger system. The JAK-STAT system consists of three main components: (1) a receptor (2) Janus kinase (JAK) and (3) Signal Transducer and Activator of Transcription (STAT).
5. The Akt/PKB signaling pathway is associated with cancer with the proteins AKT (Protein kinase B, PKB) and phosphoinositide 3-kinase (PI3K).
6. Apoptosis, programmed cell death.
7. The Hippo signaling pathway, also known as the Salvador/Warts/Hippo (SWH) pathway, controls organ size through the regulation of cell proliferation and apoptosis.
8. The PI3K/AKT/mTOR pathway is an intracellular signaling pathway important in apoptosis.
9. Integrins are receptors that mediate attachment and play a role in signaling between a cell and the tissues surrounding it, which may be other cells or the extra cellular matrix (ECM).

Among the entire cell signaling pathways, RNAi-mediated down regulation of STAT3 is found to be promising and well understood to make new approach in relation to the development of therapeutics againsts infectious diseases by down regulating the genes of interest. But the hurdle for its pharmaceutical use is to deliver siRNA exclusively into the target tissue. Development of powerful and efficient intracellular delivery methods is crucial for potential RNAi activity. So, Modifications of siRNAs mostly aim at achieving better intracellular delivery or enhanced nuclease resistance. Some chemical modifications of siRNAs include conjugation to lipids and membrane-penetrating peptides, or RNA aptamer sequences. These modified siRNAs with superior potency require lower doses for gene silencing, and specific chemical modifications reduce siRNA side-effects, such as the induction of recipient immune responses and inherent off-target effects. Successful experiments in animal models have frequently relied upon injection of chemically-modified siRNAs. Here, it is demonstrated the proof-of principle for a mechanism to provide a key-lock-approach to siRNA molecules in order to induce RNAi specifically in the target cells.

Solid-phase peptide synthesis (SPPS)

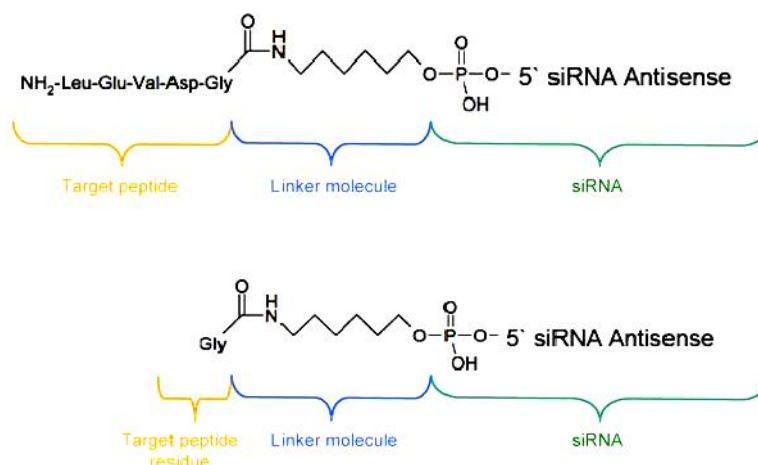
Handful of synthetic peptides plays a significant role in commercial or pharmaceutical products. Rapid, efficient and reliable methodology for the chemical synthesis of these molecules is of utmost interest. Peptide synthesis became a more potential part of present-day scientific research following the advent of SPPS. The amino group of the amino acid is protected by Fluorenyl methyloxy carbonyl chloride (Fmoc) and Trifluoro acetic acid (TFA) is used for cleavage of peptide from resin and deprotection of side chain of the amino acid. There is no need of protecting the amino group for amino acids such as leucine, glycine and alanine since they have only one carboxyl group with no side chains. But for amino acid such as aspartic acid and glutamic acid, protection of side chain carboxyl group is required. The crude pentapeptide is purified by reverse phase HPLC, eluted and the molar masses are determined by MALDI-TOF mass spectrometry. Generalised approach to SPPS is demonstrated by Gregg B. Fields 2002 as follows.



Synthesis and mechanism of peptide inhibited siRNA (PI-siRNA)

The synthesized pentapeptide carrying the protecting Fmoc-group is dissolved in dioxane. To produce PI-siRNA AS, the siRNA 5' end carrying amino-C6-linker is dissolved in triethylamine and mixed with peptide. On incubation at room temperature for 1h, because of the intrinsic peptide chemistry the peptide will only bind to the linker structure on the siRNA AS. The resultant PI-siRNA AS is purified and the concentration is determined.

siRNA SS and PI-siRNA AS are separately diluted with RNase free water and annealed by incubating at 95° C for 3 min followed by slow cooling (3° C per min) to 25° C in annealing buffer. Chemical structure of peptide inhibited and peptidase activated siRNA is described by Sandra Koehn *et al.* 2010. The target peptide for caspase 4 is bound to the siRNA AS via an aminohexyl linker structure. The activated siRNA still contains the linker molecule and a peptide residue. Caspase 4 acts between Glycine and Aspartic acid and makes siRNA active.



Cell culture and transfection

The suitable cell lines are used for studying the specific gene silencing effect of PI-siRNA. In addition, a cell line should be selected as a control with lack of peptide specific cleavage enzyme. In general, *in vivo* siRNA delivery is difficult and remains challenging to the scientific community. Known strategies for cell- or tissue-specific siRNA delivery, *e.g.*, using receptor-ligand systems or aptamer structures can also lead to the induction of RNAi in other cells by non-specific siRNA uptake. But, PI-siRNA can be used to demonstrate the proof-of principle for a mechanism to provide a key-lock-approach to siRNA molecules in order to induce RNAi specifically in the target cells.

Western blot analysis

Protein lysate is prepared and quantified by Bradford assay from the successfully transfected cell lines. Protein expression analysis is performed in order to investigate the level of PI-siRNA activation and to compare its induction kinetics with that of conventional siRNA. Western blot analysis works in the principle of Signal Transducer and Activator of Transcription (STAT3) antigen in the sample combines with STAT3 antibody which further combines anti-Rabbit IgG with HRPE, which shows luminescence under the Bio-imaging system. In addition, tubulin antigen as a universal control in the sample combines with alpha/beta Tubulin Rabbit antibody which further combines anti-Rabbit IgG with HRPE

showing luminescence. siRNA and PI-siRNA can not be used as a control since they degrade within 72h.

Conclusion

The topic explains a model for targeted, cells specific activation of peptide-inhibited siRNAs. This new tool might be useful for gene knock down applications in co-cultures *in vitro* and *in vivo*. Repeated dosing of PI-siRNA is required at frequent interval since it is degraded within 96h of delivery in the biological system. Further research is required to increase the half life of PI-siRNA. Combined with established technologies of target cell-specific siRNA delivery, this technology might reduce side-effects of siRNAs in nontarget cells and leads to increased cell-specificity in therapeutic applications.

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PRACTICALS

VISUAL INSPECTION OF SHRIMP FOR DISEASE DIAGNOSIS

S.B. Bhattacharya and R. Ananda Raja

Gross observations of clinical signs in shrimp can be easily made at the farm or pond side. Although, in most cases, such observations are insufficient for a definite diagnosis, such information is essential for preliminary compilation of a strong “case description” (or case history). Accurate and detailed gross observations also help with initiation of an action plan which can effectively reduce losses or spread of the disease, *e.g.*, destruction or isolation of affected stocks, treatments or alterations to husbandry practices (*i.e.*, feeding regimes, stocking densities, pond fertilization, *etc.*). These can all be started while waiting for more conclusive diagnostic results.

Behaviour

Abnormal shrimp behaviour is often the first sign of a stress or disease problem. Farmers or farm workers, through daily contact with their stocks, rapidly develop a subconscious sense of when “something is wrong”. This may be noticeable changes in feeding behaviour, swimming movement or unusual aggregations. Even predator activity can provide clues to more “hidden” changes such as when fish or shrimp-eating birds congregate around affected ponds. Record-keeping can provide valuable additional evidence that reinforces such observations and can indicate earlier dates when problems started to appear. It is important that farmers and workers on the farm, as well as field support staff, get to know the “normal” (healthy) behaviour of their stocks. Since some species and growing environments may demonstrate or evoke subtle differences in behaviour, these should be taken into account, especially if changing or adding species, or when information gathered from a different growing environment is used. Where any change from normal behaviour affects more than small numbers of random individuals, this should be considered cause for concern and warrants investigation. Some clues to look out for in shrimp stocks include:

- Unusual activity during the daytime – shrimps tend to be more active at night and stick to deeper water during the day.
- Swimming at or near pond surface or edges - often associated with lethargy (shrimp swimming near the surface may attract predatory birds).
- Increased feed consumption followed by going off-feed.
- Reduction or cessation of feeding.
- Abnormal feed conversion ratios, length/ weight ratios.

- General weakening - lethargy (*note: lethargy is also characteristic in crustaceans when the water temperature or dissolved oxygen levels are low, so these possibilities should be eliminated as potential causes before disease investigations are started*).

Mortalities

Mortalities that reach levels of concern to a farmer should be examined for any patterns in losses, such as:

- Relatively uniform mortalities throughout a system should be examined immediately and environmental factors determined.
- Apparently random or sporadic mortalities may indicate a within-system or stock problems. If the following conditions exist - (a) no history of stock-related mortalities, (b) all stock originate from the same source, and (c) there have been no changes to the rearing system prior to mortality problems - samples of affected and unaffected shrimp should be submitted for laboratory examination, as appropriate, and supported by gross observations and stock history.
- Mortalities that spread suggest an infectious cause and should be sampled immediately. Affected shrimp should be kept as far away as possible from unaffected shrimp until the cause of the mortalities can be established.

Feeding

Abnormalities in feeding behaviour and lack of feed in the gut are good indicators of potential problems. Daily gut content checks can be made on shrimp caught in feeding trays or bowls (where used) or, less frequently, from samples taken to determine growth. Ideally examination of feeding behaviour should be made every 1-2 weeks, even in extensive farming systems. Feeding behaviour is most easily checked by placing feed in a tray or bowl and seeing how quickly the shrimp respond, ideally after the shrimp has not been fed for at least a few hours. It is important that the feed used is attractive to the shrimp as poorly formulated, old or badly stored feeds may not be attractive to the shrimp. Gut contents can be checked by holding the shrimp against a light to show the gut in the tail segments. If these are empty, especially just after providing feed, it may indicate either of the following conditions: i) underfeeding, or ii) onset of cessation of feeding (anorexia). Feed records should be maintained to determine normal feed consumption patterns (*i.e.*, feeding activity by healthy shrimp), which can be compared with “suspect” feeding activity. In many cases of chronic loss, daily feed consumption patterns may remain stable or oscillate over periods of several weeks. These can be detected by making a graph of daily feed consumption or by comparing daily feed consumption in the record book over an extended period (*e.g.* 3-4 weeks).

Colonisation and Erosion

Colonisation of the shell (cuticle) and gills of a crustacean is an on-going process that is usually controlled by grooming. The presence of numerous surface organisms (*e.g.* “parasites” - which damage their host; or “commensals” - that do not adversely impact their host) suggests sub-optimal holding conditions or a possible disease problem. Apparent wearing away (erosion) of the cuticle or appendages (legs, tail, antennae, rostrum), or loss of appendages, with or without blackening (melanization) are also highly indicative of a disease problem. Breakage of the antennae is an early warning sign. In healthy penaeid shrimp, these should extend approximately 1/3 past the length of the body (when bent back along the body line). Likewise, erosion or swelling of the tail (uropods and telson), with or without blackening, is an early sign of disease.

Cuticle Softening, Spots and Damage

Softening of the shell, other than during a moult, may also indicate the presence of infection. Damage or wounds to the shell provide an opportunity for opportunistic infections (mainly bacterial and fungal) to invade the soft-tissues and proliferate, which can seriously impact the health of the shrimp. Certain diseases, such as White Spot Disease, directly affect the appearance of the shell; however, few changes are specific to a particular infection. In the case of white spots on the cuticle, for example, recent work has shown that bacteria can produce signs similar to those produced by White Spot Disease and Bacterial White Spot Syndrome.

Colour

Shrimp colour is another good indicator of health problems. Many crustaceans become more reddish in color when infected by a wide range of organisms, or when exposed to toxic conditions, especially those that affect the hepatopancreas. This is thought to be due to the release of yellow-orange (carotenoid) pigments that are normally stored in the hepatopancreas. This red colour is not specific for any single condition (or groups of infections), however, so further diagnosis is needed. Yellowish coloration of the cephalothorax is associated with yellowhead disease and overall reddening can be associated with gill associated virus infections, white spot disease or bacteria, as described above, or bacterial septicemia. In some cases, the colour changes are restricted to extremities, such as the tail fan or appendages and these should be examined closely. It should be noted that some shrimp brood stock, particularly those from deeper waters, can be red in colour (thought to be due to a carotenoid-rich diet). This does not appear to be related to health and its normality can be established through familiarization with the species being grown. Under certain

conditions, some crustaceans may turn a distinct blue colour. This has been shown to be due to low levels of a carotenoid pigment in the hepatopancreas (and other tissues), which may be induced by environmental or toxic conditions. Normal differences in colouration (light to dark) within a species may be due to other environmental variables. For example, *Penaeus monodon* grown in low salinities, are often much paler than *P. monodon* grown in brackish-water or marine conditions. These variations do not appear to be related to general health.

Environmental Observations

Shrimp with brown gills or soft shells (or a representative sub-sample), should be transferred to a well aerated aquarium with clean sea water at the same salinity as the pond from which they came. They should be observed every 1-2 hrs over 1 day. If the shrimp return to normal activity within a few hours, check environmental parameters in the rearing pond(s).

Soft-Tissue Surfaces

A readily observable change to soft tissues is fouling of the gill area, sometimes accompanied by brown discoloration. This can be due to disease and should trigger action since it reduces the shrimp's ability to take up oxygen and survive. Removal of the shell in the head region of shrimp allows gross examination of this region, particularly the hepatopancreas. In some conditions, the hepatopancreas may appear discoloured (*i.e.*, yellowish, pale, red), swollen or shrunken, compared with healthy shrimp. If the hepatopancreas is gently teased out of the shell, the mid-gut will become visible and permit direct examination of colour (dark - feeding; light/white/yellow - mucoid, empty or not feeding). This information is useful for determining the health of the shrimp and if infectious disease agents are present.

Environmental Parameters

Environmental conditions can have a significant effect on crustacean health, both directly (within the ranges of physiological tolerances) and indirectly (enhancing susceptibility to infections or their expression). Examples include changes to dissolved oxygen levels and/or pH which may promote clinical expression of previously latent yellowhead disease and white spot disease or the effect of salinity on the expression of necrotising hepatopancreatitis. This is especially important for species grown under conditions that bear little resemblance to the wild situation. Water temperature, salinity, turbidity, fouling and plankton blooms are all important factors. Rapid changes in conditions, rather than gradual changes, are particularly important as potential triggers for disease. Therefore, the farm manager and workers, should attempt to keep pond rearing conditions

within the optimum range for the species *and* as constant as possible within that range. High stocking rates are common in aquaculture but predispose individuals to stress so that even minor changes in environmental conditions may precipitate disease. In addition, many small changes do not affect shrimp health. However, when several of these small changes occur simultaneously, results can be far more severe.

Viral diseases of shrimp

White Spot Disease (WSD): WSD outbreaks are often characterised by high and rapid mortality of infected populations, usually shortly after the first appearance of the clinical signs. Acutely affected shrimp demonstrate anorexia and lethargy, have a loose cuticle with numerous white spots (about 0.5 to 2.0 mm in diameter) on the inside surface of the carapace. These spots are within the cuticle structure and cannot be removed by scraping. Moribund shrimp may also show a pink to red discolouration. Susceptible shrimp species displaying these clinical signs are likely to undergo high levels of mortality. Pathology is associated with systemic destruction of the ectodermal and mesodermal tissues of the gills and sub-cuticular tissues. WSD outbreaks are generally preceded by cessation of feeding followed, within a few days, by the appearance of moribund shrimp swimming near the surface at the edge of rearing ponds. These shrimp exhibit white inclusions embedded in the cuticle and often show reddish discolouration of the body. The cuticular inclusions range from minute spots to discs several mm in diameter that may coalesce into larger plaques. They are most easily observed by removing the cuticle from the cephalothorax, scraping away any attached tissue and holding the cuticle up to the light. The appearance of white spots in the cuticle can also be caused by other conditions.

Bacterial white spot syndrome (BWSS): Dull white spots are seen on the carapace and all over the body but are more noticeable when the cuticle is peeled away from the body. The white spots are rounded and not as dense as those seen in WSD. The spot center is often eroded and even perforated. During the early stage of infection, shrimp are still active, feeding and able to moult – at which point the white spots may be lost. However, delayed moulting, reduced growth and low mortalities have been reported in severely infected shrimp. Although the exact aetiology is unknown, some measures may help to reduce the risk of BWSS. Build up of high bacterial density in rearing water should be avoided. Changing water frequently is recommended. Indiscriminate use of probiotics containing *Bacillus subtilis* should also be avoided until the relationship between this bacteria and the BWSS syndrome is better understood. It has been claimed that BWSS in shrimp ponds can be treated with quick

lime (CaO) at 25 ppm, however, this is still under investigation and the use of quicklime may itself cause problems due to rapid increases in pond water pH.

Further Reading

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SAMPLING AND BASIC BACTERIOLOGICAL TECHNIQUES

Sujeet Kumar, Shyne Anand PS and R. Ananda Raja

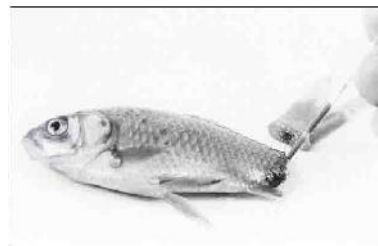
The first step in diagnostic bacteriology is to isolate the pathogen from the diseased animals as pure culture. For comparative purpose, bacteria may be isolated from infected and apparently healthy specimen at a time. As putrefying bacteria act upon the dead fish or shell fish very quickly, isolation of bacteria should be carried out from the just dead or moribund fish. Final identification is made based on its cultural, morphological, physiological, biochemical, serological and molecular characteristics.

Sampling

The first step in disease diagnosis is sampling. The sample for microbiology work should be collected in aseptic condition and better to be processed immediately or otherwise should be kept on ice or at 4⁰C. The sample for molecular biology work should be preserved in 95% ethanol and for histopathological work in 10% neutral buffer formalin. Apart from fish and shrimp, water and soil sample should also be collected from a disease affected pond.

- 1. Water sampling:** Using a sterile container, 100 ml – 500 ml of water sample should be collected keeping at least 10% of air space to allow adequate space for mixing the sample prior to analysis. Care must be taken to avoid sample contamination during collection. To make a representative sample within pond water should be collected from at least 3 places. For best results, the sample should be inoculated within 6 hours after collection and should be maintained at or below 10 °C, but not frozen.
- 2. Soil sampling:** Using a sterile container, soil sample should be collected from at least five places within a pond. Then the sub-sample should be mixed to make a composite sample. Sample should be collected in a sterile sample container. Before use the sample should be homogenised.
- 3. Sampling of shrimp and fish:** In case of disease outbreak, moribund or recently died shrimp or fish should be collected. The sample should be collected in sterile container and should be kept at 4⁰C for microbiology work, in 10% neutral buffered formalin for histopathology work, and in 95% ethanol for molecular biology work.
- 4. Haemolymph collection from shrimp:** Haemolymph should be collected from ventral sinus of shrimp using 25 – 26 Gause tuberculin syringe.
- 5. Blood collection from fish:** The method of taking blood from fish depends on the size of the fish, quantity of blood needed and whether the fish under investigation is required to be killed after taking the blood or should be spared.

5a. Tail ablation in small fish: It is used for fishes that are too small to bleed with a syringe and needle. For this, first euthanize the fish by trichlorobutanol solution (dissolve one gm of trichlorobutanol in hot water and make the volume one liter by adding water). Thereafter, disinfect the area and sever the caudal peduncle with a scalpel blade or sharp knife. Discard first few drops of blood and then collect in the sterile tube.



5b. Caudal venous puncture: Insert a needle attached to a syringe under the skin of the ventral midline of the caudal peduncle of an anaesthetized or freshly euthanized fish. Alternatively, insert the needle under the scales of the mid portion of tail just below the lateral line at a 45^0 angle to the long axis of the fish in cranial direction. Ease the needle toward the vertebral column until you reach the base of the column. Withdraw the needle a fraction of a millimetre, and obtain the blood sample.



5c. Dorsal aorta puncture: Insert a needle attached to a syringe pointing upward, along the dorsal midline of the mouth, just behind the juncture of the second gill arch.



5d. Heart puncture: This method is often performed in a euthanized fish. Hold the needle attached to a syringe perpendicular to the skin and insert the needle slightly below the tip of the V-shaped notch formed by the gill cover and the isthmus of an anaesthetized fish. Collect the blood as the needle enters the bulbous arteriosus.



General precaution in microbiology laboratory

Working in microbiology lab needs some sorts of special precaution which everyone working in microbiology lab should understand. The general precaution in microbiology lab should be:

1. Keep microbiology lab clean and wear a lab coat during the microbiological work.
2. Do not eat or drink anything in the microbiology laboratory.
3. Report all accidents even though you consider them as minor – such as cuts, pricks, abrasion to the scientist or incharge of the laboratory.
4. Place all the solid wastes (Cotton, discarded plate) in a container provided for the purpose.
5. Smear desk top with mercuric chloride (1: 1000), or alcohol (70%) prior to and after plating of cultures to reduce contamination.
6. Wash your hands with soap before and after work.
7. Plan and organise your work in advance, as microbiology work needs a lot of preliminary arrangement.

Aseptic techniques

Maintenance of aseptic conditions during all steps of microbiology is the first step for successful microbiological investigation. For ensuring this, all glassware, plasticware, media, solutions etc. are sterilized before use. Personnel care should be taken to wash the hands with 70% ethanol before start of any culture work. Sterilization methods employed for different kinds of materials used in the laboratory are given in Table 1.

Table 1. Common sterilization methods

Sl. No.	Methods of sterilisation	Condition	Materials
1	Dry sterilization by hot air oven	160 ⁰ C for two hour or 180 ⁰ C for one hour	All types of glassware like pipettes, tubes, flasks, petridish etc. and lipid like paraffin oil
2	Autoclaving	121 ⁰ C, 15 lb pressure for 15 min	Bacteriological media, discarded media, plasticware, steel items, corks, rubber materials, filter pads, distilled water, buffers, solutions and also glassware
3	Filtration	0.22 µm or 0.45 µm pore size membrane filter	Tissue culture media, antibiotics, sera, heat sensitive solutions like carbohydrate, aminoacids etc.

Autoclaving: This is used for the sterilisation of media. Usually, 15 minutes at 15 lb steam pressure which gives the temperature of about 121⁰C is sufficient for sterilisation of media.

Precaution

1. Media containing sugar should not be sterilised for more than 15 minute since prolonged heating hydrolyse sugars.
2. Flask containing media should be placed with enough spacing in the autoclave. This will facilitate the steam circulation uniformly.

3. Cotton plugged flask, tubes should be covered with heavy paper or aluminium foil before autoclaving. This prevents cotton becoming saturated with moisture.
4. Before raising pressure in the autoclave, make sure that all the air in the autoclave has been displaced by steam. This can be done by keeping the valve open for 5 minutes after the autoclave starts steaming.
5. At the close of sterilisation do not release the pressure suddenly. This results in the plugs being blown out or getting soaked by the boiling media.

Preparation of culture media

Media refers to any substance that will enable the microorganisms to grow and multiply. It may be liquid media (broth) or solid medium. Both, liquid or solid media must be sterile before use.

Procedure

1. Take known volume of distilled water (1 liter) in an Erlenmeyer flask. Add the required amount of dehydrated agar and stir with a glass rod to prevent lumping. The quantity required/ liter is always written over the media bottle.
2. Set the flask on an electric hot plate to completely dissolve the media. Then, close it with a cotton plug.
3. Autoclave at 15 lb pressure for 15 minutes.
4. After taking out from autoclave, allow the media to cool to about 50°C (the agar should be warm and melted, but not too hot to handle in its flask). Then after quickly pour the melted, sterile agar into a series of petri dishes to about one-third capacity. Replace each petri dish top as the plate is poured. When the plates are cool (agar solidified), invert them to prevent condensing moisture from accumulating on the agar surfaces.
5. Place inverted agar plates and tubes of sterilized nutrient broth at 37°C in incubator for at least 24 hours to ensure they are sterile.

Isolation of pure culture of bacteria

Isolation of pure culture of bacteria is necessary to characterize it further by cultural, biochemical, molecular methods. A number of methods are used for this purpose. These are:

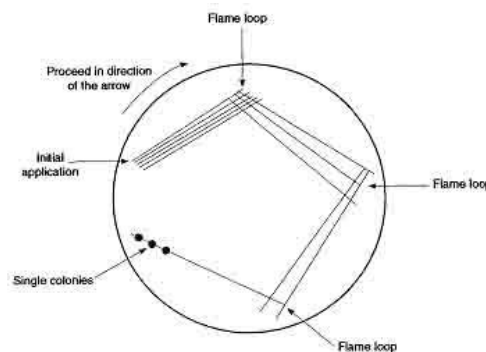
1. Streak plate technique (streaking onto solid media)
2. Pour plate technique (Incorporation into molten semi-solid media)
3. Spread plate technique

Streak plate method

This is the most commonly employed technique for obtaining pure culture of specific bacterium from mixed bacterial populations.

Procedure

1. Inoculate the infected larvae/ affected tissues/ haemolymph/water sample on the culture plates with the help of sterile bacteriological loop and streak the inoculum to get isolated colonies.
2. Incubate the inoculated agar plates at optimal temperature (28° - 30°C) for 24-48 h and observe for development of bacterial colonies.
3. Examine cultural characteristics of the bacterial colonies as given in the subsequent sections and record.
4. Obtain pure culture of bacteria by picking up morphologically distinct colonies with the help of a sterile bacteriological loop and subculture on ZMA for further characterization.



Pour plate technique

The pour plate method requires only small volume of samples like 1 ml, 0.1 ml, and 0.01 ml or 0.001 ml of sample. These dilutions are prepared by pipetting 1 ml of undiluted sample into 9 ml of buffered dilution water. The TSA medium is preferable for counting total heterotrophic bacterial population and 15-18 ml of the medium is poured to each dilution and mixed well and allowed to set with inoculums. After solidification, the plates are incubated upside-down at 35°C for 48 hours.

Spread Plate technique

The sterile TSA plates are inoculated and spread uniformly with $100\mu\text{l}$ of 10^{-1} to 10^{-7} dilutions of samples (depending upon the type of sample and amount of bacteria present) in duplicate and the plates are incubated upside-down at 35°C for 48 hours. Between spreading operations of each plate, the glass rod used for spreading is sterilized by dipping in alcohol and flaming.

After 48 hours of incubation, the colonies developing in the each plate are counted using Quebec colony counter. The sample volumes and dilutions are selected to get the total number of colonies on a plate between 30 and 300. The number of samples to be plated at any one time is limited so that not more than 20 minutes (preferable 10 minutes) elapse between the dilution of the first sample and the pouring of the last plate. The sample container is mixed thoroughly before performing dilutions (approximately 25 times) and sterile pipette is used with sterile microtips for each sample. TPC i.e. Colony-Forming Units (CFU)/g of sample is calculated as follows.

CFU/g sample = Average count x Dilution factor.

In the case of colony counts from spreaders, the average count has to be doubled before calculation of TPC. The spread plate technique is commonly used for TVC with the same procedure except using TCBS agar.

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IDENTIFICATION OF BACTERIA BY STAINING AND MICROSCOPY

Sujeet Kumar and R. Ananda Raja

Bacterial cells are very small in size, usually in the range of 0.5μ to 20μ ($1\mu = 10^{-3}\text{ mm} = 10^{-6}\text{ m}$). Limit of vision of human eye is just 100μ , so they are not visible to naked eye. Hence a microscope is used for observing bacterial cell. But bacterial cells are semi-transparent, so, suitable dyes are required to stain them to obtain better contrast. Staining and microscopy represents the first stage in microbial identification and provides the crucial information with regards to shape, size, capsule, spore etc.

Staining

Dyes used for staining purpose of living organisms are called stain. There are many stains available, and depending on our requirement, suitable ones are selected. Based upon staining principal, staining techniques are broadly classified into two categories. These are:

1. Simple staining
2. Differential staining

Simple staining: Staining of bacterial cells by the use of a single staining solution is called simple staining. Ex: Methylene blue staining, Carbol fuchsin staining, Crystal violet staining. Simple staining is done by flooding the staining solution on fixed bacterial smear over a slide. It reveals the shape and size of bacteria and distinguishes bacterial cells from non living material.

Differential staining: It involves staining of bacterial cells by more than one staining solution which may be applied separately or mixed and applied in single step. It is used to differentiate between bacterial cells or parts of bacterial cells. Ex: Grams staining, Acid-fast staining, spore staining, capsular staining.

Grams staining

This is the most commonly used staining techniques in Bacteriology. It was designed by Christian Gram in 1884. The technique broadly classifies bacteria into two categories; Gram positive and Gram negative.

Principle: The method uses crystal violet as primary stain and safranine as counter stain. Sometimes, safranine is replaced by carbol fuchsin or neutral red as counter stain. When the bacterial smear is flooded with crystal violet, the crystal violet (CV) enters the bacterial cell through the small pores on the cell wall and stains (reacts with) the cytoplasmic membrane and other negatively charged components of bacterial cells. The iodine also enters the cell

similarly. Iodine reacts with the CV and form CV-I complex within the cell. When the smear is washed with ethyl alcohol, the cell walls get dehydrated and defatted. Since the cell wall of Gram positive bacteria are more compact, the dehydration by alcohol shrinks the pores. So the CV-I complex formed inside the cell cannot be washed out by the destaining process. Hence, Gram positive bacteria retain the crystal violet and appear violet. But the cell wall of Gram negative bacteria is less compact and contains high amount of lipids (up to 11-22%). The alcohol extracts the lipids and so the porosity increases. As a result, the CV-I complex is completely washed out from the cell. So, Gram negative bacteria lose the violet dye. They appear red because the cells get stained by the counter stain, safranine.

Materials

1. Gram's Crystal violet
2. Gram's iodine
3. Safranine
4. Ethyl alcohol for destaining

Procedure

1. Take dust free, dirt free and oil free microscopic glass slide.
2. Take a speck of young culture (16-24 hr. culture) with the help of inoculation loop and emulsify with a drop of sterile water in the middle of the slide and spread uniformly.
3. Dry the smear in the air.
4. Fix the smear by passing the slide 3-4 times through the blue flame of Bunsen burner, with the smeared side on the top. Take precaution to not char the smear.
5. Flood the smear with Gram's crystal violet for 1 min. Then wash the stain with distilled water drop by drop.
6. Flood with Gram's iodine for 1 min. Then wash the stain with water drop by drop.
7. Destain with drop-wise addition of ethyl alcohol until washings are free from violet color. This should be followed by washing with water.
8. Counter stain with Safranine for 1 min. This should be followed by washing with water.
9. Dry in air.

Microscopy:

1. Observe the slides under the microscope using oil immersion objective (100X).
2. Note down whether bacteria is Gram positive or Gram negative - Cells stained violet, bluish violet or bluish purple are Gram positive. They have retained the crystal violet after de-staining. Bacteria, which appear red, are Gram negative.

3. Note the shape, size and arrangement of cells, i.e., Cocci, Bacilli, spiral or curved rod, filamentous etc. For cocci whether they are in single or pairs or chains or bunch.
4. Examine whether the culture is pure or mixed. Whether bacilli are single or in chains.
5. If the cells are Gram-positive rods, examine whether there is spore formation. If spore is present, there will be unstained area inside the cell.

Acid-fast staining

Acid fast staining is used for the staining of *Mycobacterium* spp. which is not stained by general stain as its cell wall contains very high percentage of lipids, particularly mycolic acid which prevents the entry of stain within cell. The most common acid fast staining is Zeihl - Neelson's Carbol - Fuchsin staining.

Procedure

1. Smear preparation: Just like Gram staining.
2. Stain contains phenol and basic fuchsin. The bacterial smear is flooded with the stain and heated to just boiling. With the help of phenol and heat, basic fuchsin enters the cell and stains the cellular contents and plasma membrane.
3. Wash with acid-alcohol.
4. Counter stained with methylene blue followed by washing with distilled water.

Microscopy

1. Cells appearing red are acid fast, because they retain the primary stain even after acid-alcohol washing. Non-acid fast cells appear blue because they did not retain the basic fuchsin on acid-alcohol washing.

Spore staining

Spore forming bacteria belong mainly to the two genera *Bacillus* (Aerobe) and *Clostridium* (Anaerobe). Three methods are commonly employed for spore visualization.

1. Grams staining
2. Zeihl – Neelsson's staining
3. Malachite green staining

Grams staining: When spore bearing cells are stained by Gram's staining, the body of the cells is deeply coloured, while the spore is unstained and appears as clear area inside the cell.

Zeihl - Neelson's method: By this method spores will appear red, the vegetative cell will be colourless, and background of cell will appear black.

Procedure

- i. Make a thin smear of the spore bearing bacteria on a slide.

- ii. Dry thoroughly in cool air. Then fix by passing the slide over the flame with the minimum amount of heating.
- iii. Cover the smear with blotting paper and saturate it with carbol - fuchsin. Heat the slide from below till steam evolves for 5-10 min. Keep the blotting paper moist by adding the stain drop wise during the process.
- iv. Decolourise with 95% alcohol, containing 2% nitric acid.
- v. Rinse the slide with water.
- vi. Place a drop of saturated aqueous nigrosin at one end of slide and spread it evenly with the help of another slide.
- vii. Dry the slide quickly using gentle heat.

Microscopy

10. Observe the slide under oil immersion objective. Spores will appear red, the vegetative cell colourless, and background of cell will be black.

Malachite green staining (Schaeffer and Fulton's method)

1. Make a thin smear.
2. Keep the slide over a beaker of boiling water, with smear on the topside.
3. Flood the smear with malachite green solution for one to two min.
4. Wash with distill water.
5. Counter stain with Safranin.

Microscopy: Spores are stained green and the vegetative cell red.

Capsule staining (Negative staining)

Principle: This method of staining is used to see the shape and arrangement of bacteria, including the presence of capsules. The background is stained dark but bacterial cells are remains un-stained, so the cells are made readily visible. The technique use dark field microscopy.

Procedure

1. Mix microbial suspension with India ink or nigrosin solution and prepare thin smear on a clean slide.
2. Air dry.

Microscopy

1. Cells will appear bright against a dark background. India ink staining reveals capsule which is seen as a clear space between the refractile bacterial cell and the dark India ink background. If the dried smear of the mixture of India ink and bacterial suspension is stained with methylene blue, cell will be stained blue and capsule will be more clear.

Flagella staining (Leifson's method)

Principal: Flagellae are appendages from the bacterial cells, which aid in locomotion. It is very thin and so, cannot be resolved by a compound microscope. Staining thickened the flagella by deposition of dye, so, they become visible under microscope.

Procedure

1. Take 1 ml of young broth culture and dilute to get a diluted suspension. Best results are obtained when washed cells are used.
2. Take a clean glass slide. Heat in a flame and cool on a piece of filter paper. Mark an area of about one square centimeter with a wax pencil in the middle of the slide.
3. Place a small drop of the bacterial suspension at one end of the marked area and tilt the slide to allow the liquid to run to the other end so that a very thin smear is obtained.
4. Dry in the air. Do not fix.
5. Add one ml of the flagella stain on the smear and allow to remain 5-15 minutes. When a precipitate of the stain is formed, wash in running water. Drain and dry in air.
6. Counter stain with 5-10 times diluted methylene blue stain for 1 minute. Drain and dry.

Microscopy

1. Observe under oil immersion objective. The cell will be stained blue and the flagella red.

Fluorescent staining

Principal: Development of epifluorescence microscopy revolutionised the field of marine microbiology. Acridine orange (AO) and 4',6'-diamidino-2-phenylidole (DAPI) are commonly employed for visualizing bacterial cells. The application of DAPI is advantageous being low background fluorescence compared to AO.

Sample preservation

Samples containing bacteria should be preserved before filtration by 2-4% final volume of borate-buffered formalin (20 µl/ml of sample). Alternatively, sample should be preserved with 2.5% final concentration of glutaraldehyde using 0.2 µ filtered SEM grade glutaraldehyde (100 µl/ml of sample).

Equipments

1. Epifluorescent microscope with appropriate filter sets for the fluorochromes of interest, 50 W or 100 W mercury lamp or 100 W xenon lamp.
2. Filtration apparatus for 25 mm filters
3. Forceps – blunt end
4. Adjustable pipette for quantitative delivery of water samples and fluoro-chromes
5. Stage micrometer for measuring width of microscope fields of view

6. -20°C freezer for storage of prepared slides

Supplies

1. Black-stained 0.2µ pore-size 25 mm polycarbonate membrane filters
2. Cellulosic backing filters to support the membrane filters. Backing filters should have 0.45 µ pore size.
3. Glass slides and 25 mm square No. 1 cover slips
4. Immersion oil e.g. Cargille type A or type LF or Resolve TM low viscosity immersion oil
5. Pens to mark the glass slides
6. 20 ml disposable plastic syringes
7. 1.5 ml plastic vials, e.g. microfuge tubes or cryovials

Solutions

1. **Acridine orange stock solution** (0.1% AO, 6% formalin, 0.2 µ filtered).
 - a. Dissolve 20 mg of AO in 200 ml of 0.2 µ filtered seawater or artificial seawater of the approximate salinity of the samples.
 - b. Shake or stir to completely dissolve.
 - c. Filter the AO solution through a 0.2 µ filter.
 - d. Pour the AO solution into a dark 250 ml glass bottle and add 12 ml of 37% formaldehyde (formalin).
 - e. Stored in a refrigerator, the AO stock solution is good for months.

During processing of samples, 15-18 ml of AO stock solution should be drawn into a 20 ml plastic syringe and then attach a 0.2 µ Gelman Acrodisc to the tip of the syringe. Then, freshly 0.2 µ -refiltered AO solution should be dispensed onto the filters as needed.
2. **DAPI stock solution** (200 µg/ml, 0.57 mM)
 - a. Dissolve 10 mg of DAPI in 50 ml of distilled water filtered through a 0.2 µ Gelman Acrodisc, using a plastic syringe.
 - b. Dispense 1-2 ml aliquots of the DAPI stock into labelled microcentrifuge tubes or cryovials.
 - c. Store in a -20°C (or lower temperature) freezer. DAPI stock is good for up to a year in the freezer.
3. **Borate-buffered formalin**
 - a. Fill a 500 ml dark glass bottle with about 450 ml of formalin (37% formaldehyde, reagent grade).

- b. Add sodium borate crystals until an undissolved layer of crystals lies on the bottom, i.e. a saturated solution.
- c. When preserving samples, the formalin should be freshly filtered through a 0.2 μm .

Procedures for acridine orange staining

Sample preparation

1. Prepare a glass slide. Label one end of the slide with the sample code. Put a small drop of immersion oil on the slide and smear flat with the edge of a cover slip.
2. Place a 0.45 μm cellulosic membrane backing filter onto a 25 mm filtration bottom. Wet with a few drops of deionized water. The same backing filter can be used for multiple samples. Place a black 0.2 μm membrane filter, shiny side up, onto the wetted backing filter. Pre-rinse the filtration tower with 0.2 μm filtered seawater, shake dry, and carefully place on top of the filter. Clamp tightly.
3. Filter an aliquot of preserved sample through the 0.2 μm black membrane filter. Sample volume depends on bacterial concentration. Take 5-10 ml sample if bacterial abundance of less than 10^5 /ml is expected. When bacterial abundance is greater than 10^6 /ml then 1-2 ml sample is adequate. For bacterial cultures with abundances in excess of 10^7 cells/ ml it may be necessary to dilute samples with 0.2 μm -filtered seawater. The volume filtered should be at least 1 ml to ensure even distribution on the filter.
4. Release the vacuum and gently add 1 ml of 0.2 μm filtered AO solution onto the top of the filter. Run the AO solution down the side of the filter tower to minimize disturbance of the cells on the filter. Allow to stand for 2 min. and then turn on the vacuum and filter down the AO solution.
5. With the vacuum still on, carefully remove the membrane filter from the underlying backing filter.
6. Lay the filter, sample side up, onto the film of immersion oil on the slide. Put one drop of immersion oil onto the center of the filter, and gently lay a cover slip on top of the filter, being careful to avoid air bubbles.
7. For each set of filters, prepare one or more blank filters by filtering down 1 ml of AO stock solution without a sample. These will be used to obtain a value for background counts.
8. Inspect the sample immediately, or place the slide in a slide box in a -20°C (or lower temperature) freezer. The samples are good for several months at least.

Bacterial enumeration

1. Obtain a factor for the number of microscopic grids per filter for the particular filter apparatus and microscope. Measure the diameter in millimeters of the bottom of the filter tower, and calculate the area in square millimeters. This is the area of the sample on the surface of the filter. Next, determine the width of each side of the 10 x 10 ocular grid. Put a drop of immersion oil on the mm long scale of a stage micrometer. Focus on the micrometer scale with the 100x lens. Width between lines of the micrometer is 10 μ . Calculate the area of one grid field. Divide the sample area by the grid area to arrive at the factor for number of grids per filter (usually $y \times 10^4$).
2. Use a blue light epifluorescence filter set to visualize AO stained bacteria, 450-490 excitation filter, 510 beam splitter and 520 barrier filter).
3. For each sample, count the number of bacteria in a whole grid for 7 to 10 randomly chosen fields distributed over the filter. An appropriate density of bacteria on the filter surface would result in about 30 to 50 bacterial cells. Ideally, a minimum of 300 bacteria should be counted per filter.
4. Bacterial count/ml = [(sample cells per grid - background cells per grid) x grids per filter] / volume of sample.

DAPI direct counts

Using DAPI rather than AO for bacterial enumeration has the advantages of low background fluorescence, less interference from photopigments, and fluorescence does not fade while a field is being counted. While AO stains both DNA and RNA, DAPI stains only DNA. Direct counts using DAPI require a UV light excitation filter set, 365 excitation filter, FT 395 beam splitter and LP 420 barrier filter. It is also important that the objective lens is Neofluor, i.e. does not have coatings that preclude transmission of UV light. The procedures are generally the same as for the AO method except few steps:

1. Thaw a 1-2ml vial of DAPI solution. Filter sample down to 2 ml volume, release vacuum.
2. Add 50 μ l of DAPI solution to the sample to yield a final concentration of 5 μ g DAPI/ml sample. Cover with aluminum foil to keep out light, and allow standing for 7 minutes. Filter down and process as for the AO filters.

Further Reading

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IDENTIFICATION OF PATHOGENIC BACTERIA BY BIOCHEMICAL METHODS

Sujeet Kumar, Shyne Anand PS and R. Ananda Raja

The biochemical transformations that occur both inside and outside of bacteria are governed by biological catalysts called enzymes. These enzymes degrade carbohydrates, lipids, proteins, and amino acids. The metabolism of these organic molecules often produces by-products that can be used in the identification and characterization of bacteria. Following biochemical tests are commonly employed for the identification of bacteria.

Fermentation of carbohydrate

Principle: The microorganism ferment carbohydrates and produces products like alcohols, acids, gases, or other organic molecules and are characteristic of a particular microorganism. Acid production is detected by pH indicator such as phenol red or bromocresol purple which change the medium from its original color to yellow. Gas production during the fermentation process is detected by Durham tube, a small inverted tube, by seeing the entrapped gas in the form of a bubble.

Materials

1. Phenol red broth base containing phenol red indicator.
2. Carbohydrate solution – Add 5-10 gm of carbohydrate in 20.0 ml of distil water. Prepare different carbohydrate solution such as Glucose, Lactose, Sucrose, Galactose, Maltose, Inositol, Dulcitol, Melibiose, Xylose, Salicin etc separately.
3. Durham tubes
4. Bacterial culture (e.g. *Vibrio* sps.)
5. Bunsen burner/ spirit lamp
6. Inoculating loop
7. Incubator
8. Forceps
9. Test-tube rack

Procedure

1. Place the completely media filled Durham tube in inverted position in each phenol red broth base tube.
2. Sterilise phenol red broth base at 15 lb pressure for 15 minute and carbohydrate solution at 10 lb pressure for 10 min. Add 0.2 ml of carbohydrate solution in each test tube containing 9.8 ml of phenol red broth.

3. Using aseptic technique inoculate each tube with bacterial culture.
Take the precaution to mix the tubes by rolling back and forth between the palms of the hands. Otherwise, air bubble may get trapped into the Durham tube and will give a false-positive result.
4. Place the tubes in a test-tube rack and incubate at 30°C for 24 to 48 hours.
5. Examine the tubes carefully at 4 hours, 8 hours, and 18 hours for evidence of acid (A), or acid and gas (A/G) production. Acid production is detected by the medium turning yellow and gas production by a gas bubble in the Durham tube. Purple color indicate no fermentation.

Oxidation fermentation test

Principle: Bacteria utilises glucose as source of carbon and energy through enzymatic breakdown. Some □utilize glucose oxidatively i.e., oxygen is the ultimate hydrogen acceptor and while some other bacteria are fermentative i.e., hydrogen is accepted by molecule other than oxygen. This is independent of oxygen and culture may be aerobic or anaerobic. Liquid paraffin is used to create anaerobic condition. Example: *E. coli* (fermentative) is differentiated from *Pseudomonas aeruginosa* (oxidative).

Requirement

1. Pure culture of *Vibrio* *sps.*
2. OF glucose medium
3. Sterilised glucose solution (10%).
4. Bunsen burner/spirit lamp.
5. Sterilised liquid paraffin
6. Culture tubes
7. Inoculation loop
8. Autoclave
9. Incubator

Procedure

1. Weigh OF basal medium as per the recommendation of the company.
2. Take melted agar tubes and add 1 ml of 10% sterile glucose solution. Cool the tubes.
3. Mark each tube with the name of bacteria to be tested (4 tubes per bacterium – 2 tests and 2 controls).
4. Stab each bacterial suspension into 2 culture tubes containing the above medium.
5. Add a small amount of sterile liquid paraffin in one of the culture tubes of each bacterium and a control so that it may form 1 cm thick layer.

6. Plug all the tubes.
7. Incubate all the tubes at 37⁰C for 24 hours.
8. Acid formation in the tube without paraffin indicates oxidative reaction and it appear as yellow colour. Acid formation in both the tubes indicates the fermentation reaction. Lack of acid in both the tubes indicates that the bacterial strain is not able to utilize the glucose.

Oxidase test

Principle: Oxidase activity tests is one of the basic tests to differentiate certain groups of bacteria. Some bacteria are oxidase-negative (e.g. most of the Enterobacteriaceae) whereas some others are oxidase positive (e.g. *Pseudomonas* sp.). A dye called dimethyl p-phenylenediamine hydrochloride is used in oxidase test which donates electron to cytochrome-c and makes them oxidized and produces a colour.

Requirements

1. Trypticase soya agar
2. Bacterial suspension (e.g. *Pseudomonas aeruginosa*)
3. 1% solution of dimethyl-p-phenylenediamine hydrochloride.
4. Petridish
5. Filter paper (Whatman No.2)
6. Inoculation loop/needle
7. Bunsen burner/spirit lamp
8. Incubator

Procedure

1. Pour the medium in the petridish and prepare a plate.
2. Streak the plate and prepare the culture of *Pseudomonas aeruginosa*.
3. Place one drop of dimethyl-p-phenylenediamine hydrochloride covering the colonies.
4. Oxidase is indicated by oxidation of reagent to a deep pink colour. If the colonies become first pink, then changes to dark red colour and ultimately turn black. This obviously indicate oxidase positive.

Catalase test

Principle: A large no. of aerobic or facultative anaerobic bacteria exhibits catalase activity. These organisms use oxygen to form hydrogen peroxide. Catalase produced by these microbes converts the hydrogen peroxide to water and oxygen, thus reducing the toxicity of hydrogen peroxide.

Materials

1. Bacterial culture
2. Inoculation loop
3. Slide
4. Hydrogen peroxide (3%)
5. Sterilised water

Procedure

1. Transfer a bacterial colony on the glass slide over a drop of NSS.
2. Place 2-3 drops of 3% hydrogen peroxide over the culture.
3. Examine the effervescence or bubble.
4. Observation: Release of bubble within 20 second indicate catalase positive while no bubble indicate catalase negative. *Staphylococcus* is catalase positive, while *Streptococci* is catalase negative.

Motility test: This is used for the testing of motility of microbes.

Materials

1. Motility test medium: Per liter of the motility test medium consists of Pancreatic digest of casein – 10.0 gm, Glucose – 5.0 gm, Di sodium hydrogen phosphate – 2.5 gm, Yeast extract – 2.5 gm, Agar – 3.0 gm, pH – 7.4 ± 0.2 . Mix thoroughly, gently heat and bring to boiling. Distribute into tubes for 5.0 ml volume. Autoclave and keep for 2 days prior to inoculation.
2. *Vibrio* sps.
3. Incubator

Procedure

1. Do stab inoculation of motility medium. Incubate at room tem for 24 hours.
2. Bacteria is non motile if growth is present only in the line of stab inoculation, otherwise it is motile.

IMViC test: It denotes Indole, Methyl red, Voges Proskauer, and Citrate test.

Indole test:

Principle: The amino acid tryptophan is found in nearly all proteins. Bacteria that contain the enzyme tryptophanase hydrolyze tryptophan to its metabolic products, namely, indole, pyruvic acid, and ammonia. The bacteria use the pyruvic acid and ammonia to satisfy nutritional needs while indole is not used and gets accumulated in the medium. The presence of indole can be detected by the addition of Kovacs' reagent which produces bright red color after reacting with indole.

Material

1. Kovacs reagent
p dimethyl amino benzaldehyde – 50 g
Amyl alcohol - 750 ml
Hydrochloric acid - 250 ml

Procedure

1. Prepare tryptic soys broth tube.
2. Using aseptic technique, inoculate each tube by a loopful of culture.
3. Incubate the tubes for about 24 hours at 37°C.

After 24 hours, add 0.5 ml of Kovacs' reagent to each tube, and shake the tube gently. Development of deep red color shows the presence of indole. Colorless or yellow reaction shows negative reactions.

Methyl Red Test

Principle: Enteric bacteria catabolize glucose for their energy needs; however, the end products vary depending on the enzyme pathways present in the bacteria. The test distinguishes *E. coli* (a mixed acid fermenter) from *Enterobacter aerogenes* (a butanediol fermenter). Mixed acid fermenters such as *E. coli* produce a mixture of acids and thus acidify the medium. Butanediol fermenters such as *E. aerogenes* form butanediol, acetoin, and fewer organic acids. The pH of the medium does not fall as low as during mixed acid fermentation. At a pH 4 the methyl red indicator turns red, a positive methyl red test. At a pH of 6, the indicator turns yellow, a negative methyl red test.

Materials

1. Methyl red solution

Procedure

1. Prepare the MR-VP broth media tubes.
2. Using aseptic technique, inoculate each tube with *Vibrio* sps.
3. Incubate all tubes at 37°C for 24 to 48 hours. For slow fermenters, it may take four to five days.
4. Transfer 1/3rd of each culture into an empty test tube and set these aside for the Voges-Proskauer test.
5. To the remaining 2/3rd of the culture, add 0.2 ml (about 4 to 5 drops) of methyl red indicator.
6. Carefully note any color change (a red color is positive).

Voges-Proskauer test

Principle: The Voges-Proskauer test identifies bacteria that ferment glucose, leading to 2,3-butanediol accumulation in the medium. The addition of 40% KOH and a 5% solution of alpha-naphthol in absolute ethanol (Barritt's reagent) will detect the presence of acetoin, a precursor in the synthesis of 2, 3-butanediol. In the presence of the reagents and acetoin, a cherry-red color develops.

Materials

1. Barrits reagent

Solution A – add 6 gm alpha naphthol in 100 ml of 95% ethyl alcohol

Solution B – Add 16 gm of potassium hydroxide in 100 ml of water

Procedure

1. Use the 1/3rd of aliquot from the methyl red test.
2. Add 0.6 ml of Barritt's solution A and 0.2 ml of solution B to each culture, and shake vigorously to aerate. Alternatively, about 15 drops of reagent A followed by 5 drops of reagent B works fairly well and avoids pipetting.
3. Positive reactions occur at once or within 20 minutes and are indicated by the presence of a red color.

Citrate Utilization Test

Principle: The citrate utilization test determines the ability of bacteria to use citrate as a sole carbon source for their energy needs. This ability depends on the presence of a citrate permease that facilitates transport of citrate into the bacterium. Simmons citrate agar slants contain sodium citrate as the carbon source, NH_4^{++} as a nitrogen source, and bromothymol blue serves as pH indicator. The test is done on slants since oxygen is necessary for citrate utilization. When bacteria oxidize citrate, they remove it from the medium and liberate CO_2 . Carbon dioxide combines with sodium and water to form sodium carbonate, an alkaline product. This raises the pH, turns the pH indicator to a blue color, and represents a positive citrate test. Absence of color change is a negative citrate test. Citrate-negative cultures will also show no growth in the medium.

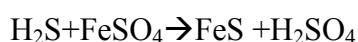
Procedure

1. Prepare Simmons citrate agar slants.
2. Inoculate each tube aseptically by means of a stab-and-streak inoculation.
3. Incubate these cultures for 24 to 48 hours at 37°C.

4. Examine the slant cultures for the presence or absence of growth and for any change in color from green to blue.

Hydrogen Sulfide Production

Principle: A large number of bacteria produce H₂S gas by the degradation of sulphur containing amino acids such as cysteine and methionine. The production of H₂S can be easily detected on Kligler's agar medium by the formation of black precipitate at the streaking or stabbing site. As the medium contains ferrous sulphate, the H₂S produced by the bacteria reacts with ferrous sulphate to form ferrous sulphide.



Requirements

1. Kligler's agar medium
2. Bacterial culture
3. Inoculating needle loop
4. Culture tubes
5. Incubator

Procedure

1. Inoculate a tube of Kligler's medium containing ferrous sulphate.
2. Incubate the tube at 37°C for 24 hrs.
3. Observe the colour change of the medium. Black precipitate will be formed due to formation of ferrous sulphide.

Further Reading

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CULTURE OF STRICT ANAEROBIC MICROBES

Debasis De and R. Ananda Raja

Anaerobic microbiology can be traced back to mid sixteen hundreds and in its early years was rooted in a clinical setting. The pioneering work of Hungate gave fresh impetus to the study of anaerobes and coupled with innovation in cultural techniques brought anaerobic microbiology within the reach of most laboratories.

The origins of anaerobic microbiology are often considered to have begun with Pasteur but in fact go further back to the time of Antoni Van Leeuwenhoek. A remarkable man in many ways he was elected a Fellow of the Royal Society in 1679. In his well detailed correspondence with the Royal Society he described in 1683 bacteria originating in the plaque from between his own teeth. The letter written in his native Dutch language can be translated as “Containing some microscopical observations about animals in the scurf of the Teeth” and contained morphological descriptions of the principal bacterial groupings including spiral forms which were no doubt the first description of anaerobic spirochaetes from the mouth.

It was some fifty years later that Pasteur observed in lactic acid fermentation that air was important to microbial growth. “There’s something new here – in the middle of the drop they are lively, going every which way, but here at the edge they are stiff as pokers.”

In 1861 Pasteur published a paper entitled, “Animalcules infusoires vivant sans gaz oxygène libre et nuisant des fermentations,” in which he concluded that not only would the organisms live in the absence of air but that air actually killed them. Pasteur believed this to be the first example of anaerobic life as he was unaware of the earlier observations of Leeuwenhoek and the conclusions of Spallanzani. Fascinated by his finding he went on to work with other anaerobes notably what we now know as the Clostridia and in 1865 he described *Clostridium septicum*. It was in 1878 that he developed the idea of micro-organisms being differentiated into those, “exclusively aerobic, at once aerobic or exclusively anaerobic” and so the world of anaerobic microbiology formally came into being.

In recent years it is the important task to maintain the cell line of anaerobic microbes in laboratory condition for further research. To isolate the anaerobic microbes and preserving them the anaerobic jar was introduced, based on that originally described by Brown in the early nineteen twenties and later modified by Brewer in which hot platinum was used to catalyse the reduction of oxygen with hydrogen to form water – a principle first applied to the culture of anaerobes in 1915 by Laidlaw. In the original jar, the catalyst was composed of

asbestos wool impregnated with finely divided palladium and enclosed in a fine-mesh brass or copper gauze envelope. The palladium was activated by heating it in a Bunsen burner flame immediately before setting up the jar, and the hydrogen was supplied from a Kipp's apparatus or from a cylinder of the gas. Electrical heating of the catalyst was introduced by Smillie in 1917, and was adopted by Fildes and McIntosh in 1921. Subsequently various minor modifications to jar design were made by Brewer in the late 1930s. But for maintaining the strict anaerobes is not totally possible with anaerobic jars, so in years the microbial and biomedical engineering introduces the anaerobic chamber which consists of automated designed system.

Anaerobic Chamber or Anaerobic hood

For culture of strict anaerobic microbes, anaerobic chamber is essential.

- The anaerobic hood is made up of stainless steel, consists of an inner and outer chamber and a control panel.
- The inner chamber is known as the work chamber consists of incubator and fitted with glove sleeves and electronic internal chamber pressure regulator Systems with a foot switch to operate the vacuum pump fitted at the back end of the system.
- Inner chamber is connected with high purity anaerobic gas cylinders controlled with regulators. The gas cylinders are of N_2/CO_2 and H_2 with minimum 99.99% purity and maximum moisture level is 3ppm, the gas will mixed automatically in a ratio of 80% N_2 , 10% CO_2 , and 10% H_2 .
- The inner chamber consists of charcoal filter, palladium catalyst wafer and desiccant wafer through which the gas mixture passes before entering the chamber.

Operation of anaerobic system:

Following steps are to be remembered for culture of strict anaerobic microbes in anaerobic chamber.

- Please check that the chamber is leak proof before the operation.
- Connect the gas cylinders and close the inner and outer door.
- Switch the catalyst fan and the equalizing knob ON.
- With the inner door securely closed and clamped, open the outer door and place the material to be transferred into the inner chamber, sterilize the outer chamber with 70% ethanol.
- Secure the outer door.

- For an automated transfer cycle press the Cycle start button, the cycle will run for three times with each time the vacuum gauge drops 20 inches.
- When the cycle is complete, the Equalize and both door lights goes out, and the cycle complete light comes ON. Then it is safe to open the inner door and transfer the materials to inner chamber.
- After removing the materials, be sure to secure the inner door. The outer door should always be kept close.
- The same procedures should be followed for removing the materials out from the chamber.

Maintenance of anaerobic microbes in anaerobic chamber:

- It is recommended to keep the incubation temperature at 35 – 37 °C.
- Use sterilized and autoclavable materials to avoid contaminations.
- Start the cycle every day to maintain the anaerobic condition.
- Periodically check the disposable anaerobic strips placed in inner chamber, the strip will start to turn blue at 0.01% O₂.
- If the strips turn blue heat the wafers at 160°C for two hrs and cool it before placing it to desired location and start the cycle for three times.

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ISOLATION OF DNA FROM SHRIMP AFFECTED WITH WHITE SPOT DISEASE

R. Ananda Raja and Sujeet Kumar

Polymerase Chain Reaction (PCR) is very often used in molecular biology with the basic principles of replicating a piece of DNA using primers for selectively amplifying a particular segment of DNA. Isolation of the genomic DNA from shrimp tissues becomes fundamental prerequisite for diagnosis of White spot disease (WSD) using PCR. So, one should get expertise with the isolation and quantification of the genomic DNA from shrimp tissue. It is the basic requirement of the PCR which is well demonstrated in this chapter with standardized method.

Materials required

Buffers and solutions

1. WSSV infected shrimp tissue samples.
2. Absolute ethanol.
3. Autoclaved Milli Q water.
4. DNA digestion buffer (50 mM Tris-HCl, pH 8; 1 mM EDTA, pH 8; 500 mM NaCl; 1 % SDS)
5. Proteinase K 5 μ g.
6. TBE buffer stock-5x (54 g Tris base, 27.5 g Boric acid and 20ml of 0.5M EDTA in 1.0 litre of autoclaved Milli Q water; pH-8.0).
7. TE buffer (100 mM Tris-HCl; 10 mM EDTA in 200 ml autoclaved Milli Q water; pH-8).
8. DNA Loading dye (0.25% bromophenol blue, 0.25% xylene cyanol FF and 15% Ficoll (Type 400) in autoclaved Milli Q water).
9. Ethidium bromide (10mg/ml in autoclaved Milli Q water as stock solution).
10. DNA marker - 100bp and 1kbp ladder.
11. Agarose.

All the chemicals and reagents should be of molecular biology grade.

Equipments

1. Adjustable micropipettes.
2. Refrigerated centrifuge.
3. Deep freezer.
4. Ice flaker.

5. Electrophoresis apparatus.
6. Micro-centrifuge tubes.
7. Glassware.
8. Homogenizer
9. Shaking water bath.
10. Spectrophotometer.
11. Timer.
12. UV transilluminator.
13. Vortex mixer.
14. Adhesive tape.
15. Disposable gloves.

Methods

Sterilized condition should be well maintained in all operations as a preliminary requisite.

I. Isolation of genomic DNA from tissue samples by Proteinase K method

- a. Homogenize 25 mg of gill or pleopod tissue from each collected samples with 500 μ l of digestion buffer.
- b. Incubate for one hour at room temperature followed by boiling at 100° C for 10 minutes in water bath.
- c. Collect the supernatant after centrifuging at 10000x g for 10 minutes at room temperature and add with 2.2 volume of 70 per cent ice-cold ethanol.
- d. Keep for 30 minutes for precipitation of DNA.
- e. After sufficient precipitation, centrifuge the sample at 10000x g for 10 minutes at 4° C.
- f. Collect the pellet, air dry and suspend with 50-100 μ l of 1x TE buffer.
- g. Store the extracted DNA sample at -70° C for further use.

II. Quantitation and Quality checking of DNA

Spectrophotometric method

Take 10 μ l of DNA and dilute in 990 μ l of TE buffer (1:100 dilutions). The ratio of absorbance at 260nm and 280nm is a useful indication of quality of DNA. The ratio is 1.8 for pure DNA. For the quantification of DNA, the O.D at 260nm should be taken and the concentration of DNA to be calculated as follows.

1 O.D of double stranded DNA at 260nm = 50 μ g/ml.

$$\text{Therefore DNA concentration } (\mu\text{g}/\mu\text{l}) = \frac{OD \times \text{Dilution factor} \times 50}{1000}$$

Agarose gel preparation and electrophoresis

1. Seal a clean, dry and perspex gel-casting mould with adhesive tape on both sides and keep on a leveling platform after putting a comb on the slot.
2. Prepare 1.5% agarose in 1x TBE electrophoresis buffer with ethidium bromide to a final concentration of 0.5 $\mu\text{g}/\text{ml}$ and mix thoroughly.
3. Pour warm agarose solution into the mold to the thickness 3mm to 5mm with out air bubbles under or between the teeth of the comb.
4. Allow the gel to get solidified and remove the comb and the tape carefully.
5. Place the gel in the electrophoresis tank filled with 1x electrophoresis buffer.
6. Mix the 3 μl of DNA with the desired gel loading dye (Bromophenol blue & Xylene cyanol) and glycerol mixture and slowly load into the wells of the submerged gel using a micropipette.
7. Load DNA markers in separate wells.
8. Connect the electrodes to the power pack and apply the voltage @ 1-5V/cm distance between the electrodes.
9. Continue the electrophoresis until the dye migrates to the appropriate distance in the gel.
10. Take out the gel from the tank and examine under UV light in a transilluminator.

Further Reading

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DIAGNOSIS OF WHITE SPOT DISEASE IN SHRIMP BY POLYMERASE CHAIN REACTION

R. Ananda Raja and Sujeet Kumar

Polymerase chain reaction (PCR) is well demonstrated in this chapter with standardized method to identify white spot disease from infected field shrimp sample preserved in 95% ethanol. The DNA isolated from the tissue (gills and/or pleopods) by proteinase K method as described in the previous chapter will be used as a template for PCR screening.

Materials required

Buffers and solutions

1. Autoclaved Milli Q water.
2. Reaction buffer without MgCl_2 .
3. MgCl_2 .
4. dNTPs (dATP, dCTP, dGTP, dTTP).
5. Specific forward primer.
6. Specific reverse primer.
7. DNA from infected shrimps.
8. DNA from healthy shrimps.
9. *Taq* DNA Polymerase.
10. TBE buffer stock-5x (54 g Tris base, 27.5 g Boric acid and 20ml of 0.5M EDTA in 1.0 litre of autoclaved Milli Q water; pH-8.0).
11. DNA Loading dye (0.25% bromophenol blue, 0.25% xylene cyanol FF and 15% Ficoll (Type 400) in autoclaved Milli Q water).
12. Ethidium bromide (10mg/ml in autoclaved Milli Q water as stock solution).
13. DNA marker - 100bp ladder.
14. Agarose.

All the chemicals and reagents should be of molecular biology grade.

Equipments

1. Adjustable micropipettes.
2. Deep freezer.
3. Ice flaker.
4. Electrophoresis system.
5. Micro-centrifuge tubes.
6. Glassware.
7. Timer.

8. UV transilluminator.
9. Vortex mixer.
10. Disposable gloves.
11. Thermal cycler.

Methods

Sterilized condition should be well maintained in all operations as a preliminary requisite.

Preparation of reaction mixture

There are three major steps in a PCR, which are repeated for 30 cycles. This is done in an automated Thermal cycler, which can heat and cool the tubes with the reaction mixture in a very short time. For the total reaction volume of 25 μ l, the following components are added in respective volumes in the PCR tube.

Sl. No	Components	Reaction volume
1.	Buffer without MgCl ₂ 10x	2.5 μ l
2.	MgCl ₂ (25mM)	1.5 μ l
3.	dNTPs (2.5mM)	2.0 μ l
4.	Forward primer	1.0 μ l (10pM)
5.	Reverse primer	1.0 μ l (10pM)
6.	DNA Template (50-100 μ g)	1.0 μ l
7.	Autoclaved Distilled Water	15.0 μ l

The Cycling Reaction

The PCR tubes are placed in the thermal cycler at 95° C for 3 min to remove all the secondary structures from DNA and immediately in ice for 5 minutes (initial denaturation). Then, 1.0 μ l (0.1-0.2U/ μ l) of the *Taq* Polymerase is added to the mixture and the thermal cycler is run as follows.

- | | | | |
|--------------------|---|-------------------|-------------|
| 1. Denaturation | - | 95° C for 30 sec. | } 30 cycles |
| 2. Annealing | - | 55° C for 30 sec. | |
| 3. Extension | - | 72° C for 30 sec. | |
| 4. Final Extension | - | 72° C for 5 min. | |

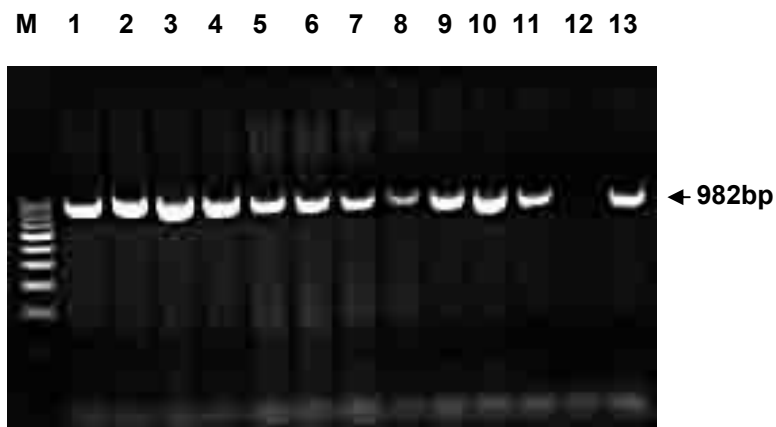
The final amplified product is run in the agarose gel electrophoresis and visualized in transillumination to know the amplification of the product of the interest.

Visualisation of PCR product by agarose gel electrophoresis

1. Seal a clean, dry and perspex gel-casting mould with adhesive tape on both sides and keep on a leveling platform after putting a comb on the slot.
2. Prepare 1.5% agarose in 1x TBE electrophoresis buffer with ethidium bromide to a final concentration of 0.5 μ g/ml and mix thoroughly.

3. Pour warm agarose solution into the mold to the thickness 3mm to 5mm with out air bubbles under or between the teeth of the comb.
4. Allow the gel to get solidified and remove the comb and the tape carefully.
5. Place the gel in the electrophoresis tank filled with 1x electrophoresis buffer.
6. Mix the 3 μ l of DNA with the desired gel loading dye (Bromophenol blue & Xylene cyanol) and glycerol mixture and slowly load into the wells of the submerged gel using a micropipette.
7. Load markers in separate wells.
8. Connect the electrodes to the power pack and apply the voltage @ 1-5V/cm distance between the electrodes.
9. Continue the electrophoresis until the dye migrates to the appropriate distance in the gel.
10. Take out the gel from the tank and examine under UV light in a transilluminator.

Gel picture



Lane M-100bp Ladder,
 Lane 1-11-*P. monodon* WSD positive,
 Lane 12- Negative control,
 Lane 13- Positive control.

LIST OF RESOURCE PERSONS

Sl. No	Name	Designation and address	Contact details
1.	Dr. J. K. Sundaray	Principal Scientist & Officer-in-charge, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 9531604400 Email: jsundaray@gmail.com
2.	Dr. T. K. Ghoshal	Senior Scientist, Nutrition, Genetics and Biotechnology Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 9434251606 Email: ghoshalciba@gmail.com
3.	Dr. Debasis De	Senior Scientist, Nutrition, Genetics and Biotechnology Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 9474728755 Email: dedebasis47@gmail.com
4.	Dr. Ashutosh D. Deo	Senior Scientist, Crustacean Culture Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 9647308131 Email: ashutoshddeo@gmail.com
5.	Dr. R. Ananda Raja	Scientist, Aquatic Animal Health and Environment Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B-743347, India.	Mob. : +91 9474987164 Email: anandarajars@gmail.com
6.	Mr. Gouranga Biswas	Scientist, Fish Culture Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture,	Email: gbis47@gmail.com

		Kakdwip, 24 Pgs (S), W.B. – 743347, India.	
7.	Dr. Sujeet Kumar	Scientist, Aquatic Animal Health and Environment Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 8016452920 Email: sujeetmicrobiol@gmail.com
8.	Mrs. Shyne Anand	Scientist, Crustacean Culture Division, Kakdwip Research Centre of Central Institute of Brackishwater Aquaculture, Kakdwip, 24 Pgs (S), W.B. – 743347, India.	Mob. : +91 9475689764 Email: shyne.anand@gmail.com
9.	Dr. S. N. Joardar	Associate Professor, Dept. of Veterinary Microbiology, Faculty of Veterinary and Animal Sciences, West Bengal University of Animal & Fishery Sciences, Kolkata, West Bengal.	Mob. : +91 9231533335 Email: sid_nj@indiatimes.com

LIST OF PARTICIPANTS

Sl. No	Name of the Participant	Designation and address	Contact details
1.	Mr. Brambhananda Chity	Fishery Extension Officer, Sahid Matangini Development Block, West Bengal-700091	Mob. : +91 9434464199 Email : bchity09@gmail.com
2.	Mr. Atanu Basak	Assistant Research Officer, D.D.F. (M&P), Pailan, West Bengal-700091	Mob. : +91 9836339934 Email : basak_a@rediffmail.com
3.	Mr. Subir Brahma	Assistant Research Officer, D.D.F. (M&P), Pailan, West Bengal-700091	Mob. : +91 9830125983 Email : ankitabrahma.96@gmail.com
4.	Mr. Mohan Chatterjee	Fishery Extension Officer, C.E.O., FFDA, North 24 Parganas, West Bengal-700091	Mob. : +91 9474727408 Email : mohanchahopadhyay@gmail.com
5.	Dr. Koushik Ghosh	Assistant Professor, The University of Burdwan, Golapbag, Burdwan, West Bengal-713104	Mob. : +91 9434251606 Email : kghoshbu@gmail.com
6.	Mr. Biraja Sankar Pattanayak	Research Scholar, Department of zoology, Ravenshaw University, Cuttack, Odisha-753003	Mob. : +91 9439142596 Email : spbiraja@gmail.com
7.	Mr. Debdas Ghosh	Aquaculture Management & Technology, Vidyasar University, West Bengal-743503	Mob. : +91 9903088834 Email : ghoshdebdas9@gmail.com
8.	Mr. Santanu Kumar Mondal	Entrepreneur, Mahabharat Aquatic Resources, Contai, West Bengal-721401	Mob. : +91 9433533144 Email : augniswar@hotmail.com
9.	Mr. Dhruvajyoti Sen	Quality Control Technologist, B. S. Sea Foods Pvt Ltd, Jassore Road, Madhyamgram, West Bengal-700035	Mob. : +91 9804714566 Email : sen.dhruvajyoti267@gmail.com
10.	Ms. Sandipta Shee	Aquaculture Management & Technology, Vidyasar University, West Bengal-712708	Mob. : +91 9804141619 Email : sandipta21@gmail.com
11.	Mr. S. Gokulakrishnan	Project Fellow, Department of Marine Studies & Coastal Resource Management, Madras	Mob. : +91 9840417187 Email : gokulswet@gmail.com gokulswet@yahoo.co.in

		Christian College, Tambaram, Tamil Nadu- 600059	
12.	Mr. Suman Kar	Contractual Technical Assistant, Outreach Activity-I on Fish Feed, KRC of CIBA, Kakdwip, South 24 Parganas, West Bengal-743347	Mob. : +91 9836386096 Email : sumankar12@gmail.com
13.	Ms. Smita Pal	M.Sc. Scholar, Department of Marine Science, Calcutta University, West Bengal-700032	Mob. : +91 9831119109 Email : smita723@gmail.com
14.	Ms. Mahua Roy Chowdhury	J.R.F, Department of Marine Science, Calcutta University, West Bengal-700019	Mob. : +91 9433393495 Email : mahua.rishra@gmail.com
15.	Ms. Ranju Chowdhury	J.R.F, Department of Marine Science, Calcutta University, West Bengal-700019	Mob. : +91 9883062678 Email : onlyrickey@hotmail.com
16.	Dr. Pramod Kumar Nanda	Scientist (SG), IVRI, ERS, Kolkata Centre, West Bengal-700037	Mob. : +91 9674417484 Email : npk700@gmail.com npk70@rediffmail.com
17.	Dr. Asok Biswas	Technical Officer (T 7-8), CIFE, Kolkata Centre, West Bengal-700091	Mob. : +91 9836359543 Email : asokbiswas2000@yahoo.co.in
18.	Dr. Dipak Kumar Mandal	Associate Professor, Visva Bharati University, Kolkata, West Bengal-731236	Mob. : +91 9474166574 Email : dkmandal.vb@rediffmail.com
19.	Mr. Kunal Mondal	JRF, DST Project, Department of Marine Science, Calcutta University, West Bengal-700019	Mob. : +91 9007794059 Email : bottu.marine@gmail.com
20.	Dr. Biplob Kumar Modak	Assistant Professor, Achhruram Memorial College, Jhalda, Purulia, West Bengal-743134	Mob. : +91 9434543692 Email : bkmodak09@gmail.com



**Central Institute of Brackishwater Aquaculture
(Indian Council of Agricultural Research)**

75 Santhome High Road, R.A. Puram, Chennai-600028, Tamil Nadu, India

Tel-No-91-44-24617523/24618817, Fax-91-44-24610311

E-Mail: director@ciba.res.in, Web: www.ciba.res.in



**Kakdwip Research Centre
Central Institute of Brackishwater Aquaculture
(Indian Council of Agricultural Research)**

Kakdwip, South 24 Parganas, West Bengal 743 347, India

Tel. No: 91-3210 255072, Fax : 91-3210 257030

E-mail: krckakdwip@yahoo.co.in

Shrimp as health Food

Brackishwater Aquaculture is not at the cost of Mangroves